# When Does Crime Enter House Prices? Evidence from Chicago

Nency Dhameja[*†]

October 2026

*Click here for the latest version*

**Abstract**

I estimate how crime affects housing prices in Chicago. Although homebuyers presumably care about crime, only four of twenty-two offenses (homicide, burglary, narcotics, and vandalism) consistently lower sale prices significantly. I argue that this may be due to limited information: crime only changes buyers' valuations if they learn of an incident and revise their expectations about future exposure. Three patterns are consistent with this interpretation. First, price effects are hyperlocal even though the predictability of crime is not. Second, price effects strengthen with real estate agents' local experience. Third, price effects vary with incident visibility: burglary discounts are larger for porch incidents than for those inside the home. These patterns suggest that buyers' awareness of nearby incidents matters for which crimes affect house prices.

**JEL classification:** R31, K42, D83, L85

**Keywords:** Crime; repeat sales; housing prices; neighborhood amenities; information frictions; real estate agents; hedonic pricing.

## 1 Introduction

A home is both most homeowners' largest asset and an important consumption good. Neighborhood crime threatens both dimensions: it undermines the safety of the environment in which families live and may reduce the value of the asset they hold. In a 2025 Redfin survey, prospective homebuyers named personal safety and a low crime rate as must-haves more often than any other feature.[1] This paper studies which crimes affect house prices, why some do and others do not, and how crime affects whether and how quickly homes sell. Three findings suggest that the observability of crime matters: (1) price effects are hyperlocal, even though crime a few blocks away predicts future crime near the property; (2) discounts are larger when buyers' agents have more local experience; and (3)

[*]Binghamton University, New York. Email: ndhamej1@binghamton.edu

[†]I am deeply indebted to David Slichter for his guidance, insightful feedback, and continued support. I am grateful to Solomon Polachek, Andreas Pape, Christopher Hanes, and Ivan Korolev for discussions and valuable suggestions.

[1]A 2025 Redfin/Ipsos survey of 1,224 prospective homebuyers finds that personal safety (78%) and a low crime rate (74%) are the two features most often rated "must-haves" among eighteen home and neighborhood features (Anderson, 2025).

for a given category of offense, more visible incidents tend to lower prices more. These patterns are consistent with Gibbons's (2003, p. 28) conjecture that "hard-to-observe crimes will have a weak impact on property prices."

I answer these questions using data from Chicago. I link 530,395 single-family, condominium, and residential-income transactions on 337,461 properties from the Chicago Multiple Listing Service, spanning 2007 to 2025, to 5.59 million geocoded crime incidents, analyzing twenty-two offense categories. The listing data track properties from initial listing to sale or withdrawal, so I observe the asking price, the final sale price, time on market, and withdrawals. The rental listings provide asking rents and listing outcomes for leases.

My identification strategy compares the same property across sales. Parcel fixed effects absorb time-invariant differences across properties, and tract-by-month fixed effects absorb time-varying shocks shared by the surrounding neighborhood, such as local price trends. Within this comparison, I contrast crime within 0.05 miles of the property, roughly the same block, with crime 0.20 to 0.30 miles away in the same neighborhood; nonparametric distance profiles motivate these bands (Section 4). Moving the comparison ring out to 0.25–0.35 or 0.30–0.40 miles leaves the estimates nearly unchanged (Table A6). The estimates are also similar when I omit the comparison ring altogether (Table OA6). I estimate this design for all twenty-two offenses and assess it using post-sale crime placebos, which cannot causally affect a price already set.

Most offenses show no detectable sale-price discount. Only four of the twenty-two carry discounts across multiple exposure windows (homicide, burglary, narcotics, and vandalism), and two more are statistically significant at a single long-horizon window (disorderly conduct and sex offenses). One of the rarest offenses carries the largest discount, about three percent per incident (homicide), while several common or widely feared offenses show no detectable discount (theft, motor-vehicle theft, assault, and robbery). An offense can therefore be severe, frequent, or widely feared and still show no detectable discount.

Why, then, are effects confined to particular offenses, when buyers are presumably averse to crime in general? A simple framework helps us think about this question. For an incident to affect the house price, three conditions must hold: (a) buyers must learn of the incident; (b) learning of it must raise their expectation of future crime at the property; and (c) buyers must be willing to pay to avert that crime at the property. The first condition concerns whether information arrives, the condition Gibbons (2004) conjectures explains why hard-to-observe crimes have weak price effects; the second, what it implies about future crime; and the third, what avoiding that crime is worth: a buyer without a car who observes a car stolen from the curb may expect more car thefts on the block and still pay no less for the house.

Three patterns inform this interpretation. The first concerns how buyers use crime information. Price effects are hyperlocal, whereas the predictive content of crime extends beyond the immediate vicinity of the property. Crime a few blocks away predicts future crime near the property for nearly every offense, yet it does not lower prices. Nor do discounts rise with how well an offense predicts its own recurrence: the most predictable offense carries a small discount (narcotics), and an

offense more predictable than burglary carries none (non-domestic assault). The second concerns observability. For the same offense, incidents at a home's frontage lower prices, while incidents inside the dwelling do not (burglary and narcotics; Table 3). Discounts are significant for incidents on the street but not at residential addresses, although the street–residential difference is significant only for homicide at twelve months (homicide and narcotics; Table A7).[2] Third, real estate agents may help this information travel: the discount is larger when the buyer's agent has more experience in the neighborhood, even after controlling for total experience (homicide; Tables A18 and A19). Since agents are not randomly assigned to transactions, this relationship is descriptive.

The transaction results show that sale-price discounts generally emerge after listing rather than in the initial asking price. Asking prices respond little on average, while closing prices show discounts. Other margins vary by offense: some offenses also lower the sale-to-list ratio (narcotics); there is some evidence that some lengthen time on market, by roughly 12 to 17 percent, or about two to three days against a median of 16 in the estimation sample, under parcel-by-era fixed effects (homicide); and some increase withdrawals despite showing no consistent closing-price discount (weapons violations). For ten of the twenty-two offenses, including all four that are priced, nearby crime raises the probability that a home is listed more than the probability that it sells. Crime therefore affects whether and how quickly a home sells as well as its price, so closing prices alone give an incomplete account of the market's response.

The estimates are robust to controls for time-varying neighborhood and housing-market conditions, including building permits, distinct locations reported vacant, active business licenses, tree canopy, 311 service requests, and recent sales and listings nearby. Inference is also robust to Conley spatial standard errors (Conley, 1999).

This paper makes three contributions.

First, I show that house prices reflect crime selectively, estimating offense-specific price responses to nearby incidents across twenty-two categories under a common empirical design. Earlier work measures either many offenses at aggregate geographies, where estimates depend on the spatial unit (the modifiable areal unit problem, MAUP) and cannot separate crime on a property's own block from broader neighborhood conditions (Thaler, 1978; Schwartz et al., 2003; Gibbons, 2004; Ihlanfeldt and Mayock, 2010), or a single crime at point resolution (Linden and Rockoff, 2008; Pope, 2008a; Congdon-Hohman, 2013; Tang and Le, 2023; Bhattacharya et al., 2021). Braakmann (2017) studies six offense categories at the street level using street fixed effects; I extend this to twenty-two offenses at the incident level with repeat sales.[3]

Second, I provide evidence on the role of observability by comparing where incidents of the same offense occur. These comparisons test Gibbons's conjecture directly. To my knowledge, this is the first test of that conjecture that holds the offense fixed and compares where incidents occur (Table 3). Moreno-Medina (2026) finds that news coverage of nearby homicides lowers house

[2] Among the twelve offense categories examined, homicide receives the most news coverage per recorded incident in the Media Cloud measure, roughly twenty times that of arson, the next-highest (Section 7). Incidents in public space are more widely observed, although I do not measure coverage by incident location.

[3] Table A5 summarizes prior estimates.

prices, identifying one channel through which buyers learn about crime. I examine a complementary dimension of observability: the location of an incident within a given offense category.

Third, I show that crime affects whether and how quickly a home sells, not only its closing price, by tracing each stage of the transaction. I examine asking prices, time on market, withdrawal, and the sale-to-list price ratio, allowing me to distinguish whether crime is incorporated into the seller's initial asking price or emerges through subsequent market adjustment (Table 4). Boggess et al. (2013) find that higher crime is followed by higher housing turnover, Wentland et al. (2014) estimate price and time-on-market effects of registered sex offenders from MLS data, and Dentler and Rossi (2024) show that the density of sales changes within 30 to 60 days of an incident. I extend this analysis to each stage of the transaction, holding the property fixed. Crime can affect whether a sale proceeds, not only its price: weapons violations increase withdrawals despite showing no consistent closing-price discount, and homicide-related closing discounts are larger when the buyer's agent has more local experience (Table 5).

I also extend the design to rental listings, a market studied far less in this literature. Most rent studies use aggregate crime rates (Rizzo, 1979; Koirala et al., 2023; Donovan et al., 2024), and property-level studies examine one or two offenses (Carrillo and Yezer, 2021; Klimova and Lee, 2014). I estimate effects of all offenses on asking rents, time on market, withdrawal, and relisting under the same design as the sale estimates (Table A22).

Economists often use house prices to infer how residents value neighborhood safety (Thaler, 1978; Gibbons, 2004; Linden and Rockoff, 2008). Whether prices reflect nearby crime therefore matters for policy. If some crime-related costs are not reflected in prices, valuations based on housing markets may miss part of the benefits of crime reduction: the absence of a price discount does not imply the absence of a cost. Whether these costs are reflected in prices depends in part on whether buyers learn of nearby crime.

The rest of the paper proceeds as follows. Section 2 develops the conceptual framework, Section 3 describes the data, and Section 4 presents the empirical strategy. Section 5 reports the results, Section 6 assesses robustness, Section 7 discusses interpretation and implications, and Section 8 concludes.

## 2 Conceptual Framework

The observed crime price effect, $\kappa_k$, is the price consequence of a change in expected crime exposure: the proportional change in a property's sale price, $P$, per additional realized incident of offense $k$ nearby, $C_k$,

$$\kappa_k \equiv \frac{\partial \ln P}{\partial C_k}. \tag{1}$$

It is negative when crime lowers prices, and it is the quantity the empirical analysis estimates.[4] A realized incident is one that occurs and is recorded near the property. A buyer, however, values the

[4] $\kappa_k$ is the effect of one additional incident. Table OA7 reports the price effect of one, two, three, and four or more nearby incidents relative to none.

property according to the incidents she expects in the future: let $q_k$ denote the number of incidents of offense $k$ she expects near the property. A realized incident lowers the price only insofar as it raises $q_k$, and the price change depends on three terms. The first is willingness to pay: the share of the property's value a buyer would give up to have one fewer expected future incident, $w_k$. The second is information arrival, or observability: the probability that the buyer learns of a realized incident, $S_k$. The third is belief updating: how much one incident the buyer learns of raises her expected number of future incidents, $b_k(S_k)$. A realized incident therefore raises expected exposure by $S_k\, b_k(S_k)$, and each expected incident lowers the price by $w_k$, so

$$\kappa_k = -\, w_k \times S_k \times b_k(S_k). \tag{2}$$

I explain each in turn.

**Willingness to pay.** Suppose that a household values occupancy at a property over a fixed horizon, which allows us to ignore resale value. Willingness to pay, $w_k$, is expressed as a fraction of the property's value before the incident, so expected crime exposure carries value $w_k q_k$. An offense may impose substantial costs on its victims yet carry little willingness to pay if those costs are not associated with occupying or owning property at that location. A buyer can also expect more incidents and attach no value to avoiding them, in which case $w_k = 0$ and $\kappa_k = 0$ regardless of observability and belief updating.

**Information arrival (observability).** Buyers enter the market with prior beliefs about crime near the property, formed from information already available about the location. A newly realized incident can change those beliefs only if information about it reaches the buyer. $S_k \in [0, 1]$ is the probability of information arrival: $S_k = 0$ means that the buyer never learns of the incident, while $S_k = 1$ means that information about it reaches the buyer with certainty.

**Belief updating.** Conditional on information arrival, the incident may change the buyer's subjective expectation of future exposure. Define

$$b_k(S_k) \equiv \frac{\partial q_k}{\partial C_k^o}, \tag{3}$$

where $C_k^o$ denotes realized incidents of offense $k$ whose information reaches the buyer. Thus $b_k(S_k)$ measures the change in expected future incidents generated by one additional observed incident. Writing $b_k$ as a function of $S_k$ allows the update per observed incident to differ across offenses with how often such incidents are learned of; if buyers know they rarely observe a type of crime, this may affect their update based on a single observation. Unlike $S_k$, $b_k(S_k)$ is not bounded between zero and one. If $b_k(S_k) = 0$, the buyer learns about the incident but does not revise expected future exposure. A value between zero and one means that one additional realized incident raises expected future exposure by less than one incident; a value of one implies a one-for-one increase;

and a value greater than one means that one realized incident leads the buyer to expect more than one additional future incident over the relevant horizon. Importantly, this is a subjective belief update, not the objective recurrence rate of the offense.

Written in units, $w_k$ is a fractional price discount per expected future incident, $S_k$ is the share of realized incidents that reach the buyer, and $b_k(S_k)$ is expected future incidents per observed incident, so the three terms multiply to the price decline per realized incident estimated empirically:

$$\underbrace{\frac{\text{price decline}}{\text{expected incident}}}_{w_k} \times \underbrace{\frac{\text{incidents observed}}{\text{incidents realized}}}_{S_k} \times \underbrace{\frac{\text{expected incidents}}{\text{incidents observed}}}_{b_k(S_k)} = \underbrace{\frac{\text{price decline}}{\text{realized incident}}}_{-\kappa_k}. \tag{4}$$

Consider a buyer who would pay 1 percent of a property's value to avoid one expected future burglary, so $w_k = 0.01$. If one realized burglary reaches the buyer ($S_k = 1$) and raises expected future burglaries over the horizon from a prior belief of 0.5 to 2.5, the belief update is $b_k(S_k) = 2$ and the implied effect is $\kappa_k = -\,w_k \times S_k \times b_k(S_k) = -\,0.01 \times 1 \times 2 = -0.02$: the sale price falls by 2 percent per realized burglary. A value of $b_k(S_k)$ above one simply means one realized burglary leads the buyer to expect more than one additional future burglary. Had the incident not reached the buyer, or reached the buyer without changing expected exposure, this channel would generate no price response.

A zero observed discount can therefore arise through three distinct channels: willingness to pay to reduce expected exposure may be zero ($w_k = 0$); information about the realized incident may not reach the buyer ($S_k = 0$); or the incident may be fully observed but leave expected future exposure unchanged ($b_k(S_k) = 0$). The last case is particularly important. A buyer may place a high value on avoiding future homicide and learn that a homicide occurred nearby, yet view the incident as isolated and therefore not revise the expected risk of future homicide. The price response is then zero despite high willingness to pay and full information. Thus, full information ensures that the incident is known, but not that it changes beliefs about future exposure; a zero estimated discount therefore need not imply low willingness to pay.

Prior studies find that housing prices respond to a risk when information about it is put in front of buyers. Requiring sellers to disclose a hazard lowers prices, even when the information was already public (Pope, 2008b), while hazards that buyers must assess themselves, such as flood risk, are priced only partly or unevenly (Bernstein et al., 2019; Bakkensen and Barrage, 2022; Gourevitch et al., 2023). Asymmetric information about a property's condition and its surroundings also affects prices and who buys (Garmaise and Moskowitz, 2004; Kurlat and Stroebel, 2015; Stroebel, 2016). For crime, house prices fall when a sex offender's address enters a public registry (Linden and Rockoff, 2008) and rebound when the offender moves away (Pope, 2008a), fall near registered gun offenders without any rise in crime (Tang and Le, 2023), and carry large, persistent discounts where deaths in a home must be disclosed (Bhattacharya et al., 2021). Gibbons (2004) finds that vandalism, but not burglary, lowers house prices in London, and suggests that visible crime matters because buyers can see it. Most crime, by contrast, is publicly recorded but not shown where buyers

search for homes.

Evidence from other markets shows that information can be available and still not enter a decision. Chetty et al. (2009) show that consumers respond more strongly to taxes included in posted prices, even when they know that additional taxes apply. Bordalo et al. (2012, 2013) show how salient payoffs and attributes receive disproportionate weight in choice. These mechanisms motivate distinguishing information arrival from the response to information.

**Prediction 1: Observability.** An incident is priced only if it reaches the buyer ($S_k > 0$), so for a given offense, incidents more likely to be observed should carry larger discounts. I test this by splitting incidents of the same offense by where they occurred and whether they are visible from outside the dwelling. Buyers whose agents have more local experience should also be more likely to learn of a nearby incident. I test this with the local experience of the buyer's agent for closing prices and of the listing agent for asking prices; because agents are not randomly assigned, these comparisons are associations.

**Prediction 2: Recurrence.** The belief update $b_k(S_k)$ is subjective and is not observed directly. I therefore ask whether discounts track how well each offense predicts future crime. Holding valuation and information arrival fixed, stronger predictive content should generate a larger discount; across offenses these other components may differ, so recurrence alone need not determine the ranking of discounts. I use a forecasting regression of whether current crime predicts recurrence to measure what an incident objectively implies about future crime, not what buyers believe it implies. If discounts do not rise with recurrence, recurrence alone does not determine which offenses are priced.

**Prediction 3: Price Formation.** The framework describes a change in buyers' valuation but does not say where in the transaction that change appears. If sellers anticipate the effect on buyers' willingness to pay, asking prices should adjust. If they do not, the valuation change can still appear after listing: in a lower sale-to-list ratio, or in non-price margins such as a longer time on the market or a greater chance the listing is withdrawn. I use these margins to trace where the discount enters the transaction.

More generally, the framework is not specific to crime: the price response to any local disamenity with discrete events can be characterized by what buyers would pay to avoid it, whether information about an event reaches them, and how much it changes their expectations.

# 3 Data

The analysis combines three data sources: residential listings from the Chicago Multiple Listing Service, incident-level crime reports from the Chicago Police Department, and measures of neighborhood conditions matched to each parcel. This section describes each source and the construction

of the estimation sample.

Chicago is well suited for this analysis because it provides variation along both dimensions the design requires. First, the volume and breadth of crime incidents allow effects to be estimated separately across the 22 offense categories reported here: the data contain 10,766 homicides alongside more than 1.3 million thefts over the sample period. Second, crime varies substantially at fine spatial scales: roughly 55 percent of the variation in crime exposure within 0.05 miles occurs within tract-months, against 14 percent for crime in the surrounding 0.20–0.30 mile ring. Near-ring crime therefore provides substantially more within-neighborhood variation than crime in the surrounding ring.

**Housing transactions.** The housing data are the universe of Chicago Multiple Listing Service (MLS) closed transactions in single-family, condominium, and residential-income properties. I keep transactions closing between 2007 and 2025 with a recorded close price between $10,000 and $10,000,000, and parcels whose recorded parcel count is the same at every sale, which excludes properties that were split or consolidated between sales (Table OA1). The resulting sample is 530,395 sales on 337,461 parcels, of which 136,866 parcels sell at least twice (329,800 sales) and so identify the within-property comparison. Single-family homes make up 88 percent of these repeat sales, two-to-four-unit buildings 11 percent, and condominiums 1 percent. Each record carries the property identifier, close date, and close price and the full listing history: original and revised list prices, days on market, and whether the listing ended in sale or withdrawal. The sale-price sample consists of closed transactions; the withdrawal analysis instead uses all MLS listings, whether or not they ended in a sale. That history lets me trace how crime affects asking prices, price revisions, time on market, and eventual transaction outcomes.

For the quantity margin, I build an annual panel of all Chicago dwelling parcels in the Cook County Assessor file, excluding vacant land, garages, and minor improvements, 2007–2025. The *listing rate* is the share of parcels listed in the MLS in a given year, and the *turnover rate* is the share that sell. At the parcel level, each is an indicator for whether the parcel is listed, or sells, that year; their sample means are 6.0 and 3.8 percent. Near- and far-ring crime counts are measured over the same calendar year.

**Rental listings.** The rental analysis uses MLS lease listings subject to the same property-type, location, and parcel-count filters as the sale sample: single-family, condominium, and residential-income units in Chicago, on parcels with a constant parcel count, listed between 2007 and 2025 with monthly asking rents of $500 to $10,000. These restrictions yield 191,155 lease listings. Only 0.1 percent record a closing rent, so the rental price outcome is the asking rent, with crime exposure measured relative to the list date. The design otherwise mirrors the sale analysis, including days on market and withdrawal. Applying the same near–far comparison to leases allows me to estimate how nearby crime affects asking rents.

**Property and listing characteristics.** Table A1 summarizes both samples. In the sale sample, the median property has three bedrooms and two bathrooms, and the median sale price is $279,000; the median rental unit has two bedrooms and a monthly asking rent of $1,950. Physical characteristics (living area, bedrooms, bathrooms, and year built) come from standardized listing fields and are available for most transactions. Two-thirds of sold listings record at least one asking-price reduction before closing.

**Crime incidents.** Incident-level crime data come from the Chicago Police Department's public "Crimes, 2001 to present" file. Over 2007–2025, 5,657,404 offenses were reported citywide, of which 98.8 percent carry coordinates for the location of the incident; coverage varies by offense, from 78.8 percent for embezzlement to above 99 percent for most offenses (Table 1). Panel B of Figure 1 maps their density across the city, alongside residential sales in Panel A. Incident locations are released at block-face resolution: the published point is shifted from the actual location for partial redaction but falls on the same block, so distance between a parcel and an incident is measured with error of up to a block face. For homicide the file records one row per victim rather than per incident, so homicide counts, and the per-incident homicide estimates below, are per victim record.[5]

I report estimates for twenty-two offenses, each at every window in Table A3; Appendix A.1 describes the remaining offense groups, which are too sparse or too imprecise to be informative. Online Appendix Table OA3 lists the police offense codes in each group. Assault combines the department's assault codes (threats without physical contact) and battery codes (physical contact), and is split into domestic and non-domestic incidents using the department's domestic flag.

For each listing I count incidents of each offense type within concentric rings of the listing's location over pre-transaction windows of one to twelve months, matching incidents to properties by distance,[6] and time the counts to the listing's close date (to its list date on the lease side). Chicago's largely regular street grid makes straight-line distance a useful approximation to local proximity, although it does not capture barriers or differences in street connectivity.[7] Table 1 reports, for each offense, its citywide count, geocoding rate, and near exposure, together with the raw cross-sectional price gap between listings with and without a nearby incident. These raw gaps are large, exceeding 40 percent for homicide and narcotics, and reflect differences between blocks with and without nearby crime rather than the effect of crime itself. Listings with an incident within 0.05 miles in the three or six months before sale are also cheaper, smaller, and slower to sell (Table OA2).

[5] To protect victim privacy, the Chicago Police Department releases each incident's address "at the block level only" and does not identify the specific location (City of Chicago Data Portal, *Crimes – 2001 to Present*); the recorded point is placed on the correct block, so it can differ from the true location by up to a block face. Chicago blocks are about 330 feet east–west and 660 feet north–south, against a near-ring radius of about 264 feet, so the shift can move an incident into or out of the near ring. Location masking can assign incidents to the wrong distance ring, introducing measurement error into the crime counts. The direction of the resulting bias is not determined a priori.

[6] Distances are great-circle distances from each listing's MLS coordinates.

[7] Distance-based rings do not account for physical barriers such as expressways, rail lines, or rivers. Across-barrier incidents constitute approximately 1–2% of near-ring exposure for the four priced offenses. I do not separately estimate

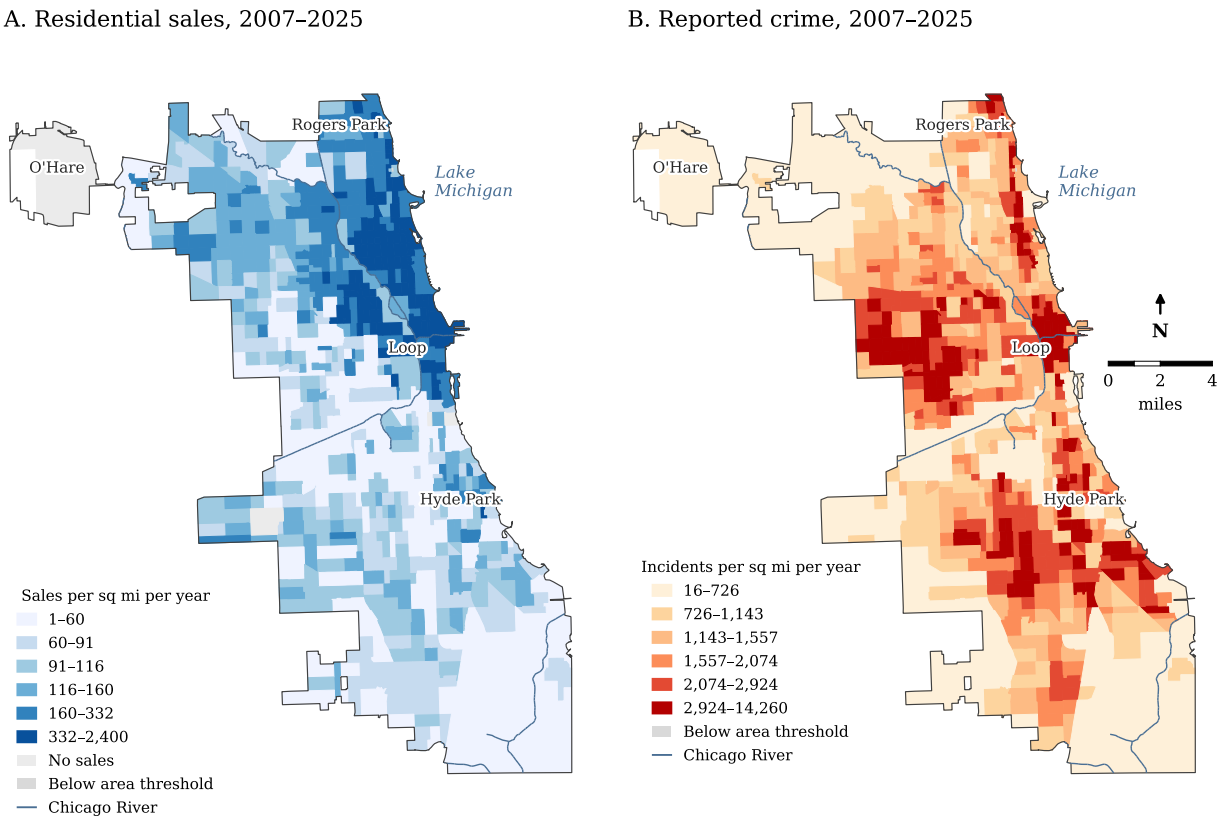



Figure 1: Residential sales and reported crime, Chicago, 2007–2025.

*Notes.* Census-tract densities are expressed as annual averages over 2007–2025 per square mile of land within the city boundary. Panel A shows closed MLS sales in the analysis sample (529,449 of 530,395 sales counted; 30 lack coordinates and 916 lie outside the city boundary); it uses the full sales data rather than the repeat-sales estimation sample. Panel B shows all reported Chicago Police Department incidents with coordinates inside the city boundary (5,575,987 of 5,657,404). Shading denotes sextiles of tract density, calculated separately for each panel; light-grey tracts in Panel A have no sales. Tract portions smaller than 0.01 square miles are shown in grey and excluded from the density classification. Tracts are clipped to the city's land boundary.

Table 1: Crime Incidence and Near Exposure

| Offense | Incidents (city) | Geocoded (%) | Mean near count | % any near | Raw price gap (%) |
|---|---|---|---|---|---|
| Theft | 1,364,912 | 97.8 | 5.56 | 83.2 | 13.7 |
| Assault (non-domestic) | 739,961 | 99.7 | 1.92 | 56.3 | -20.9 |
| Assault (domestic) | 670,887 | 99.7 | 1.71 | 55.5 | -35.6 |
| Criminal damage/vandalism | 632,166 | 99.6 | 1.88 | 67.1 | -18.2 |
| Narcotics | 439,526 | 97.9 | 0.83 | 25.3 | -45.5 |
| Other offenses | 360,851 | 99.0 | 1.09 | 49.5 | -24.1 |
| Burglary | 296,724 | 99.6 | 0.97 | 47.1 | -15.5 |
| Motor vehicle theft | 292,290 | 99.6 | 0.87 | 46.5 | -9.9 |
| Robbery | 213,364 | 99.8 | 0.55 | 32.3 | -12.2 |
| Trespass | 138,862 | 99.7 | 0.41 | 23.8 | -1.3 |
| Fraud | 136,957 | 97.5 | 0.71 | 33.4 | 18.0 |
| Weapons | 114,818 | 99.6 | 0.23 | 14.4 | -36.1 |
| Sex offenses | 63,549 | 89.2 | 0.18 | 14.6 | -9.8 |
| Disorderly conduct | 39,056 | 99.7 | 0.09 | 7.4 | -24.2 |
| Forgery | 32,711 | 97.1 | 0.11 | 8.9 | 15.9 |
| Prostitution | 31,173 | 99.6 | 0.07 | 2.2 | -14.9 |
| Family (nonviolent) | 29,131 | 99.0 | 0.07 | 5.9 | -35.2 |
| Homicide | 10,766 | 100.0 | 0.02 | 2.0 | -42.9 |
| Arson | 9,488 | 99.7 | 0.02 | 2.1 | -35.8 |
| Bad checks | 9,424 | 97.3 | 0.03 | 3.0 | 12.8 |
| Gambling | 8,105 | 99.4 | 0.02 | 1.2 | -72.6 |
| Liquor violations | 7,941 | 99.5 | 0.03 | 2.2 | 2.5 |
| Kidnapping | 5,921 | 99.5 | 0.01 | 1.4 | -34.0 |
| Extortion | 3,281 | 98.7 | 0.01 | 1.1 | 4.4 |
| Child pornography/obscenity | 2,677 | 89.8 | 0.01 | 0.6 | -4.3 |
| Stolen property | 1,586 | 98.6 | 0.00 | 0.4 | 1.3 |
| Embezzlement | 1,055 | 78.8 | 0.00 | 0.4 | 45.4 |

*Notes:* Offense groups ordered by citywide incident count (Chicago Police Department, 2007–2025). Near exposure: 0–0.05 miles, twelve months before sale ($N$ = 530,365). Raw gap: cross-sectional log-price difference (percent) between listings with and without a near incident; descriptive only.

**Neighborhood conditions.** To assess whether the estimated discounts reflect physical decline or disinvestment rather than crime, I construct five measures of neighborhood conditions around each property: building permits, distinct locations reported vacant, 311 service requests, active business licenses, and tree canopy.[8] Each measure is included as a control twice, once for the near ring and once for the far ring, as crime is. Permits, distinct locations reported vacant, and 311 requests are counted over the twelve months preceding the longest crime-exposure window (361 to 720 days before closing),[9] and business licenses are those active at the start of this period; these counts and the active-license snapshot are included as $\ln(1 + x)$. The 311 measure uses the same three request types before and after the City's 2019 change of 311 system, with an indicator for windows overlapping 2019. Tree canopy is included as percent cover, from the raster for the latest calendar year ending before the longest crime-exposure window begins. Where the required data are unavailable, a measure is set to zero with a missing-value indicator, so the sample is unchanged. Appendix A.2 describes the construction of each measure.

Foreclosures can lower nearby prices and increase nearby crime (Campbell et al., 2011; Ellen et al., 2013; Anenberg and Kung, 2014), and vacant-location reports capture only part of this distress. I also construct five measures of local housing-market conditions from the MLS over the same twelve months: the mean repeat-sale price change of neighboring homes, the share of neighboring sales that were distressed (bank-owned real estate, short sale, or foreclosure), the mean sale-to-list ratio, mean days on market, and the number of new sale listings. All housing-market measures exclude the property's own building. Sales-based measures are coded as missing, with an indicator, when no eligible neighboring sales are observed. All neighborhood and housing-market controls are measured before the start of the twelve-month crime-exposure window.

# 4 Empirical Strategy

## 4.1 Design and Identification

The design is a spatial difference-in-differences with two-way fixed effects and continuous treatment. Crime exposure within a narrow ring around the property constitutes the near treatment, while exposure in a far ring around the same property provides the spatial comparison. Identification combines three comparisons: within parcel across transactions, within tract and month, and near relative to farther crime exposure (Figure 2). Parcel fixed effects absorb permanent differences across properties, and tract-by-month fixed effects absorb contemporaneous shocks shared by the surrounding neighborhood. This division of variation is borne out in the data: only 14 percent of far-ring crime exposure varies within tract-months, so the far ring is almost entirely a neighborhood-

their effects.

[8] Tree canopy is from the U.S. Geological Survey National Land Cover Database; business licenses, 311 service requests, building permits, and vacant-building reports are from the City of Chicago Data Portal.

[9] For a sale closing on June 15, 2020, the longest crime-exposure window covers roughly June 2019 to June 2020; permits, vacant locations, and 311 requests are counted from June 2018 to June 2019, licenses are those active in June 2018, and tree canopy is from the 2018 raster.

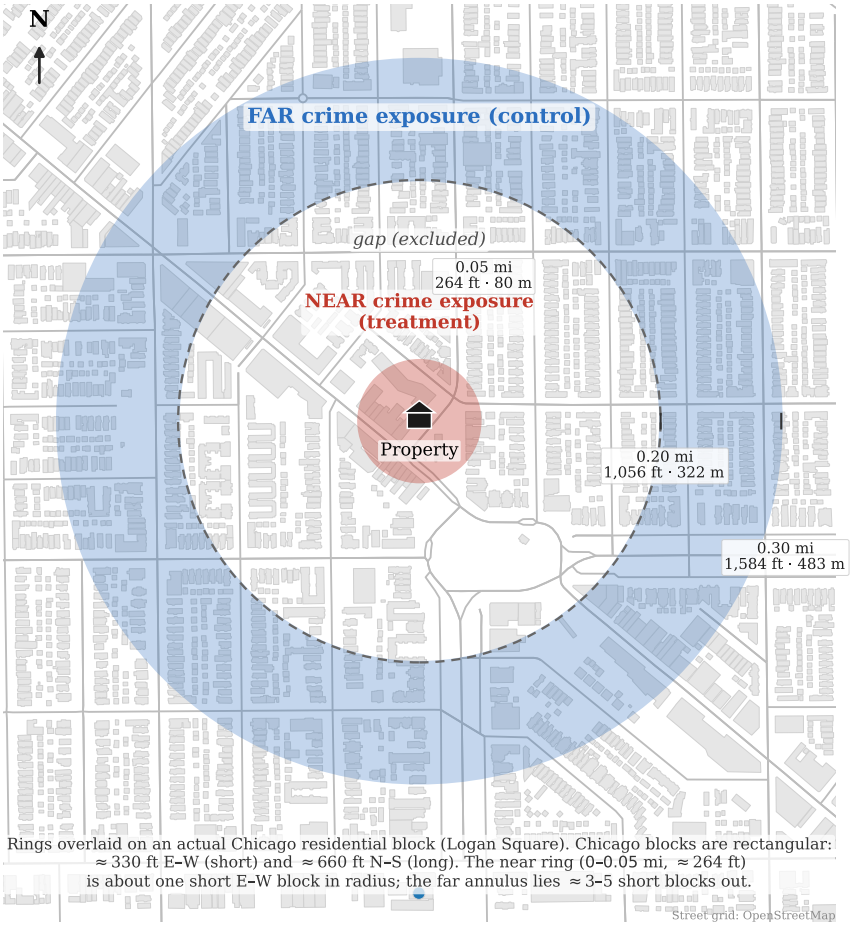



Figure 2: The near–far design. Crime within the near ring (0–0.05 miles, about 260 feet, roughly one short east–west Chicago block) is the treatment; crime in the far ring (0.20–0.30 miles, about three to five short blocks out) is the spatial comparison, separated from the near ring by a gap so the two rings are non-adjacent. The near–far difference $\delta = \beta_{\text{near}} - \beta_{\text{far}}$ differences out conditions common to the surrounding neighborhood. Near ring in red, far ring in blue; incidents in the gap and beyond the far ring (grey) are excluded.

level quantity that the fixed effects absorb nearly in full, whereas a majority of near-ring exposure survives within tract-months (Section 3). The near-relative-to-far contrast is therefore identified from within-neighborhood differences in exposure. The parameter of interest is the differential price response,

$$\delta = \beta_{\text{near}} - \beta_{\text{far}}.$$

The estimate $\delta_k$ is the empirical counterpart of $\kappa_k$ in Equation (1), under the linear approximation and the far-ring restriction.

For property $i$ transacting at date $t$, let $\text{Near}_{it}$ be the count of incidents of a given offense in the near ring, within 0.05 miles of the parcel, in the $m$ months before the sale, and $\text{Far}_{it}$ the count in the far ring, 0.20–0.30 miles, over the same window, the same comparison ring as Tang and Le (2023); Table A6 reports alternative far rings. Stale counts $\text{SNear}_{it}$ and $\text{SFar}_{it}$, near- and far-ring incidents in the part of a twelve-month lookback that falls *beyond* the $m$-month window,

and therefore non-overlapping with $\text{Near}_{it}$ and $\text{Far}_{it}$, absorb slower-moving local conditions; they are empty when $m = 12$. The window length $m$ is not fixed a priori; I estimate across $m = 1, \ldots, 12$ months and examine the choice in Section 4.2. I report estimates for all twelve exposure windows to show how the results vary with the period over which nearby crime is measured. The estimating equation is

$$\ln P_{it} = \beta_{\text{near}}\text{Near}_{it} + \beta_{\text{far}}\text{Far}_{it} + \gamma_n\text{SNear}_{it} + \gamma_f\text{SFar}_{it} + \mu_i + \lambda_{c(i)t} + \varepsilon_{it}, \tag{5}$$

where $\mu_i$ is a parcel fixed effect and $\lambda_{c(i)t}$ a tract×month fixed effect.

Identification requires that, after accounting for parcel and tract-by-month fixed effects and the included controls, the difference in crime exposure between the near ring and the far ring is unrelated to other factors that affect the sale price. This does not require near and far rings to be alike in levels, or unconditional parallel trends across high- and low-crime properties. Parcel fixed effects absorb time-invariant property differences, and tract-by-month fixed effects absorb shocks shared within a tract and month. What remains is block-level variation across a property's own transactions. Formally, let $\mathbf{X}_{it} = (\text{Near}_{it}, \text{Far}_{it})'$. Identification requires

$$E\left[\widetilde{\mathbf{X}}_{it}\,\widetilde{\varepsilon}_{it}\right] = \mathbf{0}, \tag{6}$$

where tildes denote residualization with respect to the fixed effects and controls in Equation (5). Under this condition $\beta_{\text{near}}$ and $\beta_{\text{far}}$ are identified, and so is $\delta = \beta_{\text{near}} - \beta_{\text{far}}$.

Two threats to this assumption, and the interpretation of $\delta$, are addressed directly.

1. *No omitted shocks correlated with the near–far crime difference.* A local event, such as a neighboring building becoming vacant, could raise crime on the block relative to the far ring while independently lowering the property's price. The near–far comparison removes shocks shared equally by the two rings, but not those confined to the block, so the estimate could attribute such an event's price effect to crime. I assess sensitivity to these conditions by adding predetermined neighborhood and lagged housing-market controls (Section 6). These controls capture observable pre-existing conditions but cannot rule out unobserved shocks during the exposure period.

2. *No reverse causality or confounding price dynamics.* Crime could respond to earlier prices, or both could reflect a changing block: a temporary decline in local property values could coincide with rising crime, so the estimate would capture that trajectory rather than the effect of an incident. Measuring crime before the sale establishes temporal order but does not rule this out. I assess the concern with a mirror placebo for every estimate, replacing pre-sale crime with post-sale crime. A significant placebo raises concerns about confounding dynamics or anticipated risk; a null is consistent with the identifying assumption but does not establish it.

The interpretation requires that the far ring is beyond the effect's reach. $\delta$ compares marginal

responses to near and far exposure and equals the near-ring effect only under the far-ring restriction, that crime a fifth to three tenths of a mile out has no price effect of its own. Crime at that distance is the benchmark for what would have happened on the block anyway; if an incident there still moved the price, the contrast would recover a relative effect rather than the full near-ring effect. If $\beta_{\text{far}} < 0$, subtracting it makes $\delta$ less negative than $\beta_{\text{near}}$ alone. The spatial decay results are consistent with the restriction: the discount is no longer detectable at the adjacent block.

The design does not assume that crime is randomly distributed across the city, across neighborhoods, or over time; only that the residual near–far variation is unrelated to unobserved price shocks. Section 6 reports each check.

The parcel fixed effect restricts identification to the 136,866 parcels that transact more than once (329,800 sales). The tract-by-month fixed effect then drops any transaction that is the only sale in its cell, together with the singletons this creates, leaving the 242,815 transactions that identify the two-way model and form the sample in the results tables (Table OA1).

Enforcement intensity affects recorded crime directly (Levitt, 1998) and can bias the estimate in either direction. Where enforcement deters crime and improves the block, detection may still raise recorded counts, so a block that is getting safer shows more incidents and the discount is attenuated. Where enforcement is aimed at blocks that are already deteriorating, recorded counts rise alongside falling prices and the discount is inflated. Tract-by-month fixed effects absorb enforcement changes shared within each tract and month.[10] Uneven deployment within tracts, including enforcement targeted at particular addresses, remains a concern. This matters for offenses recorded through enforcement rather than victim complaint, which here means narcotics, prostitution, gambling, and liquor violations, and I describe those measures as enforcement-recorded offenses rather than as a census of crime.

### 4.2 Choice of Spatial Bands and Exposure Windows

The rings and windows in (5) are chosen from the data rather than imposed. I estimate the price effect as a smooth function of distance and time since incident by a tensor-product spline[11] (Wood, 2017; Korolev, 2026). The resulting distance profile, reported in Section 5.2, guides the choice of the 0–0.05 mile near ring. The 0.20–0.30 mile far ring serves as the reference region for the spline (Figure 4; Online Appendix J). Margaretic and Sosa (2025) find that estimated crime spillovers depend on the spatial scale of aggregation. Administrative boundaries need not match the area over which households experience crime. Estimates may therefore depend on the geographic unit chosen,

[10] A Chicago police beat averages about 0.8 square miles and spans roughly three census tracts; the 791 tracts in the estimation area average about 0.29 square miles. Beats nest within sectors and sectors within the city's roughly two dozen districts. Beat and tract boundaries cross rather than nest, so beat-level deployment is absorbed only in part.

[11] A tensor-product spline models a function of two variables as a weighted sum of products of one-dimensional basis functions, here a basis in distance from the parcel and a basis in months since the incident. Each product term lets the profile in one dimension vary with the other, so the distance profile can differ across lags without imposing a parametric form. The basis sizes are $c(6, 12)$, six in distance and twelve in lag, and the smoothness penalty is chosen by restricted maximum likelihood (Wood, 2017).

the modifiable areal unit problem (MAUP). I report estimates across nine near–far ring definitions (Table A6) and all one-to-twelve-month windows. Summary tables use the main window among those with significant discounts; Table 2 reports estimates across all twelve windows. Significant-window counts are descriptive; the windows are nested, not independent.

Section 6 reports four robustness checks. A post-transaction placebo addresses reverse causality and mean reversion; predetermined near–far measures of neighborhood conditions address hyper-local confounding; a full-universe model of sale propensity and a complementary selection correction assess endogenous transaction selection; and a mass-point diagnostic following Caetano (2015) and Caetano et al. (2021) addresses selection into zero crime exposure. The full set of diagnostics is reported in the robustness section. The comparisons that follow use incident location and recurrence to distinguish among the mechanisms in the conceptual framework.

The parameter of interest is the effect of a recent incident on the property's block, relative to the surrounding neighborhood in the same month. The spatial scale is not imposed: the distance profile in Figure 4, estimated without rings, locates the response within 0.05 miles, and the rings follow it. The estimates do not capture the long-run co-movement of crime and prices across neighborhoods.

**Main window.** I report estimates for all twelve exposure windows. When a table requires one estimate per offense, I use the median window among those with a negative estimate significant at five percent, choosing the lower median when their number is even. This gives ten months for homicide and narcotics, eight for burglary, and six for vandalism. For offenses with no significant discount, I report the twelve-month estimate.

# 5 Results

## 5.1 Sale Prices

Four of the twenty-two offenses, homicide, burglary, narcotics, and vandalism, carry a significant sale-price discount in at least two of the twelve exposure windows, which count crime from one to twelve months before the sale (Table 2; Appendix Table A2; Figure 3). At each offense's main window, one additional incident within a radius of 0.05 miles of a home, relative to one 0.20–0.30 miles away, lowers its sale price by about 3 percent for homicide ($-3.02$, s.e. 1.38), 0.24 percent for burglary (s.e. 0.12), 0.29 percent for narcotics (s.e. 0.14), and 0.17 percent for vandalism (s.e. 0.08).[12] I refer to these as *priced offenses*. Disorderly conduct and sex offenses are *suggestive*, with a significant discount in only one window. The remaining sixteen offenses are *nulls*, with no significant discount in any window; for arson, gambling, stolen property, and weapons, however, the estimates are too imprecise to rule out sizable effects. Table A3 reports full results for every offense at all twelve windows.

[12] The main window is the median of the windows with a negative estimate significant at five percent, taking the lower median when their number is even. The 0–0.05 mile near ring and the 0.20–0.30 mile comparison ring follow the distance profile of the price response (Sections 4.2 and 5.2); Table A6 reports alternative rings.

Table 2: Sale Prices

| Offense | Identifying parcels | Sig. discount windows | Min $\delta$ (%) (s.e.) [mo] | Max $\delta$ (%) (s.e.) [mo] | "Median" $\delta$ (%) (s.e.) [mo] |
|---|---|---|---|---|---|
| *Priced* | | | | | |
| Homicide | 3,053 | 2/12 | $-5.86^{*}$ (3.35) [1] | $-1.24$ (1.43) [8] | $-2.23$ (2.60) [2] |
| Narcotics | 35,801 | 5/12 | $-0.93$ (0.75) [1] | $-0.18$ (0.19) [6] | $-0.33^{***}$ (0.11) [12] |
| Burglary | 64,661 | 6/12 | $-0.34^{**}$ (0.14) [6] | $-0.18^{*}$ (0.09) [12] | $-0.22^{**}$ (0.10) [11] |
| Vandalism | 77,053 | 2/12 | $-0.17^{**}$ (0.08) [6] | 0.26 (0.22) [1] | $-0.10$ (0.07) [9] |
| *Suggestive* | | | | | |
| Disorderly conduct | 13,670 | 1/12 | $-1.33^{*}$ (0.78) [2] | $-0.46$ (0.54) [4] | $-0.65$ (0.63) [3] |
| Sex offenses | 26,831 | 1/12 | $-0.47^{**}$ (0.23) [12] | 0.11 (0.35) [3] | $-0.26$ (0.25) [9] |
| *Precise nulls* | | | | | |
| Larceny/theft | 86,303 | 0/12 | 0.01 (0.02) [12] | $0.08^{***}$ (0.03) [1] | 0.02 (0.01) [8] |
| Motor vehicle theft | 63,142 | 0/12 | $-0.02$ (0.09) [12] | $0.51^{*}$ (0.30) [1] | 0.07 (0.09) [11] |
| Assault (non-dom.) | 67,192 | 0/12 | $-0.05$ (0.06) [7] | 0.01 (0.11) [3] | $-0.03$ (0.09) [4] |
| Bad checks | 7,580 | 0/12 | $0.66^{*}$ (0.35) [11] | $2.20^{**}$ (1.03) [1] | $1.08^{**}$ (0.43) [7] |
| Family (non-violent) | 10,002 | 0/12 | $1.00^{*}$ (0.51) [12] | $2.16^{***}$ (0.82) [5] | $1.49^{**}$ (0.60) [9] |
| Fraud | 53,093 | 0/12 | $-0.36$ (0.23) [1] | $-0.04$ (0.09) [6] | $-0.08$ (0.08) [10] |
| Assault (domestic) | 64,422 | 0/12 | $-0.19^{*}$ (0.10) [12] | 0.02 (0.15) [5] | $-0.03$ (0.12) [7] |
| Trespass | 41,169 | 0/12 | $-0.15$ (0.14) [6] | 0.18 (0.22) [3] | $-0.09$ (0.38) [1] |
| Robbery | 48,458 | 0/12 | $-0.06$ (0.14) [9] | 0.30 (0.42) [1] | 0.01 (0.13) [11] |
| Forgery | 19,519 | 0/12 | $-0.07$ (0.59) [1] | 0.47 (0.45) [2] | 0.23 (0.18) [11] |
| Liquor law | 5,074 | 0/12 | 0.08 (1.20) [1] | 1.34 (0.91) [2] | 0.51 (0.36) [10] |
| Prostitution | 4,671 | 0/12 | $-1.00$ (0.82) [1] | 0.08 (0.12) [12] | $-0.18$ (0.35) [5] |
| *Inconclusive* | | | | | |
| Weapons | 20,182 | 0/12 | $-0.62$ (0.47) [5] | 0.11 (0.58) [3] | $-0.17$ (0.35) [9] |
| Arson | 3,515 | 0/12 | $-0.20$ (1.06) [11] | 1.27 (3.58) [1] | 0.07 (1.63) [6] |
| Gambling | 1,333 | 0/12 | $-4.77$ (4.33) [3] | 1.99 (8.32) [1] | $-1.79$ (1.77) [11] |
| Stolen property | 1,012 | 0/12 | $-2.23$ (2.63) [2] | 0.81 (1.20) [8] | $-0.67$ (1.75) [6] |

*Notes.* $\delta = (\beta_{\text{near}} - \beta_{\text{far}}) \times 100$: percent change in sale price per incident within 0–0.05 miles, relative to 0.20–0.30 miles. Parcel and tract × year-month fixed effects; standard errors clustered by tract; $N = 242{,}815$ repeat sales, 2007–2025. Estimated at each cumulative window, one to twelve months. Sig. windows: negative and significant at 5%. "Median" $\delta$: the sixth-lowest of the twelve window estimates (one of the two medians); Min and Max: most negative and most positive estimates across the twelve windows (standard error; window in brackets). Identifying parcels: exposure varies within parcel, counted at the twelve-month window. Priced: a discount significant at 5% in two or more consecutive windows. Suggestive: a discount significant at 5% in exactly one window and at 10% in an adjacent window. Precise nulls: no discount significant at 5% in any window, and the 95 percent intervals rule out a one-percent discount in at least nine of twelve windows; for larceny, motor vehicle theft, non-domestic assault, bad checks, and family offenses they also rule out a discount as large as burglary's main-window estimate (0.24 percent, eight months) in at least nine of twelve windows, for the last two because their estimates are positive. Inconclusive: the remaining offenses with no discount significant at 5% in any window. Kidnapping, embezzlement, extortion, child pornography, and other offenses have too few incidents to estimate.
$^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

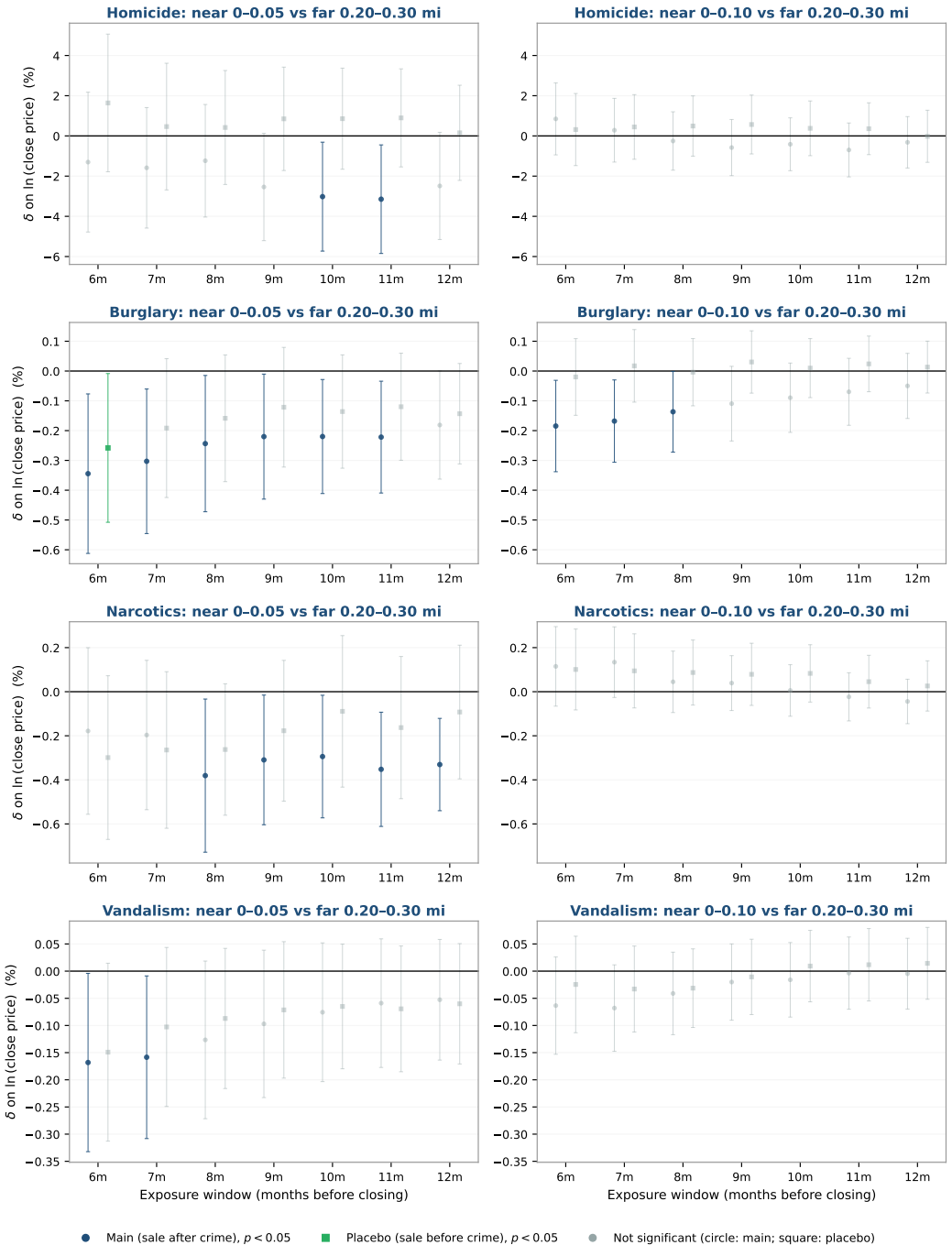



Figure 3: Sale-Price Discounts and Post-Sale Placebos by Exposure Window. Left: near ring 0–0.05 miles (baseline). Right: 0–0.10 miles. Both against the 0.20–0.30 mile comparison ring; counts; parcel and tract×month fixed effects; no controls; clustered by tract; $N = 242{,}815$. Circles: crime in the window before the sale. Squares: post-sale placebo (crime in the same window after closing). 95 percent intervals; each row shares a scale.

For larceny, motor vehicle theft, and non-domestic assault, the intervals also rule out, in at least ten of twelve windows, a discount as large as burglary's eight-month estimate (0.24 percent per incident, s.e. 0.12). The estimates for bad checks and family offenses are positive. Tables OA9 and OA10 split assault and the other offenses by domestic status.

Fraud, forgery, and bad checks impose direct monetary losses, yet none carries a detectable discount in the pooled estimates; embezzlement has too few incidents to estimate (Appendix B.1).

I next examine how discounts vary with the number of nearby incidents, replacing the near-ring count with indicators for one, two, three, and four or more incidents, relative to zero (Table OA7; Figure OF1). Four or more burglaries and vandalism incidents are associated with discounts of 1.85 (s.e. 0.76) and 1.05 percent (s.e. 0.52) at eight and six months, respectively. No priced offense shows a significant discount for exactly one incident at any window. A joint test of linearity over zero to three nearby incidents rejects the restriction for burglary at eight months ($p = 0.02$), but not for the other priced offenses at their main windows.

### 5.2 Distance Profile of Discounts

The distance profile of the price response supports the bands chosen in Section 4.2. Discounts decline with distance, consistent with studies of sex-offender arrivals that find price effects concentrated within about a tenth of a mile (Linden and Rockoff, 2008; Pope, 2008a). A tensor-product spline in distance and lag, estimated as a fixed-effects signal regression following Korolev (2026), shows this decay (Figure 4). For burglary and narcotics, the effect is significantly negative within 0.05 miles and not significantly different from zero at the adjacent-block ring. Vandalism's distance profile is not significantly different from zero within 0.05 miles, consistent with its small per-incident discount (Table 2). Homicide's confidence band is roughly ten times wider than those of the other three offenses and includes zero within 0.05 miles, consistent with sparse homicide exposure (Online Appendix J). In the time dimension, the burglary effect is concentrated among incidents about three to seven months before the sale and the vandalism effect three to six months before, while the narcotics profile shows no significant pattern; Appendix C.1 compares recent with earlier exposure.

### 5.3 The Four Priced Offenses

**Homicide.** The homicide estimate is significant at ten and eleven months; shorter windows contain few homicides and are imprecise. This discount is similar in magnitude to estimates from other studies of individual murders (Table A5). Klimova and Lee (2014) find a 3.9 percent decline in sale prices within 0.2 miles of a murder in Sydney during the following year, while Gautier et al. (2009) find a decline of about 3 percent in asking prices in Amsterdam neighborhoods with large Muslim populations following the van Gogh murder. Discounts are substantially larger for the property where the incident occurs: in Hong Kong, such units sell for roughly a third less, with discounts of about 19 percent for other units on the same floor and 10 percent for units elsewhere in the building, and no statistically significant recovery over time (Bhattacharya et al., 2021). At the neighborhood level, Ihlanfeldt and Mayock (2010) find no detectable effect of murder on house

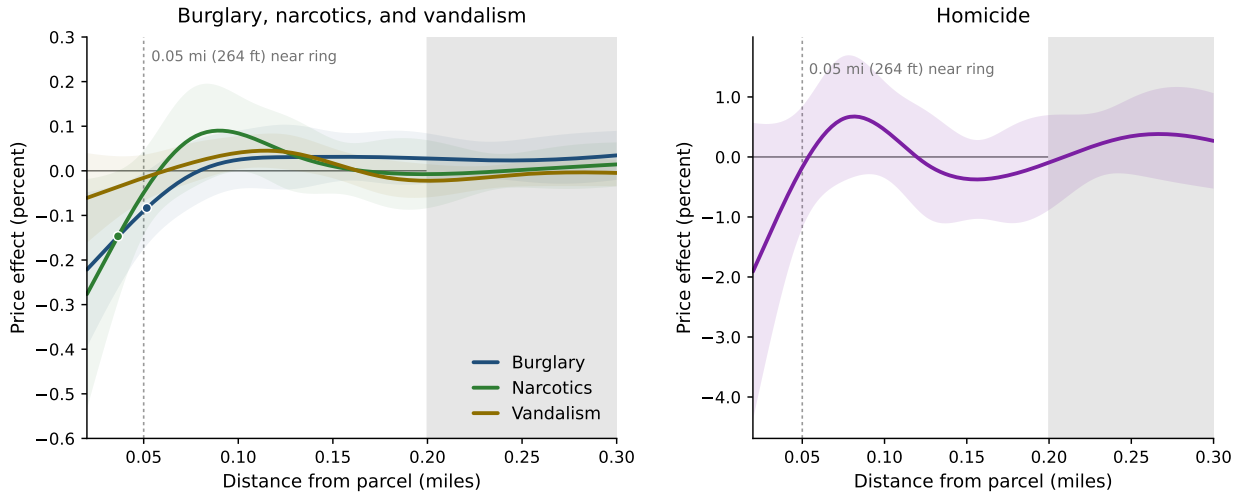



Figure 4: Price response by distance from the parcel. Distance profiles from a tensor-product spline in distance and lag, estimated as a fixed-effects signal regression in `mgcv` following Korolev (2026), fit on 0–0.30 miles and centered on their mean, averaged over the one-to-twelve-month windows; MLS close-date sample, 2007–2025, with 95 percent cluster-robust bands. *Left*: burglary, narcotics, and vandalism. *Right*: homicide, whose band is wider because incidents are rare and includes zero within 0.05 miles. The dotted line marks the 0.05-mile near-ring edge; the shaded band is the 0.20–0.30 mile comparison ring. The profile locates where the response is negative and where it is flat; its level is not the near–far discount in Table 2 (estimation details in Online Appendix J).

prices, while effects of aggravated assault and robbery emerge with a lag of two to four years. The contrast suggests that the geographic scale of crime measurement may matter for the estimated magnitude and timing of price effects, although differences in settings and research designs also limit direct comparisons. Homicide discounts also appear when the near ring is 0.10–0.15 miles (significant at six to nine months, e.g., −2.86 percent at six months, s.e. 0.89, with no significant post-sale placebo; Figure 5), although the estimates do not decline smoothly with distance: the 0.05–0.10 mile ring shows no discount.

**Burglary.** The burglary discount is significant from six to eleven months. At the baseline 0–0.05 mile ring, sales with zero nearby burglaries do not follow the price trend among sales with one or more (Caetano, 2015). Part of the burglary discount may therefore reflect how sales with zero exposure differ, rather than burglary itself. The post-sale placebo is also significant at six months. Widening the near ring to 0–0.10 miles addresses both concerns: the zero-exposure test does not reject, the placebo is not significant in any window, and the discount remains significant at six to eight months (−0.14 percent at eight months, s.e. 0.07; Figure 3; Section 6).[13]

**Narcotics.** The narcotics discount is significant from eight to twelve months and stable across them, between −0.29 percent (s.e. 0.14, ten months) and −0.38 percent (s.e. 0.18, eight months) per incident, and it is confined to 0.05 miles (Appendix C). Narcotics incidents are recorded largely

[13] With a near ring of 0–0.15 miles, close to the 250-metre radius in Gibbons (2004), the burglary estimate is not significant at any window from six to twelve months (−0.10 percent at six months, s.e. 0.06, to −0.02 percent at twelve, s.e. 0.04), consistent with his finding of no burglary effect in London.

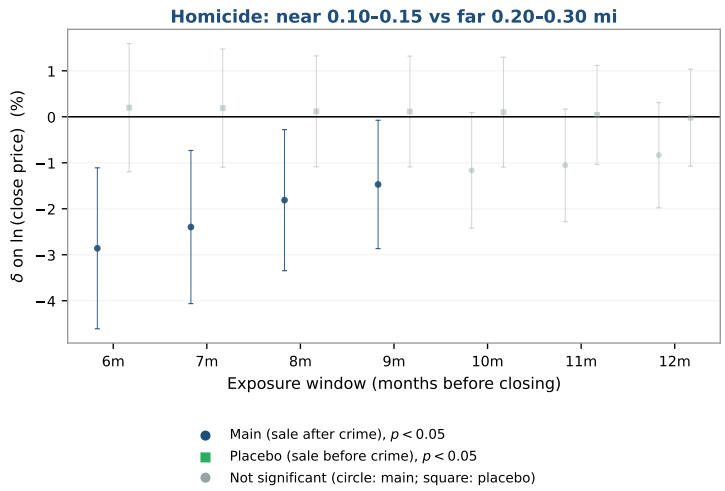



Figure 5: Homicide: Discount at a Near Ring of 0.10–0.15 Miles. Near ring 0.10–0.15 miles against the 0.20–0.30 mile comparison ring; counts; parcel and tract×month fixed effects; no controls; clustered by tract; $N = 242{,}815$. Circles: crime in the window before the sale. Squares: post-sale placebo (crime in the same window after closing). 95 percent intervals.

through police enforcement rather than victim reports, so the estimate captures the price response to recorded drug activity and enforcement on the block (Section 4.2). The per-incident discount is far smaller than the 10 to 19 percent decline that Congdon-Hohman (2013) find near a discovered methamphetamine laboratory, a rarer and more conspicuous event.

**Vandalism.** The vandalism discount is significant only at six and seven months and is confined to 0.05 miles (Appendix C). Unlike the narcotics discount, it shrinks as older incidents enter the window, from −0.17 percent at six months (s.e. 0.08) to −0.05 percent at twelve (s.e. 0.06), consistent with damage that is visible for a time and then repaired. The spline places the vandalism effect among incidents three to six months before the sale, and four or more incidents are associated with a discount of 1.05 percent (s.e. 0.52) at six months.

## 5.4 Mechanisms

Sixteen of twenty-two offenses carry no detectable discount, and two more (disorderly conduct and sex offenses) are significant only at a single long-horizon window.

**Location of the incident** Prediction 1 says that, holding the offense fixed, incidents more likely to be observed carry larger discounts. I split incidents of the same offense in two ways: (1) by the type of place where they were recorded (a residential address, the street, or a commercial premises), and (2) among incidents at residential addresses, by whether they occurred inside the dwelling or

at its frontage (the garage, porch, yard, driveway, or entryway).[14]

The homicide and narcotics discounts are significant for incidents on the street (in 4 and 5 of 12 windows) but not for incidents at residential addresses; the two estimates differ significantly only for homicide at the twelve-month window ($p = 0.046$); the vandalism discount is concentrated at residential addresses (8 of 12 windows) and differs significantly from street vandalism in 7 of 12 (Tables A7 and A8).[15] At its main six-month window, residential vandalism is associated with a 0.56 percent lower sale price per incident (significant at one percent), compared with an estimate of −0.01 percent for street vandalism.

The interior–frontage split compares incidents of the same offense recorded at residential addresses but in locations that differ in their likely visibility from the street.[16] For burglary, frontage incidents carry a discount (−0.74 percent, s.e. 0.33), significant in 7 of 12 windows, while interior incidents have no detectable effect at any window (Table 3); the difference is significant at twelve months ($p = 0.030$) but not at the main window ($p = 0.109$).[17] For narcotics, frontage discounts are significant in seven of twelve windows, while no significant discount is detected for interior incidents. The frontage–interior difference is significant in six of twelve windows. For vandalism, the discount is concentrated instead in interior incidents (−0.73 percent, s.e. 0.25; 9 of 12 windows).[18] As a comparison, larceny shows no detectable discount at either location, and the difference is not statistically significant (Table OA15).

**Target of the incident** Among frontage incidents, damage to property can leave a visible trace at the site, whereas damage to a car can leave with the vehicle. Detailed vandalism codes distinguish property damage from vehicle damage.[19] Among frontage vandalism, property-damage incidents carry significant discounts in six of twelve windows, while vehicle-damage incidents show

[14]Interior incidents are those whose recorded location is a residence, apartment, Chicago Housing Authority apartment, house, coach house, rooming house, or trailer; frontage incidents are those recorded at a residential garage, porch, yard, driveway, or entryway, or in a hallway or common area (Table OA14). The residence and apartment codes record that an incident occurred at the dwelling without a more specific exterior location, so the interior category may include some incidents outside the living space.

[15]The pooled residential burglary estimate has a comparable post-sale placebo estimate (Table 3 note), so the frontage–interior split, whose placebos are clean, provides the relevant test for burglary.

[16]Bhattacharya et al. (2021) similarly find that deaths of residents occurring outside the home carry no discount, while deaths inside the unit do. Their comparison varies whether the event is attached to the property; mine varies whether an incident near the property is observable from the street. Frontage also includes porch, hallway, and common-area codes, which make up 39 percent of frontage-coded incidents across all offenses (Table OA14), so frontage is an imperfect proxy for street visibility.

[17]Interpreting this difference as evidence of observability assumes that interior and frontage burglaries signal similar future risks and that buyers value avoiding them similarly. The most evident difference between them works against the result: an intrusion into the dwelling is more harmful, so buyers who know of it should demand a larger discount, yet interior burglaries carry no detectable discount.

[18]CPD masks each incident to the hundred-block, so an interior record cannot be separated between the sold parcel and a neighbor's. For burglary this matters little, since the loss is the stolen contents and any entry damage is minor; for vandalism an interior record is damage to the property itself, which lowers its value as repair cost rather than through information. For vandalism, the location split therefore separates damage to the dwelling from damage outside it rather than visible from hidden incidents.

[19]Vandalism is the only priced offense for which detailed police offense codes distinguish property damage from vehicle damage; homicide targets a person, while burglary and narcotics codes record the method of entry or the drug, not the target.

Table 3: Within-Offense Comparisons by Incident Location and Target

| | Exposure window (months before sale) | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 |
| *Panel A. Observability: residential incidents, frontage versus interior* | | | | | | | | | | | | |
| Burglary | | | | | | | | | | | | |
| Frontage (visible) | −0.37 | −0.45 | −0.54 | −0.68 | −0.54 | −0.74** | −0.76** | −0.74** | −0.62** | −0.70** | −0.71*** | −0.73*** |
| | (0.77) | (0.59) | (0.49) | (0.46) | (0.41) | (0.36) | (0.35) | (0.33) | (0.29) | (0.27) | (0.25) | (0.24) |
| Interior (hidden) | −0.17 | −0.27 | −0.12 | −0.11 | −0.20 | −0.30 | −0.22 | −0.14 | −0.18 | −0.19 | −0.20 | −0.13 |
| | (0.47) | (0.30) | (0.26) | (0.22) | (0.20) | (0.19) | (0.18) | (0.16) | (0.15) | (0.13) | (0.13) | (0.13) |
| Frontage = interior ($p$) | 0.824 | 0.796 | 0.457 | 0.258 | 0.462 | 0.299 | 0.183 | 0.109 | 0.182 | 0.098 | 0.081 | 0.030 |
| Narcotics | | | | | | | | | | | | |
| Frontage (visible) | −7.45 | −3.68 | −7.08** | −3.94 | −4.63** | −4.63** | −3.17* | −3.54** | −2.59* | −2.92** | −2.86** | −2.35** |
| | (6.31) | (3.57) | (2.97) | (2.69) | (2.07) | (1.93) | (1.83) | (1.70) | (1.52) | (1.29) | (1.12) | (1.01) |
| Interior (hidden) | −1.20 | −0.44 | 0.42 | 0.72 | 0.39 | 0.92 | 0.76 | 0.73 | 0.31 | 0.37 | 0.27 | −0.28 |
| | (2.74) | (1.94) | (1.41) | (1.20) | (1.06) | (1.00) | (0.88) | (0.80) | (0.78) | (0.75) | (0.74) | (0.72) |
| Frontage = interior ($p$) | 0.344 | 0.424 | 0.023 | 0.118 | 0.035 | 0.014 | 0.063 | 0.025 | 0.087 | 0.029 | 0.025 | 0.111 |
| Vandalism | | | | | | | | | | | | |
| Frontage (visible) | 0.76 | 1.02* | 0.70* | 0.32 | 0.15 | −0.14 | −0.45 | −0.54* | −0.54* | −0.60** | −0.54** | −0.48** |
| | (0.74) | (0.55) | (0.42) | (0.37) | (0.33) | (0.33) | (0.30) | (0.30) | (0.29) | (0.28) | (0.26) | (0.24) |
| Interior (hidden) | −0.34 | −0.70 | −0.79* | −0.66** | −0.70** | −0.73*** | −0.67*** | −0.62*** | −0.50** | −0.45** | −0.38** | −0.44** |
| | (0.67) | (0.51) | (0.40) | (0.31) | (0.28) | (0.25) | (0.23) | (0.23) | (0.22) | (0.20) | (0.19) | (0.18) |
| Frontage = interior ($p$) | 0.246 | 0.024 | 0.010 | 0.047 | 0.056 | 0.170 | 0.557 | 0.847 | 0.892 | 0.671 | 0.626 | 0.883 |
| *Panel B. Vandalism, frontage incidents, by target* | | | | | | | | | | | | |
| Damage to property | 0.85 | 1.06 | 0.80 | 0.22 | 0.03 | −0.49 | −0.89** | −0.95** | −0.91** | −1.03*** | −0.92** | −0.89*** |
| | (1.12) | (0.83) | (0.63) | (0.58) | (0.51) | (0.47) | (0.45) | (0.43) | (0.41) | (0.39) | (0.36) | (0.34) |
| Damage to vehicle | 0.70 | 0.89 | 0.50 | 0.33 | 0.20 | 0.21 | 0.05 | −0.07 | −0.13 | −0.08 | −0.09 | 0.02 |
| | (0.93) | (0.67) | (0.56) | (0.53) | (0.48) | (0.45) | (0.42) | (0.40) | (0.40) | (0.38) | (0.37) | (0.34) |
| Property = vehicle ($p$) | 0.916 | 0.874 | 0.724 | 0.896 | 0.818 | 0.298 | 0.143 | 0.142 | 0.168 | 0.078 | 0.102 | 0.057 |

*Notes.* $\delta$ as in Table 2, at each window; standard errors below. Panel A splits residential burglary, narcotics, and vandalism incidents into frontage (garage, porch, yard, driveway, entryway, porch/hallway and common areas; Table OA14) and interior incidents. Panel B splits frontage vandalism by target. Last row of each panel: Wald $p$ for equal effects. Post-sale placebos for the frontage components are significant in no window for any of the three offenses, and for the interior components in at most one window, except narcotics interior (six windows), which carries no discount; the pooled residential burglary coefficient, by contrast, is matched window-for-window by post-sale crime (Online Appendix C); both fixed-effect designs in Online Appendix D. The burglary frontage estimate rests on about 45,000 incidents; its post-sale placebo is not significant at five percent in any window.

$^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

no significant discount at any window. The difference is significant at the ten-percent level at ten and twelve months (Table 3; Online Appendix D).[20]

**Risk of future crime** Prediction 2 asks whether discounts track what an incident reveals about future crime at the property: the updating term $b_k(S_k)$ in the framework (Section 2). These tests measure predictability in recorded crime, not buyers' subjective beliefs, which are unobserved. I use two measures: how well crime 0.20–0.30 miles away forecasts future crime within 0.05 miles, and how strongly crime within 0.05 miles predicts its own recurrence near the same parcel.

Neither measure shows a consistent correspondence with the size of the discounts. At the six-month horizon, fraud, weapons violations, domestic assault, and motor vehicle theft are forecast at least as well as burglary and vandalism but carry no detectable discount (Table A13). Homicide carries the largest discount yet is among the least forecastable offenses and has a small, negative recurrence coefficient ($\rho = -0.026$, s.e. 0.012). Narcotics is the most self-predicting offense in Table A12 ($\rho = 0.40$), yet carries a much smaller discount. Homicide's weak recurrence is consistent with Carrillo and Yezer (2021), who find it less persistent from year to year than other violent crime. Caetano and Maheshri (2018) document offense-specific persistence over shorter horizons. These measures of predictability therefore do not, on their own, explain the pattern of discounts (Appendix D).

### 5.5 Price Formation

Prediction 3 asks where in the transaction the discount appears. I examine asking prices, mid-listing price cuts, the sale-to-list ratio, time on market, and withdrawal to distinguish price responses present at listing from those that emerge later in the transaction. The housing-market microstructure literature studies asking prices, price revisions, selling time, and transaction prices as jointly determined outcomes of search and bargaining (Knight, 2002; Merlo and Ortalo-Magné, 2004; Han and Strange, 2015).

**Where the discount appears** The discount is detectable in closing prices but not in asking prices. All four priced offenses show significant closing-price discounts, and, across all listings, none of the four has a statistically significant asking-price response, although the burglary estimate is too imprecise to rule out a decline in asking prices as large as its closing discount (Table 4). For narcotics, the sale-to-list ratio shows the gap directly: homes close further below their asking price ($-0.16$, s.e. 0.07). A nearby homicide during the listing is associated with a higher probability that the home is cut below its original asking price (8.24 percentage points, s.e. 3.30; Table 4), although the cut may precede the incident within the listing. Together, these findings are consistent with discounts that emerge after listing rather than in the initial asking price.

[20] Because the police records mask each incident to the hundred-block, a frontage damage-to-property record can be the sold parcel itself, so this contrast carries an own-property component.

Table 4: Transaction Outcomes: From Asking Price to Sale

| Offense | Ask Price % | Sale-to-list % | Price cut pp | Days on market % | Withdrawal pp | Close % |
|---|---|---|---|---|---|---|
| Homicide | 0.38 | 0.53 | $8.24^{**\ddagger}$ | $14.54^{**}$ | $2.24^{**}$ | $-3.02^{**}$ |
| | (0.57) | (0.45) | (3.30) | (6.85) | (0.96) | (1.38) |
| Narcotics | 0.21 | $-0.16^{**}$ | 0.43 | −0.78 | 0.04 | $-0.29^{**}$ |
| | (0.17) | (0.07) | (0.35) | (0.98) | (0.04) | (0.14) |
| Burglary | −0.15 | −0.01 | −0.33 | 0.23 | −0.10 | $-0.24^{**}$ |
| | (0.12) | (0.05) | (0.33) | (1.25) | (0.10) | (0.12) |
| Vandalism | 0.05 | 0.00 | −0.12 | −0.10 | 0.00 | $-0.17^{**}$ |
| | (0.07) | (0.03) | (0.21) | (0.69) | (0.05) | (0.08) |
| Disorderly conduct | $-0.43^{**}$ | $-0.28^{**}$ | $2.55^{***}$ | −1.97 | 0.10 | $-0.65^{**}$ |
| | (0.21) | (0.12) | (0.95) | (3.86) | (0.23) | (0.29) |
| Weapons violations | −0.14 | 0.11 | $1.35^{**}$ | −0.33 | $0.59^{***}$ | 0.10 |
| | (0.20) | (0.25) | (0.53) | (3.52) | (0.21) | (0.28) |
| Parcel FE | Yes | Yes | Yes | Yes | Yes | Yes |
| Tract × month FE | Yes | Yes | Yes | Yes | Yes | Yes |

*Notes.* Near–far estimates as in Table 2, pooled 2007–2025. Columns: log asking price, log sale-to-list ratio, mid-listing price cut (percentage points), log days on market, withdrawal (percentage points), and log closing price. Closing price at each offense's main window (homicide 10, burglary 8, narcotics 10, vandalism 6, disorderly conduct 12, weapons violations 12 months); listing margins at the median significant window. Days on market uses parcel × era fixed effects; withdrawal uses all listings.

$^{\ddagger}$The price cut compares the last asking price with the original asking price of the same listing. It does not record when the cut was made, so it cannot show whether the cut followed the crime. The price-cut column uses crime during the listing, with fixed effects for the number of months listed.

$^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

**Market frictions** The response to nearby crime can also appear outside the price: crime may affect how quickly homes sell and whether listings reach a sale at all. The estimates suggest that nearby homicide raises time on market by about 15 percent at eight months under parcel-by-era fixed effects (s.e. 6.8), equivalent to roughly 2.4 days at the sample median of 16 days; this estimate is sensitive to specification, with statistical significance confined to the seven-to-nine-month window under parcel-by-era fixed effects (Table A15). Homicide withdrawal estimates are consistently positive but significant only at two months. Weapons violations show the more consistent withdrawal response: it is positive in all twelve windows under both specifications and significant in several, despite no consistent closing-price discount (Table 4; Table A16). Withdrawals also select the sample of closing prices: if the listings that withdraw would have faced the largest discounts, the price estimates are biased toward zero (Section 6). More broadly, price estimates alone may understate the response to crime, part of which operates through withdrawals and longer time on market (Appendix F).

**Real estate agent experience** Prior work shows that real estate agents' incentives and information shape transaction outcomes (Rutherford et al., 2005; Levitt and Syverson, 2008; Hendel et al., 2009; Han and Hong, 2011; Barwick and Pathak, 2015; Gilbukh and Goldsmith-Pinkham, 2024). I ask whether agents with more local experience, measured by the number of listings the agent has

previously handled in the same census tract (in logs), respond more to nearby crime (Table 5). For homicide, they do on both sides of the transaction. Asking prices fall more when the listing agent has more local experience, significantly in seven of twelve windows. Closing prices fall more when the buyer's agent has more local experience: comparing the same agent across transactions, a one-standard-deviation increase in the buyer's agent's local experience deepens the homicide discount by 1.14 percentage points (s.e. 0.53; significant in six of twelve windows; Table A17), and this result holds when the agent's total experience is controlled for. For narcotics, the pattern is weaker: asking prices fall more with the listing agent's local experience in seven of twelve windows, but the difference disappears once total experience is controlled for (Appendix F.1). Since real estate agents are not randomly assigned to listings, these estimates are associations rather than causal effects.

Table 5: Agent Local Experience: Ask (Listing) and Close (Buyer)

| Offense | Most negative (s.e.) | Most positive (s.e.) | Sig. windows |
|---|---|---|---|
| *Panel A. Listing agent, asking price (ln list)* | | | |
| Homicide | $-1.16^{***}$ (0.40) | $-0.04$ (0.62) | 7/12 |
| Burglary | $-0.02$ (0.04) | 0.08 (0.06) | 0/12 |
| Narcotics | $-0.13^{**}$ (0.06) | $-0.06$ (0.10) | 7/12 |
| Vandalism | $-0.07^{**}$ (0.03) | $-0.02$ (0.08) | 3/12 |
| *Panel B. Buyer agent, closing price (ln close)* | | | |
| Homicide | $-4.45^{**}$ (1.85) | $-0.68$ (0.63) | 6/12 |
| Burglary | 0.02 (0.10) | $0.38^{**}$ (0.19) | 0/12 |
| Narcotics | $-0.04$ (0.33) | $0.27^{*}$ (0.15) | 0/12 |
| Vandalism | $-0.03$ (0.10) | $0.10^{**}$ (0.04) | 0/12 |

*Notes.* Near–far effect interacted with agent experience, $E = \ln(1 +$ prior same-agent same-tract listings). Panel A: listing agent, asking price. Panel B: buyer's agent, closing price. Parcel, tract × month, and agent fixed effects; clustered by tract. Gradient: change in the near–far effect per standard deviation of $E$ (s.d. 1.20 for listing agents and 0.71 for buyer's agents, each on its estimation sample) (negative: experienced agents price crime more), shown as the most negative and most positive estimates across the twelve windows and the number of significantly negative windows. Table A17 reports the same specification at the windows given in its notes.
$^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

**The quantity margin** Using the annual parcel panel, I estimate each offense's effect on the listing rate and the turnover rate, with parcel and tract × year fixed effects (Online Appendix Table OA18). Ten of the twenty-two offenses, including all four priced offenses, raise the listing rate significantly more than the turnover rate; for domestic assault and weapons offenses, this quantity response appears without a detectable price response. Since crime and listings are measured within the same calendar year, these estimates do not establish that crime precedes the listing.[21]

[21] The turnover rate is the parcel-level counterpart of the tract-level measure studied by Boggess et al. (2013), who find that higher neighborhood crime is followed by more housing sales, with sales inhibited only where crime and vacancy are already high.

### 5.6 Heterogeneity

Housing-market conditions changed sharply over the sample, so I test whether the discounts differ between the 2009–2013 foreclosure crisis and the 2014–2025 recovery. I interact near–far exposure with era indicators, with 2007–2008 as the base, and estimate each era's discount in one regression. The point estimates differ: burglary and vandalism discounts are larger in the crisis, and homicide's in the recovery. For none of the four priced offenses, however, does a Wald test reject equal discounts across the two eras ($p$ between 0.10 and 0.63; Appendix I).

The discounts also vary with neighborhood socioeconomic status. When I interact the near–far term with each of ten standardized, predetermined tract characteristics, one at a time and controlling for the tract's predetermined crime level, all four priced offenses carry larger discounts in lower-income and less-educated tracts (Table OA20). A one-standard-deviation increase in tract income is associated with a 2.28-percentage-point smaller homicide discount (s.e. 1.07), with smaller differences for burglary, narcotics, and vandalism. The effect of nearby crime varies with tract income for several offenses with no average discount as well, such as trespass and fraud, so this variation is not specific to the priced offenses. Income, poverty, education, home values, and rents are strongly correlated. The results suggest a common socioeconomic gradient rather than distinct effects of each characteristic. Joint specifications do not consistently separate income from racial composition or segregation; detailed results appear in Online Appendix G.

## 6 Robustness

### 6.1 Threats to Identification

**Reverse Causality and Mean Reversion** Prices could drive crime rather than the reverse, or repeat-sales properties transacting at temporarily depressed prices could appreciate mechanically. I test both using a post-date placebo that replaces pre-transaction crime with crime occurring after the transaction, which cannot affect a price already set. The priced offenses pass this placebo at their main windows and in almost all specifications: post-sale crime is significant in none of the 63 specifications for homicide and vandalism, one for narcotics, and four for burglary (bottom panels of Figure 6 and Figures A4–A6). The exception is burglary, where post-sale crime is significant at six months (these are the placebo hits in Figure A6); six months is also where burglary's pre-sale estimate is largest ($-0.34$, s.e. 0.14; Table A3), so the placebo does not fully separate burglary's peak discount from post-sale crime.

**Prior Neighborhood and Housing-Market Conditions** I assess robustness to prior neighborhood and housing-market conditions (Table A20). I add five neighborhood measures, each entered separately for the near and far rings: building permits, the number of distinct locations reported vacant, three types of 311 service requests (graffiti, sanitation, and abandoned vehicles), active business licenses, and tree canopy. Permits, distinct locations reported vacant, and 311 requests are counted over the twelve months before the longest crime-exposure window, 361 to

720 days before closing; business licenses are those active at the start of that period, and tree canopy comes from the last annual raster completed before the crime window opens. I also add five housing-market measures from the MLS over the same twelve months: neighboring repeat-sale price changes, the distressed-sale share, the sale-to-list ratio, days on market, and the number of new listings.

Point estimates and standard errors change little with the additional controls (Table A20). All four discounts remain significant at the five-percent level when all ten controls are included.

**Cross-Crime Confounding** A discount attributed to one offense could reflect another offense that occurs nearby. Homicide is nearly uncorrelated with other offenses in the near ring (correlations of at most 0.18); burglary and narcotics are moderately correlated with them (at most 0.47 and 0.43). Entering all twenty-two offenses' near–far terms jointly at a common twelve-month window leaves their discounts nearly unchanged relative to separate estimates at that window (Online Appendix K; Table OA27). Vandalism is more strongly correlated with other offenses, especially domestic and non-domestic assault (0.55 and 0.51). Its joint estimate is −0.02 percent (s.e. 0.06), but its separate twelve-month estimate is already small and insignificant (−0.05 percent, s.e. 0.06). This comparison therefore does not establish whether its six-month discount survives adjustment for other offenses. Disorderly conduct retains a discount in the joint specification (−0.58 percent, s.e. 0.29).

**Sample Selection** The repeat-sales estimates may be affected by selection if crime changes which properties sell and those changes are related to unobserved determinants of sale prices (Online Appendix H.1 examines distressed sales). I examine this in three ways. First, nearby crime barely changes whether a parcel sells: among the priced offenses, only burglary has a statistically significant positive effect on the annual sale probability, by 0.02 percentage points (s.e. 0.01) against a base rate of 3.8 percent (Table OA18). This does not rule out changes in the composition of properties sold. Second, a Heckman correction, which uses mortgage-rate lock-in to predict whether an owner sells, leaves the crime coefficients largely unchanged (Online Appendix L). Third, a cross-sectional hedonic regression on all sales, including properties that sell only once, replaces the parcel fixed effect with observed characteristics and keeps the four priced discounts negative and significant (Table A21). For narcotics, the estimated discount is concentrated in distressed sales: at ten months, the discount is 0.64 percent (s.e. 0.26) for distressed transactions and approximately zero for regular sales (0.00, s.e. 0.13), with a statistically significant difference ($p = 0.03$; Online Appendix H.1).

The hedonic estimates are larger than the repeat-sales estimates for burglary, narcotics, and vandalism and smaller for homicide. Robbery, domestic and non-domestic assault, and motor-vehicle theft also have significant negative estimates in the hedonic specification, but these shrink substantially under repeat sales, with comparable or smaller standard errors. This pattern is consistent with these offenses being concentrated near properties with persistently lower values, differences that parcel fixed effects absorb but observed hedonic controls may not fully capture.

The hedonic sample also includes properties that sell only once, so part of the difference may reflect sample composition.

**Reporting Endogeneity** Recorded incidents reflect both underlying crime and reporting or detection. The near–far comparison mitigates reporting differences shared by nearby locations, and the largest discount is for homicide, one of the most reliably measured crimes (Fox and Zawitz, 2003). Estimates remain nearly unchanged after controlling for predetermined 311 requests (Table A20), while post-sale placebos provide additional reassurance (Section 6.3). Bias could remain if reporting or enforcement changes not captured by the controls differ between the rings and covary with unobserved determinants of sale prices.

### 6.2 Exposure Specification and Inference

**Selection into Zero Exposure** Properties with no nearby crime could differ from exposed properties in unobserved ways. I apply the zero-exposure test of Caetano (2015) and Caetano et al. (2021) to the three priced offenses with sufficient observations at positive exposure levels to estimate the dose–response relationship. About 98 percent of sales have no nearby homicide within the ten-month window. For narcotics and vandalism, the raw discontinuity is no longer statistically significant after including the baseline fixed effects. Adding an indicator for zero exposure leaves the narcotics estimate nearly unchanged and makes the vandalism estimate more negative. For burglary, a discontinuity appears once parcel fixed effects are included ($-0.98$, s.e. 0.32), and adding an indicator for zero exposure moves the discount from $-0.24$ (s.e. 0.12) to $-0.55$ (s.e. 0.17). With a wider 0–0.10 mile near ring, the test does not reject and the burglary discount remains significant ($-0.14$, s.e. 0.07; Table A6; Online Appendix M).

**Functional Form** I compare three ways of measuring crime exposure: raw counts, an indicator for any incident, and the inverse hyperbolic sine (IHS) transformation (Table OA26). Homicide's estimated discount remains significant under all three at its main window. For burglary, narcotics, and vandalism, only the raw-count estimates are significant at five percent, although narcotics remains negative under all three forms. The count-indicator specification allows a more flexible relationship and shows significant discounts for four or more burglaries and vandalism incidents (Table OA7; Online Appendix I).

**Spatial Standard Errors** Baseline standard errors are clustered by census tract (Cameron and Miller, 2015). Conley standard errors that allow correlation across tract boundaries, with cutoffs of 0.5 and 1.0 km (Conley, 1999), give similar inference: homicide, burglary, and vandalism remain significant at five percent under both cutoffs, and narcotics at five percent at 0.5 km and ten percent at 1.0 km (Table OA23). Clustering instead by the 77 community areas, a much coarser unit, leaves homicide and burglary significant at five percent and vandalism at ten percent, while narcotics is not significant at ten percent ($p = 0.12$).

### 6.3 Specification Curves and Spatial Checks

**Specification curves: sale prices** Figure 6 reports the sale-price specification curve for vandalism (narcotics, homicide, and burglary appear in Figures A4–A6; theft, an offense with no detectable discount, appears in Figure 7 as a comparison: 0 of 63 specifications are significant at five percent); each curve sorts the near–far discount $\delta = (\beta_{\text{near}} - \beta_{\text{far}}) \times 100$ across 63 specifications. The specifications vary the exposure window, the fixed effects, the comparison ring, the functional form, and the predetermined controls; the notes to Figure 6 list the full grid. The window begins at six months because shorter windows fall inside the listing and under-contract period, where measured pre-sale exposure is sparse and partly contemporaneous with the sale itself.

Nearly all specifications are negative for the four priced offenses; the exceptions are the presence-indicator estimates for burglary (0.54, s.e. 0.98) and vandalism (about 0.46), both imprecise. The discount is significant at five percent in 33 of 63 specifications for vandalism, 29 for narcotics, 29 for burglary, and 15 for homicide, whose point estimates range from about −3.6 percent to about zero with wide intervals. The post-sale placebo is significant in none of the 63 for homicide and vandalism, one for narcotics (the presence indicator at ten months), and four for burglary (the linear count at six months under each of the three far rings, and the inverse hyperbolic sine at eight months).[22] The full curve over all one-to-twelve-month exposure windows (103 specifications) is reported in Online Appendix OF4–OF7.

**Spatial bands** Moving the comparison ring out to 0.25–0.35 or 0.30–0.40 miles leaves the estimates nearly unchanged (Table A6). A falsification test that moves the treatment band outward to 0.05–0.10, 0.10–0.15, or 0.15–0.20 miles yields significantly negative estimates in one of seventy-two specifications for burglary, none for narcotics, and four for vandalism. All four vandalism estimates occur in the 0.15–0.20-mile band. This pattern is consistent with effects concentrated within 0.05 miles.

Homicide behaves differently. It is significant in 11 of 72 specifications outside the near ring, all in the 0.10–0.15 band and at windows of three to nine months (for example, −2.86 percent, s.e. 0.89, at six months under the baseline fixed effects and comparison ring), while the intervening 0.05–0.10 band shows no significant discount. At homicide's ten-month main window, the band estimate is −1.16 percent (s.e. 0.64), compared with −3.02 percent (s.e. 1.38) within 0.05 miles. Homicide is therefore robust at the doorstep but, unlike the other three offenses, not localized to it. In Section 5.4, the homicide discount is significant for incidents on the street but not at residential addresses; the two estimates differ significantly only at the twelve-month window ($p = 0.046$). Homicide's estimated price effects extend beyond the near ring, but the comparison-ring coefficient is small and statistically insignificant at the main window (0.20, s.e. 0.35; Table OA6). Omitting the far-ring control leaves the estimated discount similar: −2.83 percent (s.e. 1.35), compared with

[22] The specification curve shows how stable the estimates are across modeling choices. The specifications share data and most modeling choices. Their estimates are therefore dependent, and the number of significant estimates is descriptive rather than a formal test.

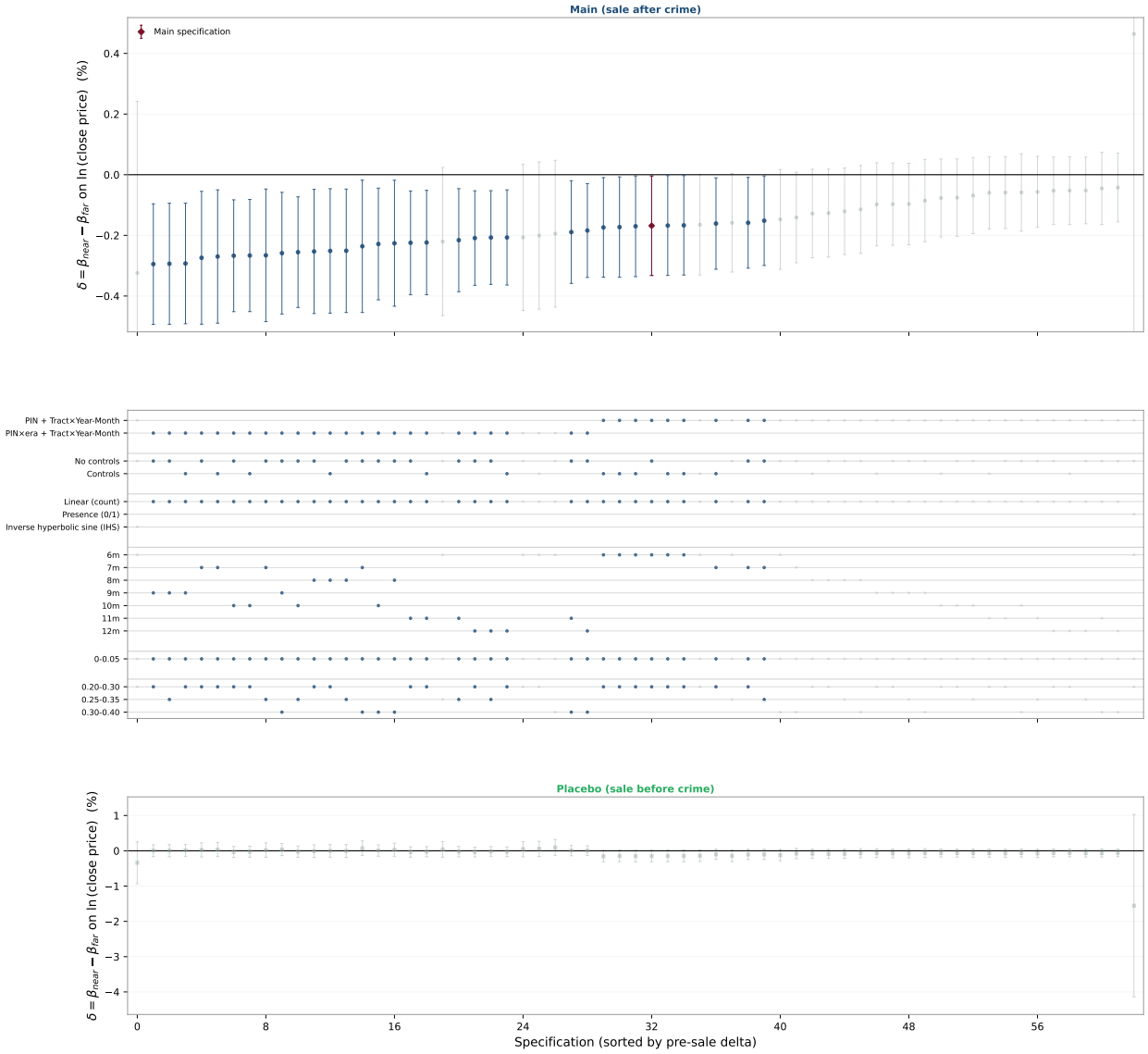



Figure 6: Vandalism: Sale-Price Discount Across Specifications

*Notes.* Common to this figure, Figure 7, and Figures A4–A6. The 63 specifications vary the exposure window from six to twelve months, the fixed-effect design (parcel + tract×month, or parcel×era + tract×month), the comparison ring (0.20–0.30, 0.25–0.35, 0.30–0.40 mile), the functional form (linear count, presence indicator, inverse hyperbolic sine), and the predetermined controls (none, all five jointly, or one at a time). Top panel: $\delta$ (%) with 95 percent intervals, sorted from most to least negative. The baseline specification is the linear count with parcel + tract×month fixed effects, the 0.20–0.30 mile comparison ring, and no controls, at each offense's main window: ten months for homicide and narcotics, eight for burglary, and six for vandalism. Middle panel: specification choices. Bottom panel: the post-sale placebo, which replaces pre-sale crime with crime after the sale. All years pooled (2007–2025); standard errors clustered by tract. Vandalism: 33/63 significant at five percent; placebo 0/63.

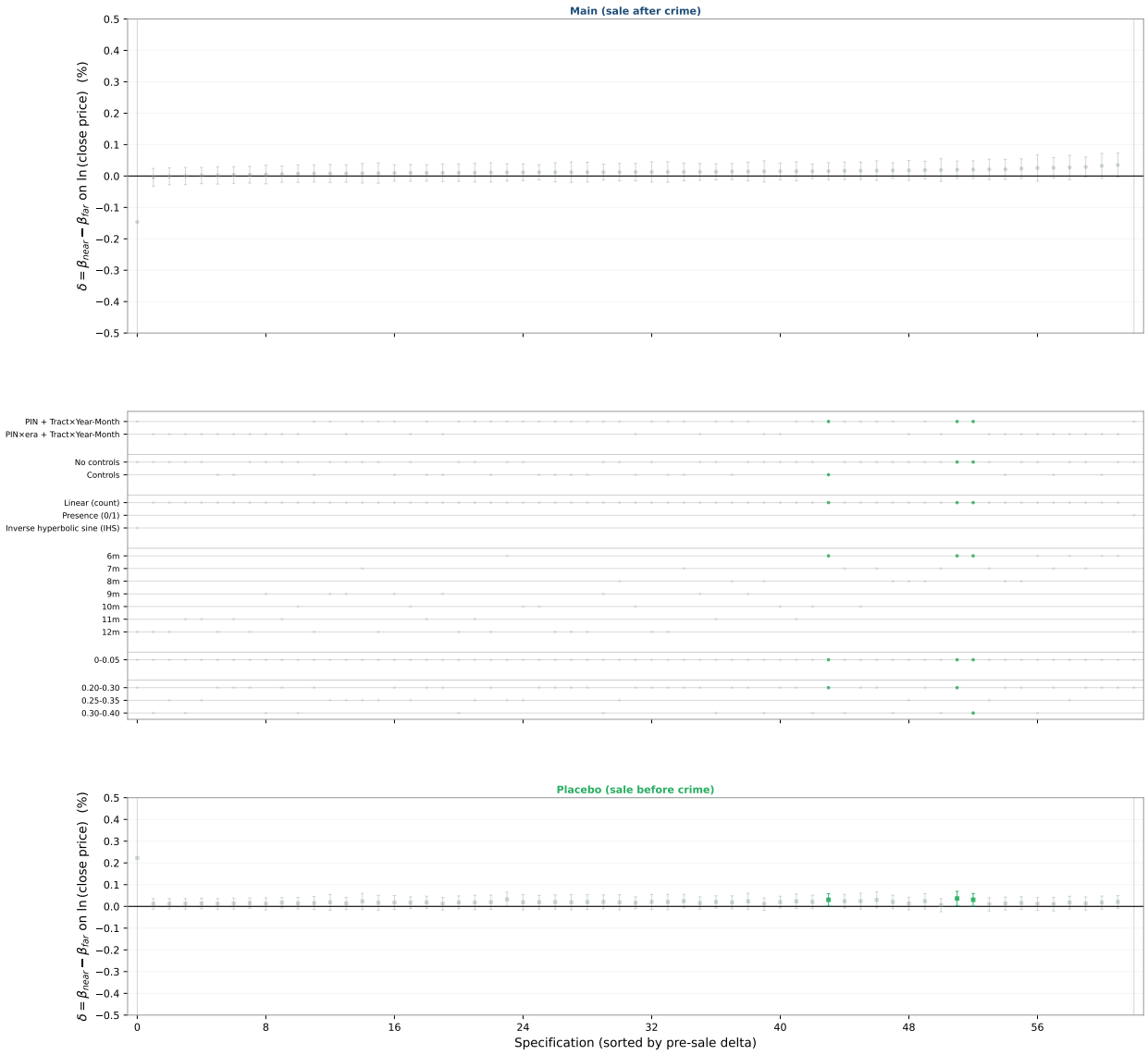



Figure 7: Theft: Sale-Price Discount Across Specifications

*Notes.* The 63 specifications vary the exposure window (six to twelve months), the fixed effects (parcel + tract×month, or parcel×era + tract×month), the comparison ring (0.20–0.30, 0.25–0.35, or 0.30–0.40 mile), the functional form (linear count, presence indicator, or inverse hyperbolic sine), and the predetermined controls (none, all five jointly, or one at a time); see Section 6.3. Top panel: $\delta$ (%) with 95 percent intervals, sorted from most to least negative. The baseline specification is the linear count with parcel + tract×month fixed effects, the 0.20–0.30 mile comparison ring, no controls, and a twelve-month window. Middle panel: specification choices. Bottom panel: post-sale placebo. Theft: 0/63 significant at 5%, placebo 3/63. The vertical axis is cut at ±0.5 percent.

the baseline near–far estimate of −3.02 percent (s.e. 1.38).

Significantly negative post-sale placebo estimates outside the near ring number 4 of 72 for narcotics, 5 of 72 for burglary and vandalism, and none for homicide.

**Exposure Concentration** Leaving out each census tract in turn leaves the four priced estimates similar in magnitude, with jackknife standard errors close to the baseline clustered estimates. Excluding the twenty most exposed tracts produces larger changes for homicide, whose estimated discount declines from 3.02 (s.e. 1.38) to 2.03 percent (s.e. 1.53; $p = 0.18$); the other three estimates remain of comparable magnitude (Online Appendix H.3).

# 7 Discussion

## 7.1 Why most offenses are not priced

House prices can respond to localized signals of risk, such as a registered offender moving nearby, even without a new crime (Section 2). My findings suggest the converse: realized crime need not lower prices, even when it predicts future incidents. Together, these findings suggest that price effects depend partly on whether buyers learn of an incident and regard it as relevant to the property.

Neither frequency nor severity alone explains the pattern. Homicide is rare but carries the largest discount, while the estimates for larceny, motor vehicle theft, and non-domestic assault rule out discounts as large as burglary's. Domestic assault also has a near-zero price effect despite its severity (Table A3). Limited buyer awareness of incidents inside homes is one plausible explanation.

Price effects may also depend on whether buyers regard the harm as tied to the property. Under the classification proposed in Section 2, all four priced offenses are anchored to an address; among the eleven offenses involving harm to a person, money, or a movable item, only sex offenses show a suggestive discount (Table OA16). This classification differs from NIBRS, under which the priced offenses span crimes against persons, property, and society (Table A4). The pattern is consistent with relevance to the property, but the grouping is interpretive, and the groups also differ in severity, frequency, and visibility.

Measured predictability does not explain the ranking of discounts (Section 5.4). This is consistent with differences in attention to crime information (Bordalo et al., 2012; Gabaix, 2014), although forecasts from recorded crime need not match buyers' expectations. Within-offense comparisons provide further evidence consistent with observability: burglary discounts are concentrated among frontage incidents, with the frontage–interior difference significant at five percent only at twelve months (Table 3). This may help explain the contrast with Gibbons (2004), whose burglary measure pools incidents within 250 meters without distinguishing their visibility.

### 7.2 Why asking prices respond little

Asking-price responses are generally small in the full listing sample (Table 4). This alone does not establish that crime discounts emerge between listing and closing: the asking- and closing-price estimates use different samples. The within-listing comparison provides evidence of a lower closing price relative to asking for narcotics, but not for the other three priced offenses (Table 4).

Several mechanisms could explain muted asking-price responses. Sellers may anchor on their purchase price or an earlier valuation. Genesove and Mayer (2001), for example, show that owners facing prospective nominal losses set higher asking prices and sell more slowly. Asking prices may also rely on comparable sales that predate the incident. More negative asking-price responses among locally experienced listing agents are consistent with better information, although agent assignment is not random and the estimates do not distinguish information from other sources of expertise.

Sellers may also respond to nearby crime by withdrawing rather than lowering prices. Among listings that sell, homicide's asking-price estimate is negative at every window, though significant only at one month (Table A14). Sellers who maintain high asking prices may instead withdraw. Homicide withdrawal estimates are consistently positive but rarely significant; weapons violations show a more consistent withdrawal response despite no consistent closing-price discount. These findings suggest that housing markets can adjust through withdrawals as well as prices, although they do not establish why sellers withdraw.

Sellers' bargaining positions may also affect how nearby crime enters the sale price. At the main windows, burglary, narcotics, and vandalism discounts do not differ detectably between soft and tight markets ($p > 0.4$; Table OA30; Online Appendix N), while homicide estimates are too imprecise for an informative comparison. Narcotics discounts are concentrated in distressed sales, with a significant distressed–regular difference ($p = 0.03$; Online Appendix H.1). This is consistent with sellers' circumstances affecting the price response, but does not isolate bargaining from differences in properties or exposure.

### 7.3 Information and beliefs

The estimates do not reveal what buyers knew about nearby crime or how they revised their expectations. In the framework, $b_k$ is the change in expected future incidents from one additional observed incident (Section 2). Because beliefs are subjective, $b_k$ need not equal the change in risk predicted by recorded crime. Across offenses, small discounts cannot distinguish limited awareness from low willingness to pay. Within-offense differences by incident location and agents' local experience are consistent with limited information, but location may also change an incident's relevance to the property, and agent assignment is not random. These comparisons do not establish what buyers knew or whether they updated correctly.

Offenses that lower perceived safety and offenses that lower sale prices only partly overlap. Higher recorded homicide, burglary, and narcotics counts are associated with lower safety scores, but so are domestic assault and trespass, which carry no consistent sale-price discount. Vandalism is

priced yet is not associated with lower perceived safety (Appendix E). These sector-level perceptions describe residents' views, not buyers' knowledge of individual incidents.

Public records do not ensure buyer awareness: finding and interpreting information takes effort (Stigler, 1961). Agents may help bridge that gap. For homicide, the buyer-agent gradient in closing prices survives adjustment for total experience, while total experience has no detectable gradient. The corresponding listing-agent gradient survives at only one of twelve windows. For narcotics, the asking-price gradient instead reflects listing agents' total experience, consistent with general pricing expertise. These patterns relate to evidence that agents' information advantages affect transaction outcomes (Levitt and Syverson, 2008), although the gradients do not identify causal effects of expertise.

News coverage may help buyers learn about nearby crime. In the Media Cloud measure, homicide receives roughly twenty times as many stories per recorded incident as arson, the next-highest of the twelve categories examined, and carries the largest sale-price discount.[23] This comparison measures relative media attention, not whether buyers learned of a particular incident. Existing evidence also differs: Klimova and Lee (2014) find no additional price effect from more heavily covered murders in Sydney, whereas Moreno-Medina (2026), using sporting events that displace local crime news, finds that biased crime coverage lowers house prices in minority neighborhoods. My data do not separate the contributions of media, agents, and local contacts.

Information may also be interpreted through neighborhood reputation. Vandalism discounts are larger in more segregated tracts even after controlling for income (Table OA21). Sampson and Raudenbush (2004) similarly find greater perceived disorder in neighborhoods with more Black residents, holding observed disorder constant. The price pattern is consistent with stigma amplifying responses to visible disorder, but does not establish that mechanism.

Incident-level variation in news coverage could provide a stronger test of information transmission, and measuring buyers' awareness and expectations directly would help distinguish whether information reaches them from how they interpret it.

### 7.4 Welfare and policy

The estimates measure price responses among completed transactions, not average household willingness to pay to avoid crime. Buyers most averse to crime may avoid exposed properties, leaving a selected group of purchasers. Negotiated prices also reflect sellers' valuations and bargaining positions, while buyers may consider resale prospects as well as current occupancy. The data do not separate these influences.

Housing prices alone may miss benefits of crime reduction, and reducing different offenses need not produce similar price gains. Longer marketing periods can entail carrying costs and search effort; withdrawals may represent forgone moves, delays, or alternative uses of a property. Their

[23] The Media Cloud measure divides deduplicated articles mentioning Chicago and an offense, from thirty Chicago news outlets in 2013–2024, by recorded Chicago incidents over the same period (Online Appendix A.1). The ratios are 4.5 for homicide and 0.22 for arson. Comparisons depend on search terms and do not measure coverage of individual incidents or buyers' awareness.

welfare implications depend on what follows. Price effects also need not rank offenses by the harm they impose on victims (McCollister et al., 2010). When buyers are unaware of incidents, the harm those incidents cause may not be reflected in what they pay.

Information disclosure raises a distinct policy question. If observability contributes to price responses, crime maps or disclosure rules may change property values without changing underlying crime. A resulting price decline need not indicate a welfare loss; it may reflect better information about an existing risk. Conversely, higher prices following the removal of crime information would not establish that a neighborhood had become safer. The findings motivate distinguishing changes in safety from changes in information, but do not establish the welfare effects of disclosure.

### 7.5 Scope

For enforcement-sensitive offenses, price responses may reflect policing as well as underlying crime. Enforcement affects recorded offense counts (Levitt, 1998) and can itself affect local well-being (Ang, 2021; Legewie and Fagan, 2019). If buyers regard concentrated drug or vice enforcement as a disamenity, the design cannot separate that response from the response to crime. Among the four priced offenses, this concern applies most directly to narcotics.

The geographic reach of crime's price effects may differ across cities. The estimates measure the price response per additional incident within 0.05 miles relative to one 0.20–0.30 miles away. The design can be applied elsewhere, but the same radius may encompass different numbers of properties and blocks. Generalizability depends on density, street layout, housing-market conditions, and how buyers learn about incidents. Differences in price effects across neighborhood income and education levels within Chicago also suggest that socioeconomic composition matters when comparing findings across settings (Section 5.6).

Sale and rental estimates differ in magnitude and sometimes in sign, but these differences cannot be attributed solely to resale exposure. The smaller lease sample generally yields less precise estimates and consists of formal, MLS-listed units. Differences in properties, household characteristics, and preferences may contribute to the gap. Purchasing also exposes owners to future resale values, while renting primarily buys occupancy. Moreover, the rental outcomes are landlords' asking rents, whereas sale prices reflect completed negotiations. The rental findings therefore apply to posted rents in the MLS market and may not extend to rentals outside it.

## 8 Conclusion

Homebuyers care about safety, yet most offenses carry no detectable sale-price discount. Comparing repeated sales of the same property and distinguishing nearby crime from crime a few blocks away, I find discounts across multiple exposure windows for four of twenty-two offense categories: homicide, burglary, narcotics, and vandalism. Neither severity nor measured recurrence alone explains which offenses are priced. Three patterns suggest that the observability of crime matters. Price effects are more localized than the predictive relationships in recorded crime; discounts are larger when buyers'

agents have more local experience; and, for a given category of offense, more visible incidents tend to lower prices more. Price effects may therefore depend partly on which incidents buyers learn about.

The market's response is not confined to closing prices. Asking prices respond little, while closing prices are discounted, although the two are estimated on different samples. Some offenses also affect time on market or withdrawals, and for ten of the twenty-two offenses, nearby crime raises the probability that a home is listed more than the probability that it sells. Closing prices alone therefore give an incomplete picture of how the market responds.

These findings matter for how housing prices are used to value safety. Observed discounts need not equal willingness to pay to avoid crime or rank offenses by the harm they impose. Housing-market valuations may therefore miss benefits of crime reduction when buyers have limited information about local crime. Changes in platform crime maps offer a setting to test whether information affects how crime enters prices, controlling for underlying crime.[24]

[24] Zhang et al. (2022) propose a related design based on the introduction of public crime maps.

## References


Dana Anderson. One in five house hunters is willing to compromise personal safety to afford a home. Redfin News, Aug 2025. Survey fielded May 2025. `https://www.redfin.com/news/survey-homebuyers-compromise-safety/`.

Elliot Anenberg and Edward Kung. Estimates of the size and source of price declines due to nearby foreclosures. *American Economic Review*, 104(8):2527–2551, 2014. doi: 10.1257/aer.104.8.2527.

Desmond Ang. The effects of police violence on inner-city students. *The Quarterly Journal of Economics*, 136(1):115–168, 2021.

Laura A. Bakkensen and Lint Barrage. Going underwater? Flood risk belief heterogeneity and coastal home price dynamics. *The Review of Financial Studies*, 35(8):3666–3709, 2022. doi: 10.1093/rfs/hhab122.

Panle Jia Barwick and Parag A. Pathak. The costs of free entry: An empirical study of real estate agents in Greater Boston. *RAND Journal of Economics*, 46(1):103–145, 2015. doi: 10.1111/1756-2171.12082.

Asaf Bernstein, Matthew T. Gustafson, and Ryan Lewis. Disaster on the horizon: The price effect of sea level rise. *Journal of Financial Economics*, 134(2):253–272, 2019. doi: 10.1016/j.jfineco.2019.03.013.

Utpal Bhattacharya, Daisy Huang, and Kasper Meisner Nielsen. Spillovers in prices: The curious case of haunted houses. *Review of Finance*, 25(3):903–935, 2021.

Kelly C. Bishop and Alvin D. Murphy. Estimating the willingness to pay to avoid violent crime: A dynamic approach. *American Economic Review: Papers and Proceedings*, 101(3):625–629, 2011. doi: 10.1257/aer.101.3.625.

Lyndsay N. Boggess, Robert T. Greenbaum, and George E. Tita. Does crime drive housing sales? Evidence from Los Angeles. *Journal of Crime and Justice*, 36(3):299–318, 2013. doi: 10.1080/0735648X.2013.812976.

Pedro Bordalo, Nicola Gennaioli, and Andrei Shleifer. Salience theory of choice under risk. *Quarterly Journal of Economics*, 127(3):1243–1285, 2012. doi: 10.1093/qje/qjs018.

Pedro Bordalo, Nicola Gennaioli, and Andrei Shleifer. Salience and consumer choice. *Journal of Political Economy*, 121(5):803–843, 2013. doi: 10.1086/673885.

Nils Braakmann. The link between crime risk and property prices in England and Wales: Evidence from street-level data. *Urban Studies*, 54(8):1990–2007, 2017. doi: 10.1177/0042098016634611.

Carolina Caetano. A test of exogeneity without instrumental variables in models with bunching. *Econometrica*, 83(4):1581–1600, 2015. doi: 10.3982/ECTA11231.

Carolina Caetano, Gregorio Caetano, Hao Fe, and Eric Nielsen. A dummy test of identification in models with bunching. Finance and Economics Discussion Series 2021-068, Board of Governors of the Federal Reserve System, 2021.

Gregorio Caetano and Vikram Maheshri. Identifying dynamic spillovers of crime with a causal approach to model selection. *Quantitative Economics*, 9(1):343–394, 2018. doi: 10.3982/QE756.

A. Colin Cameron and Douglas L. Miller. A practitioner's guide to cluster-robust inference. *Journal of Human Resources*, 50(2):317–372, 2015. doi: 10.3368/jhr.50.2.317.

John Y. Campbell, Stefano Giglio, and Parag Pathak. Forced sales and house prices. *American Economic Review*, 101(5):2108–2131, 2011. doi: 10.1257/aer.101.5.2108.

Paul E. Carrillo and Anthony M. Yezer. What do hedonic house price estimates tell us when capitalization rates vary? IIEP Working Paper 2021-03, Institute for International Economic Policy, George Washington University, 2021.

Raj Chetty, Adam Looney, and Kory Kroft. Salience and taxation: Theory and evidence. *American Economic Review*, 99(4):1145–1177, 2009. doi: 10.1257/aer.99.4.1145.

Joshua M. Congdon-Hohman. The lasting effects of crime: The relationship of discovered methamphetamine laboratories and home values. *Regional Science and Urban Economics*, 43(1):31–41, 2013.

Timothy G. Conley. GMM estimation with cross sectional dependence. *Journal of Econometrics*, 92(1):1–45, 1999.

Alexander Dentler and Enzo Rossi. Residents' willingness to pay to avoid crime. *Journal of Housing Economics*, 2024. Earlier version: Swiss National Bank Working Paper 2021-20, "Shooting Up Liquidity: The Effect of Crime on Real Estate".

Stuart Donovan, Thomas de Graaff, Henri L. F. de Groot, and Aaron Schiff. An urban overhead? Crime, agglomeration, and amenity. *Journal of Housing Economics*, 64:101994, 2024. doi: 10.1016/j.jhe.2024.101994.

Ingrid Gould Ellen, Johanna Lacoe, and Claudia Ayanna Sharygin. Do foreclosures cause crime? *Journal of Urban Economics*, 74:59–70, 2013. doi: 10.1016/j.jue.2012.09.003.

James Alan Fox and Marianne W. Zawitz. Homicide trends in the United States: 2000 update. Crime Data Brief NCJ 197471, Bureau of Justice Statistics, U.S. Department of Justice, 2003.

Xavier Gabaix. A sparsity-based model of bounded rationality. *Quarterly Journal of Economics*, 129(4):1661–1710, 2014. doi: 10.1093/qje/qju024.

Mark J. Garmaise and Tobias J. Moskowitz. Confronting information asymmetries: Evidence from real estate markets. *The Review of Financial Studies*, 17(2):405–437, 2004. doi: 10.1093/rfs/hhg037.

Pieter A. Gautier, Arjen Siegmann, and Aico van Vuuren. Terrorism and attitudes towards minorities: The effect of the Theo van Gogh murder on house prices in Amsterdam. *Journal of Urban Economics*, 65(2):113–126, 2009.

David Genesove and Christopher Mayer. Loss aversion and seller behavior: Evidence from the housing market. *Quarterly Journal of Economics*, 116(4):1233–1260, 2001.

Stephen Gibbons. The costs of urban property crime. CEP Discussion Paper 574, Centre for Economic Performance, London School of Economics, 2003.

Stephen Gibbons. The costs of urban property crime. *Economic Journal*, 114:F441–F463, 2004.

Sonia Gilbukh and Paul Goldsmith-Pinkham. Heterogeneous real estate agents and the housing cycle. *The Review of Financial Studies*, 37(11):3431–3489, 2024. doi: 10.1093/rfs/hhae048.

Jesse D. Gourevitch, Carolyn Kousky, Yanjun Liao, Christoph Nolte, Adam B. Pollack, Jeremy R. Porter, and Joakim A. Weill. Unpriced climate risk and the potential consequences of overvaluation in US housing markets. *Nature Climate Change*, 13(3):250–257, 2023. doi: 10.1038/s41558-023-01594-8.

Lu Han and Seung-Hyun Hong. Testing cost inefficiency under free entry in the real estate brokerage industry. *Journal of Business & Economic Statistics*, 29(4):564–578, 2011. doi: 10.1198/jbes.2011.08314.

Lu Han and William C. Strange. The microstructure of housing markets: Search, bargaining, and brokerage. In *Handbook of Regional and Urban Economics*, volume 5, pages 813–886. Elsevier, 2015.

Igal Hendel, Aviv Nevo, and François Ortalo-Magné. The relative performance of real estate marketing platforms: MLS versus FSBOMadison.com. *American Economic Review*, 99(5):1878–1898, 2009. doi: 10.1257/aer.99.5.1878.

Keith Ihlanfeldt and Tom Mayock. Panel data estimates of the effects of different types of crime on housing prices. *Regional Science and Urban Economics*, 40(2–3):161–172, 2010.

Anastasia Klimova and Adrian D. Lee. Does a nearby murder affect housing prices and rents? The case of Sydney. *Economic Record*, 90(S1):16–40, 2014. doi: 10.1111/1475-4932.12118.

John R. Knight. Listing price, time on market, and ultimate selling price: Causes and effects of listing price changes. *Real Estate Economics*, 30(2):213–237, 2002.

Bishwa Koirala, Rik Chakraborti, and Gyan Pradhan. Urban crime and its net implicit price to North Carolina households. *American Business Review*, 26(1):83–94, 2023. doi: 10.37625/abr.26.1.83-94.

Ivan Korolev. Estimating semiparametric and nonparametric fixed effects panel data models with `mgcv`, 2026. Working paper, arXiv:2606.12739.

Pablo Kurlat and Johannes Stroebel. Testing for information asymmetries in real estate markets. *The Review of Financial Studies*, 28(8):2429–2461, 2015. doi: 10.1093/rfs/hhv028.

Joscha Legewie and Jeffrey Fagan. Aggressive policing and the educational performance of minority youth. *American Sociological Review*, 84(2):220–247, 2019.

Steven D. Levitt. Why do increased arrest rates appear to reduce crime? Deterrence, incapacitation, or measurement error? *Economic Inquiry*, 36(3):353–372, 1998.

Steven D. Levitt and Chad Syverson. Market distortions when agents are better informed: The value of information in real estate transactions. *The Review of Economics and Statistics*, 90(4): 599–611, 2008.

Leigh Linden and Jonah E. Rockoff. Estimates of the impact of crime risk on property values from Megan's Laws. *American Economic Review*, 98(3):1103–1127, 2008.

Paula Margaretic and Juan Bautista Sosa. How local is the crime effect on house prices? *The Journal of Real Estate Finance and Economics*, 70:754–793, 2025. doi: 10.1007/s11146-023-09941-4.

Kathryn E. McCollister, Michael T. French, and Hai Fang. The cost of crime to society: New crime-specific estimates for policy and program evaluation. *Drug and Alcohol Dependence*, 108 (1–2):98–109, 2010.

Antonio Merlo and François Ortalo-Magné. Bargaining over residential real estate: evidence from england. *Journal of Urban Economics*, 56(2):192–216, 2004.

Jonathan Moreno-Medina. Crime news bias: Extent and effects on housing markets. Working paper, University of Texas at San Antonio, January 2026.

Jaren C. Pope. Fear of crime and housing prices: Household reactions to sex offender registries. *Journal of Urban Economics*, 64(3):601–614, 2008a.

Jaren C. Pope. Buyer information and the hedonic: The impact of a seller disclosure on the implicit price for airport noise. *Journal of Urban Economics*, 63(2):498–516, 2008b.

Mario J. Rizzo. The effect of crime on residential rents and property values. *The American Economist*, 23(1):16–21, 1979.

Ronald C. Rutherford, Thomas M. Springer, and Abdullah Yavaş. Conflicts between principals and agents: Evidence from residential brokerage. *Journal of Financial Economics*, 76(3):627–665, 2005. doi: 10.1016/j.jfineco.2004.06.006.

Robert J. Sampson and Stephen W. Raudenbush. Seeing disorder: Neighborhood stigma and the social construction of "broken windows". *Social Psychology Quarterly*, 67(4):319–342, 2004.

Amy Ellen Schwartz, Scott Susin, and Ioan Voicu. Has falling crime driven New York City's real estate boom? *Journal of Housing Research*, 14(1):101–135, 2003.

George J. Stigler. The economics of information. *Journal of Political Economy*, 69(3):213–225, 1961.

Johannes Stroebel. Asymmetric information about collateral values. *The Journal of Finance*, 71 (3):1071–1112, 2016. doi: 10.1111/jofi.12288.

Cheng Keat Tang and Thao Le. Crime risk and housing values: Evidence from the gun offender registry. *Journal of Urban Economics*, 134:103526, 2023. doi: 10.1016/j.jue.2022.103526.

Richard Thaler. A note on the value of crime control: Evidence from the property market. *Journal of Urban Economics*, 5(1):137–145, 1978.

Scott A. Wentland, Bennie D. Waller, and Raymond T. Brastow. Estimating the effect of crime risk on property values and time on market: Evidence from Megan's Law in Virginia. *Real Estate Economics*, 42(1):223–251, 2014. doi: 10.1111/1540-6229.12028.

Simon N. Wood. *Generalized Additive Models: An Introduction with R*. Chapman and Hall/CRC, 2 edition, 2017.

Meng Le Zhang, Monsuru Adepeju, and Rhiannon Thomas. Estimating the effect of crime (maps) on house prices using a (un)natural experiment. SocArXiv preprint 9zupw, 2022. Study protocol.

# Appendix

This appendix reports supporting tables and figures, grouped in the order of the main text. Unless a note says otherwise, each reports $\delta = (\beta_{\text{near}} - \beta_{\text{far}}) \times 100$, the percent sale-price change per incident within 0–0.05 miles of the property against 0.20–0.30 miles, with parcel and tract × year-month fixed effects and standard errors clustered by census tract; $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. All other supporting material is in the Online Appendix.

# A Data and Offense Coverage

Table A1: Summary Statistics

| | Mean | SD | p10 | Median | p90 | Obs. |
|---|---|---|---|---|---|---|
| *A. Sale listings* ($N = 530,395$) | | | | | | |
| Close price ($) | 366,777 | 382,539 | 74,000 | 279,000 | 700,000 | 530,395 |
| Log close price | 12.44 | 0.92 | 11.21 | 12.54 | 13.46 | 530,395 |
| Original list price ($) | 384,828 | 410,804 | 85,000 | 289,900 | 725,000 | 529,956 |
| Days on market | 96 | 147 | 5 | 42 | 248 | 529,995 |
| Any list-price cut | 0.68 | 0.47 | 0.00 | 1.00 | 1.00 | 530,395 |
| Bedrooms | 3.0 | 1.7 | 1.0 | 3.0 | 5.0 | 517,982 |
| Bathrooms | 2.1 | 1.1 | 1.0 | 2.0 | 3.0 | 530,395 |
| Living area (sq. ft.) | 1,175 | 1,115 | 0 | 1,100 | 2,483 | 454,018 |
| Total rooms | 7.1 | 3.2 | 4.0 | 6.0 | 11.0 | 517,753 |
| Year built | 1954 | 41 | 1902 | 1954 | 2007 | 453,343 |
| Garage spaces | 1.6 | 0.6 | 1.0 | 2.0 | 2.0 | 351,747 |
| Fireplaces | 0.9 | 0.7 | 0.0 | 1.0 | 2.0 | 206,694 |
| *B. Lease listings* ($N = 191,155$) | | | | | | |
| Monthly rent ($) | 2,208 | 1,086 | 1,250 | 1,950 | 3,400 | 191,155 |
| Log monthly rent | 7.61 | 0.41 | 7.13 | 7.58 | 8.13 | 191,155 |
| Days on market | 43 | 57 | 6 | 25 | 102 | 191,122 |
| Bedrooms | 1.8 | 1.0 | 1.0 | 2.0 | 3.0 | 191,031 |
| Bathrooms | 1.6 | 0.8 | 1.0 | 1.0 | 2.0 | 191,155 |
| Living area (sq. ft.) | 883 | 675 | 0 | 900 | 1,691 | 191,041 |
| Total rooms | 4.7 | 1.4 | 3.0 | 5.0 | 6.0 | 191,032 |
| Year built | 1972 | 37 | 1912 | 1982 | 2008 | 150,337 |

*Notes:* Chicago MLS single-family, condominium, and residential-income listings, 2007–2025, on parcels with a constant parcel count. Panel A: closed sales ($10,000–$10,000,000). Panel B: leases (monthly rent $500–$10,000). Any list-price cut: at least one reduction before sale.

## A.1 Offense Coverage

I estimate separate effects for 27 of the 29 offense groups, omitting human trafficking and nonforcible sex offenses (222 incidents combined) as too sparse.[25] Of these 27, I report 22 in the main analysis. I set aside the residual "all other offenses" category, which names no specific offense, and four offenses, kidnapping, embezzlement, extortion, and child pornography, whose estimates are too imprecise to be informative.[26]

## A.2 Neighborhood Control Data

Tree canopy is percent canopy cover from the U.S. Geological Survey National Land Cover Database (NLCD) Tree Canopy Cover product (30 m resolution). Near and far canopy are averages over the 30 m cells within the 0–0.05 mile disk and the 0.20–0.30 mile ring. Canopy comes from the last annual raster completed before the crime window opens; sales for which that year lies outside 2011–2021 receive a missing-value indicator. Business licenses, 311 service requests, and building permits are geocoded records from the City of Chicago Data Portal (`https://data.cityofchicago.org`). Before the City's unified 311 system launched in December 2018, the portal releases requests in category-specific historical files; the 311 control uses the same three request types from those files for 2011–2018 and from the unified system for 2019 onward: graffiti removal, sanitation code complaints, and abandoned vehicles. Vacant-building complaints are excluded, so the 311 control does not overlap the vacancy measure, and an indicator marks windows that overlap 2019, the first year of the unified system. Vacant locations are "Vacant and Abandoned Buildings Reported" records from the same portal, available for 2010–2018; within each sale's window and ring, reports at the same location, with coordinates rounded to four decimal places, are counted once. Business licenses are those active at the start of the twelve-month period, each license counted once across renewals.

[25] The pornography/obscene-material group is predominantly child pornography rather than adult material: about two-thirds of the Chicago Police incidents so classified are child pornography (IUCR 1582) or related offenses involving children, and roughly one-quarter is adult obscene material (obscene telephone calls are classified separately, under all other offenses). I therefore label this group "child pornography/obscenity" throughout. It is sparse, returns nulls in every market, and its estimates appear only in the appendix.

[26] Their standard errors run from roughly 0.6 to 1.3 percentage points, too wide to rule out either a null or a large discount.

# B Sale-Price Estimates

Table A2: Sale-Price Effects, All Twenty-Two Estimable Offenses

| Offense | Identifying parcels | Sig. discount windows | Min $\delta$ (s.e.) [mo] | Max $\delta$ (s.e.) [mo] |
|---|---|---|---|---|
| *Priced* | | | | |
| Homicide | 3,053 | 2/12 | $-5.86^{*}$ (3.35) [1] | $-1.24$ (1.43) [8] |
| Narcotics | 35,801 | 5/12 | $-0.93$ (0.75) [1] | $-0.18$ (0.19) [6] |
| Burglary | 64,661 | 6/12 | $-0.34^{**}$ (0.14) [6] | $-0.18^{*}$ (0.09) [12] |
| Vandalism | 77,053 | 2/12 | $-0.17^{**}$ (0.08) [6] | 0.26 (0.22) [1] |
| *Suggestive* | | | | |
| Disorderly conduct | 13,670 | 1/12 | $-1.33^{*}$ (0.78) [2] | $-0.46$ (0.54) [4] |
| Sex offenses | 26,831 | 1/12 | $-0.47^{**}$ (0.23) [12] | 0.11 (0.35) [3] |
| *Precise nulls* | | | | |
| Theft | 86,303 | 0/12 | 0.01 (0.02) [12] | $0.08^{***}$ (0.03) [1] |
| Motor vehicle theft | 63,142 | 0/12 | $-0.02$ (0.09) [12] | $0.51^{*}$ (0.30) [1] |
| Assault (non-dom.) | 67,192 | 0/12 | $-0.05$ (0.06) [7] | 0.01 (0.11) [3] |
| Bad checks | 7,580 | 0/12 | $0.66^{*}$ (0.35) [11] | $2.20^{**}$ (1.03) [1] |
| Family (nonviol.) | 10,002 | 0/12 | $1.00^{*}$ (0.51) [12] | $2.16^{***}$ (0.82) [5] |
| Fraud | 53,093 | 0/12 | $-0.36$ (0.23) [1] | $-0.04$ (0.09) [6] |
| Assault (domestic) | 64,422 | 0/12 | $-0.19^{*}$ (0.10) [12] | 0.02 (0.15) [5] |
| Trespass | 41,169 | 0/12 | $-0.15$ (0.14) [6] | 0.18 (0.22) [3] |
| Robbery | 48,458 | 0/12 | $-0.06$ (0.14) [9] | 0.30 (0.42) [1] |
| Forgery | 19,519 | 0/12 | $-0.07$ (0.59) [1] | 0.47 (0.45) [2] |
| Liquor | 5,074 | 0/12 | 0.08 (1.20) [1] | 1.34 (0.91) [2] |
| Prostitution | 4,671 | 0/12 | $-1.00$ (0.82) [1] | 0.08 (0.12) [12] |
| *Inconclusive* | | | | |
| Weapons | 20,182 | 0/12 | $-0.62$ (0.47) [5] | 0.11 (0.58) [3] |
| Arson | 3,515 | 0/12 | $-0.20$ (1.06) [11] | 1.27 (3.58) [1] |
| Gambling | 1,333 | 0/12 | $-4.77$ (4.33) [3] | 1.99 (8.32) [1] |
| Stolen property | 1,012 | 0/12 | $-2.23$ (2.63) [2] | 0.81 (1.20) [8] |

*Notes.* All twenty-two estimable offenses under the specification of Table 2; kidnapping, embezzlement, extortion, child pornography, and other offenses have too few incidents to estimate and are excluded; columns as there, with Min $\delta$ and Max $\delta$ the most negative and most positive estimates across the twelve windows; offense groups as in Table 2: *priced*, a significant discount at two or more windows; *suggestive*, a significant discount at only one window; *precise nulls*, no significant discount and intervals that rule out a one-percent discount in at least nine of twelve windows; *inconclusive*, no significant discount but intervals too wide to rule out a one-percent discount. Sig. discount windows counts windows with a significantly negative estimate at five percent, so positive significant estimates (bad checks, family offenses) count as zero.
$^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table A3: Sale-Price Effects at All Twelve Exposure Windows

| | **Exposure window (cumulative months)** | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Offense | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 |
| *Priced* | | | | | | | | | | | | |
| Homicide | −5.86* | −2.23 | −1.45 | −2.04 | −1.28 | −1.30 | −1.59 | −1.24 | −2.54* | −3.02** | −3.15** | −2.49* |
| | (3.35) | (2.60) | (2.46) | (2.05) | (1.82) | (1.77) | (1.53) | (1.43) | (1.36) | (1.38) | (1.38) | (1.36) |
| Narcotics | −0.93 | −0.37 | −0.35 | −0.32 | −0.28 | −0.18 | −0.20 | −0.38** | −0.31** | −0.29** | −0.35*** | −0.33*** |
| | (0.75) | (0.42) | (0.38) | (0.27) | (0.23) | (0.19) | (0.17) | (0.18) | (0.15) | (0.14) | (0.13) | (0.11) |
| Burglary | −0.20 | −0.23 | −0.20 | −0.21 | −0.25* | −0.34** | −0.30** | −0.24** | −0.22** | −0.22** | −0.22** | −0.18* |
| | (0.31) | (0.20) | (0.17) | (0.16) | (0.15) | (0.14) | (0.12) | (0.12) | (0.11) | (0.10) | (0.10) | (0.09) |
| Vandalism | 0.26 | 0.13 | −0.06 | −0.11 | −0.16* | −0.17** | −0.16** | −0.13* | −0.10 | −0.08 | −0.06 | −0.05 |
| | (0.22) | (0.15) | (0.12) | (0.10) | (0.09) | (0.08) | (0.08) | (0.07) | (0.07) | (0.07) | (0.06) | (0.06) |
| *Suggestive* | | | | | | | | | | | | |
| Disorderly conduct | −1.22 | −1.33* | −0.65 | −0.46 | −0.71 | −0.66 | −0.58 | −0.50 | −0.63* | −0.60* | −0.57* | −0.65** |
| | (1.03) | (0.78) | (0.63) | (0.54) | (0.48) | (0.44) | (0.38) | (0.35) | (0.33) | (0.32) | (0.29) | (0.29) |
| Sex offenses | −0.08 | 0.10 | 0.11 | −0.02 | −0.29 | −0.31 | −0.25 | −0.20 | −0.26 | −0.39 | −0.43* | −0.47** |
| | (0.64) | (0.42) | (0.35) | (0.32) | (0.31) | (0.27) | (0.26) | (0.25) | (0.25) | (0.25) | (0.24) | (0.23) |
| *Precise nulls* | | | | | | | | | | | | |
| Larceny/theft | 0.08*** | 0.04* | 0.04* | 0.03** | 0.03* | 0.02 | 0.02 | 0.02 | 0.01 | 0.02 | 0.01 | 0.01 |
| | (0.03) | (0.02) | (0.02) | (0.02) | (0.01) | (0.01) | (0.02) | (0.01) | (0.02) | (0.01) | (0.02) | (0.02) |
| Motor vehicle theft | 0.51* | 0.28 | 0.17 | 0.10 | 0.05 | 0.04 | 0.09 | 0.09 | 0.05 | 0.06 | 0.07 | −0.02 |
| | (0.30) | (0.22) | (0.18) | (0.16) | (0.14) | (0.12) | (0.12) | (0.11) | (0.10) | (0.10) | (0.09) | (0.09) |
| Assault (non-dom.) | −0.05 | −0.02 | 0.01 | −0.03 | −0.04 | −0.04 | −0.05 | −0.03 | −0.02 | −0.01 | −0.03 | −0.04 |
| | (0.20) | (0.14) | (0.11) | (0.09) | (0.08) | (0.07) | (0.06) | (0.06) | (0.05) | (0.05) | (0.04) | (0.04) |
| Bad checks | 2.20** | 2.00*** | 1.36** | 1.38** | 1.09** | 1.18*** | 1.08** | 0.97** | 0.77* | 0.72** | 0.66* | 0.66** |
| | (1.03) | (0.74) | (0.61) | (0.57) | (0.53) | (0.45) | (0.43) | (0.41) | (0.39) | (0.37) | (0.35) | (0.33) |
| Family (non-violent) | 1.99 | 1.07 | 1.16 | 1.68* | 2.16*** | 1.77** | 1.89*** | 1.68*** | 1.49** | 1.28** | 1.07* | 1.00* |
| | (1.87) | (1.33) | (1.09) | (0.92) | (0.82) | (0.75) | (0.66) | (0.62) | (0.60) | (0.57) | (0.55) | (0.51) |
| Fraud | −0.36 | −0.29* | −0.11 | −0.07 | −0.06 | −0.04 | −0.05 | −0.05 | −0.05 | −0.08 | −0.11 | −0.11 |
| | (0.23) | (0.17) | (0.12) | (0.10) | (0.10) | (0.09) | (0.10) | (0.09) | (0.09) | (0.08) | (0.08) | (0.08) |
| Assault (domestic) | 0.02 | 0.00 | −0.02 | 0.02 | 0.02 | 0.01 | −0.03 | −0.09 | −0.17 | −0.13 | −0.18* | −0.19* |
| | (0.37) | (0.26) | (0.21) | (0.17) | (0.15) | (0.13) | (0.12) | (0.12) | (0.11) | (0.10) | (0.10) | (0.10) |
| Trespass | −0.09 | 0.16 | 0.18 | 0.00 | −0.10 | −0.15 | −0.14 | −0.12 | −0.06 | −0.05 | −0.09 | −0.14 |
| | (0.38) | (0.29) | (0.22) | (0.18) | (0.15) | (0.14) | (0.13) | (0.12) | (0.11) | (0.10) | (0.10) | (0.10) |
| Robbery | 0.30 | 0.15 | 0.09 | 0.08 | 0.04 | 0.02 | −0.02 | −0.03 | −0.06 | −0.01 | 0.01 | −0.03 |
| | (0.42) | (0.30) | (0.24) | (0.22) | (0.19) | (0.18) | (0.16) | (0.15) | (0.14) | (0.13) | (0.13) | (0.13) |
| Forgery | −0.07 | 0.47 | 0.12 | 0.15 | 0.05 | 0.38 | 0.44* | 0.39* | 0.30 | 0.30* | 0.23 | 0.21 |
| | (0.59) | (0.45) | (0.38) | (0.32) | (0.28) | (0.24) | (0.22) | (0.20) | (0.19) | (0.18) | (0.18) | (0.17) |
| Liquor law | 0.08 | 1.34 | 0.71 | 0.72 | 0.47 | 0.39 | 0.61 | 0.65 | 0.64* | 0.51 | 0.36 | 0.36 |
| | (1.20) | (0.91) | (0.81) | (0.60) | (0.52) | (0.48) | (0.43) | (0.40) | (0.38) | (0.36) | (0.35) | (0.35) |
| Prostitution | −1.00 | −0.43 | −0.23 | −0.23 | −0.18 | −0.28 | −0.14 | −0.04 | 0.01 | 0.01 | 0.05 | 0.08 |
| | (0.82) | (0.51) | (0.48) | (0.39) | (0.35) | (0.33) | (0.30) | (0.27) | (0.22) | (0.18) | (0.15) | (0.12) |
| *Inconclusive* | | | | | | | | | | | | |
| Weapons | −0.47 | 0.07 | 0.11 | −0.51 | −0.62 | −0.39 | −0.18 | −0.12 | −0.17 | 0.04 | 0.09 | 0.10 |
| | (0.92) | (0.71) | (0.58) | (0.48) | (0.47) | (0.44) | (0.40) | (0.37) | (0.35) | (0.31) | (0.30) | (0.28) |
| Arson | 1.27 | 0.32 | 0.27 | 0.10 | −0.13 | 0.07 | 1.07 | 0.57 | −0.15 | −0.13 | −0.20 | 0.07 |
| | (3.58) | (2.75) | (2.33) | (2.01) | (1.78) | (1.63) | (1.49) | (1.31) | (1.22) | (1.15) | (1.06) | (1.03) |
| Gambling | 1.99 | 1.69 | −4.77 | −1.39 | −1.35 | −0.76 | −1.17 | −3.04 | −2.84 | −2.91* | −1.79 | −1.94 |
| | (8.32) | (5.78) | (4.33) | (3.99) | (3.39) | (3.12) | (2.74) | (2.43) | (2.10) | (1.75) | (1.77) | (1.75) |
| Stolen property | −2.23 | −2.23 | −1.78 | −2.02 | −1.14 | −0.67 | 0.26 | 0.81 | 0.08 | 0.19 | −0.25 | −0.43 |
| | (4.06) | (2.63) | (2.18) | (1.95) | (1.83) | (1.75) | (1.59) | (1.20) | (1.06) | (1.02) | (1.05) | (1.05) |

*Notes.* $\delta$ at every cumulative window from one to twelve months, under the specification of Table 2; offense groups as there: *priced*, a significant discount at two or more windows; *suggestive*, a significant discount at only one window; *precise nulls*, no significant discount and intervals that rule out a one-percent discount in at least nine of twelve windows; *inconclusive*, no significant discount but intervals too wide to rule out a one-percent discount. Covers the twenty-two estimable categories, including the white-collar offenses, prostitution, gambling, and stolen property. Standard errors in parentheses.

$^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table A4: Priced Offenses under Two Classifications

| Offense | Priced | What is threatened | NIBRS crimes against |
|---|---|---|---|
| Homicide | Yes | Place (violent event) | Persons |
| Burglary | Yes | Dwelling | Property |
| Narcotics | Yes | Place (environment) | Society |
| Vandalism | Yes | Structure or land | Property |
| Disorderly conduct | Suggestive | Place (environment) | Group B (none) |
| Sex offenses | Suggestive | Person | Persons |
| Assault (domestic) | No | Person | Persons |
| Assault (non-domestic) | No | Person | Persons |
| Robbery | No | Person | Property |
| Theft | No | Portable asset | Property |
| Motor vehicle theft | No | Portable asset | Property |
| Stolen property | No | Portable asset | Property |
| Fraud | No | Financial | Property |
| Forgery | No | Financial | Property |
| Bad checks | No | Financial | Group B (none) |
| Arson | No | Structure or land | Property |
| Trespass | No | Structure or land | Group B (none) |
| Prostitution | No | Place (environment) | Society |
| Weapons | No | Place (environment) | Society |
| Gambling | No | Place (environment) | Society |
| Liquor law | No | Place (environment) | Group B (none) |
| Family offenses | No | Person | Group B (none) |

*Priced offenses by group*
Anchored to the address: 4 of 11; harms a person, money, or a movable item: 0 of 11
NIBRS (Group A): persons 1 of 4; property 2 of 9; society 1 of 4; Group B offenses (5) have no crime-against category.

*Notes.* "What is threatened" follows Table OA16. "NIBRS crimes against" is the Federal Bureau of Investigation's National Incident-Based Reporting System category for each offense. "Priced" follows Table 2. NIBRS assigns a crime-against category only to Group A offenses.

## B.1 Offenses with Monetary Losses

Fraud, forgery, and bad checks impose direct monetary losses, yet none carries a detectable discount in the pooled estimates (Table OA4); embezzlement has too few incidents to estimate. Bad checks instead carry a positive and significant estimate (0.66, s.e. 0.33), positive at five percent in ten of twelve windows (Table A3). Splitting each offense's incidents by recorded location (residential, street, and commercial) leaves the picture unchanged (Table A9). At the 12-month window, bad checks have no detectable effect in any location category, and forgery's only significant estimate is positive and occurs at commercial addresses.

Fraud is associated with a residential discount of 0.40 percent (s.e. 0.15) that survives the post-sale placebo. This estimate is one of sixty-six location-specific comparisons, does not survive a Bonferroni correction, and appears only at the longest exposure windows, so I treat it as suggestive.

Table A5: Effects of Crime on Housing Prices in the Prior Literature, Compared with This Paper

| Study | Setting | Design | Offense | Spatial unit of crime | Estimate as reported |
|---|---|---|---|---|---|
| *Sale prices, aggregate crime rates* | | | | | |
| Thaler (1978) | Rochester, NY | Cross-sectional hedonic | Property crime | Census tract | ≈ 3% per SD† |
| Gray & Joelson (1979) | Minneapolis | Cross-sectional hedonic, tract | Eight offense types | Tract | Only vandalism and burglary significant: +1% burglary −$336; +1% vandalism −$117† |
| Burnell (1988) | Chicago suburbs | IV, community fiscal and demographic instruments | Property crime | Community (71) | Elasticity −0.10† |
| Gibbons (2004) | London | Hedonic, deviations from district means, IV | Criminal damage | 250 m radius | ≈ 1% per 0.1 SD |
| Gibbons (2004) | London | Hedonic, deviations from district means, IV | Burglary | 250 m radius | Null |
| Schwartz, Susin & Voicu (2003) | New York City | Hedonic and repeat-sales price models | Violent (murder, rape, robbery, assault) | Precinct (74) | Elasticity 0.13; ≈ one third of the post-1994 boom |
| Schwartz, Susin & Voicu (2003) | New York City | Hedonic and repeat-sales price models | Property (burglary, larceny, auto theft) | Precinct (74) | Negative, insignificant |
| Tita, Petras & Greenbaum (2006) | Columbus, OH | Hedonic, IV (murder rate), by neighborhood income | Violent, property, total | Tract (189) | Violent crime most costly; effects vary by income |
| Ihlanfeldt & Mayock (2010) | Miami-Dade | First differences of tract repeat-sales index, IV | Murder, assault, robbery | Tract | Assault −0.15, robbery −0.11 (elasticities); murder null |
| Ihlanfeldt & Mayock (2010) | Miami-Dade | First differences of tract repeat-sales index, IV | Property | Tract | Small or null |
| Bishop & Murphy (2011) | San Francisco Bay Area | Dynamic hedonic (structural) | Violent (homicide, rape, robbery, assault) | City (79), distance-weighted | WTP ≈ $472/yr per 10% less |
| Pope & Pope (2012) | ≈3,000 U.S. zip codes | Zip-code fixed effects and IV, 1990s crime drop | Violent and property | Zip code | Elasticity −0.15 to −0.35 |
| *Sale prices, point-located events* | | | | | |
| Linden & Rockoff (2008) | Mecklenburg Co., NC | DiD, neighborhood-year FE | Sex-offender arrival | 0.1 mile | ≈ −4% |
| Pope (2008a) | Hillsborough Co., FL | DiD around offender arrival and departure | Sex-offender arrival | 0.1 mile | −2.3% |
| Wentland, Waller & Brastow (2014) | Central Virginia | 3SLS (price and time on market jointly) | Sex-offender proximity | 0.1 mile | Price ≈ −7%; time on market +88 days (≈ +80%) |
| Congdon-Hohman (2013) | Akron, OH | DiD, near vs. slightly farther | Methamphetamine-lab discovery | 1/8 mile (vs. 1/8–1/4) | −10% to −19% |
| Dealy, Horn & Berrens (2017) | Linn County, OR | Spatial difference-in-differences | Meth-lab discovery and decontamination | 0.1 mile (near vs. far) | −6.5% on discovery; +5% after decontamination |
| Gautier, Siegmann & van Vuuren (2009) | Amsterdam | DiD on asking prices | Murder (van Gogh) | Neighborhood (Muslim share > 25%) | ≈ −3% within 10 months |

*Continued on next page*

*Table A5, continued*

| Study | Setting | Design | Offense | Spatial unit of crime | Estimate as reported |
|---|---|---|---|---|---|
| Bhattacharya, Huang & Nielsen (2021) | Hong Kong | Unit FE (repeat sales), before/after death | Murder in unit | Unit, floor, building | Murder: unit ≈ −34%; floor −19%; building −10% (Table VI); all unnatural deaths: unit −20% |
| Tang & Le (2023) | Baltimore | Before/after offender arrival | Ex-gun-offender arrival | 0.1 mile | −8.0% ($12,632); no rise in crime |
| Braakmann (2017) | England and Wales | Street FE, regional time trends | Six categories (anti-social behaviour, violent, other, burglary, robbery, vehicle) | Street | Violent −0.6 to −1.6%, anti-social −0.6 to −0.8% per case per 10 residents; robbery, burglary, vehicle ≈ 0 or positive |
| Dentler & Rossi (2021) | Rochester, NY | Discontinuity in time (±30–60 days) within a distance band | Robbery | 0.3–0.6 km | −8 to −10% |
| | | | Aggravated assault | | −7 to −9% |
| | | | Burglary | | ≈ −6% |
| | | | Shootings | | Insignificant |
| *Rents* | | | | | |
| Rizzo (1979) | Chicago | Cross-section, IV | Total crime | Community area | Value elasticity ≈ −1.67†; rent elasticity −0.24† |
| Koirala, Chakraborti & Pradhan (2023) | North Carolina cities | Seemingly unrelated regressions of wages and rents | Violent and property rates | City | ≈ $52/month net compensation |
| Donovan et al. (2024) | New Zealand | Cross-section of urban areas, spatial equilibrium, control-function IV | Victimizations per 1,000 | Urban area (134) | Rent elasticity −0.15 to −0.44 |
| Klimova & Lee (2014) | Sydney | Hedonic DiD | Murder | 0.1–0.3 mile | Rents: no effect; values −3.9% within 0.2 mile |
| Carrillo & Yezer (2021) | Washington, DC | Rent and value hedonics, tract FE | Homicide, violent | 0.25 mile | Not directly comparable |
| *This paper* | | | | | |
| This paper | Chicago | Repeat sales, near–far, parcel + tract×month FE | Homicide | 0.05 mile vs. 0.20–0.30 mile | −3.0% per incident (−3.0 to −3.1%, 10–11 months) |
| | | | Burglary | | −0.24% (−0.22 to −0.34%, 6–11 months) |
| | | | Narcotics | | −0.29% (−0.29 to −0.38%, 8–12 months) |
| | | | Vandalism | | −0.17% (−0.16 to −0.17%, 6–7 months) |
| | | | Sex offenses; disorderly conduct | | −0.47%; −0.65% (12 months only; suggestive) |
| | | | Other offenses | | No detectable discount |
| | | | Rents (burglary; sex offenses; liquor law) | | −0.34 to −0.07% (6 of 12 windows); −0.32 to −0.16% (7 of 12, parcel×era); −0.30 to 0.09% (4 of 12, parcel×era) |

*Notes.* Estimates are in the units of the original study and are not on a common scale. Prior estimates can be compared with this paper's estimates for the same offense in the final panel; ranges cover the windows significant at five percent (Table A3); rent ranges are the minimum and maximum across the twelve windows with the number of significant windows (Table A23). Distance bands for Pope (2008a) are inferred from the design. Dentler & Rossi (2021): Model (4) of their Table 3; their simpler models are mostly insignificant. Outcomes are sale prices except asking prices (Gautier, Siegmann & van Vuuren, 2009), a tract repeat-sales price index (Ihlanfeldt & Mayock, 2010), owner-reported values and rents (Rizzo, 1979), owner-estimated median values (Gray & Joelson, 1979; Burnell, 1988), sale prices and asking rents (Klimova & Lee, 2014), rents and wages (Koirala, Chakraborti & Pradhan, 2023; Donovan et al., 2024), rents and values (Carrillo & Yezer, 2021), and housing values (Tang & Le, 2023); Bishop & Murphy (2011) estimate willingness to pay from sale prices in a dynamic model. This paper uses sale prices, asking rents, and transaction margins. †Reported in a secondary source.

# C Spatial and Temporal Structure

**Spatial decay.** The distance over which discounts appear differs across offenses. Table A6 sets the near ring to each successive ring while holding the far ring at 0.20–0.30 miles. Widening burglary's near ring to 0–0.10 miles gives −0.14 percent (s.e. 0.07) at eight months, significant in 5 of 12 windows, with no significant post-sale placebo; the 0.05–0.10 ring alone is significant only at one month. Homicide discounts are significant in the 0.10–0.15 mile ring in seven of twelve windows (months three to nine), but not at ten months, where the estimate is −1.16 percent (s.e. 0.64). The post-sale placebo is not significant in that ring at any window, and homicide is not significant in the 0.05–0.10 ring at any window. For narcotics, the pre-sale estimate in the 0.05–0.10 ring is not significant at any window, but the post-sale placebo is significant there at windows four to eight.

Table A6: The Near–Far Discount Across Nine Ring Definitions

| Near vs. far ring (miles) | Homicide | Burglary | Narcotics | Vandalism |
|---|---|---|---|---|
| 0–0.05 vs. 0.20–0.30 *(baseline)* | -3.02 (1.38) | -0.24 (0.12) | -0.29 (0.14) | -0.17 (0.08) |
| 0–0.05 vs. 0.25–0.35 | -2.75 (1.39) | -0.24 (0.12) | -0.32 (0.14) | -0.16 (0.08) |
| 0–0.05 vs. 0.30–0.40 | -2.66 (1.35) | -0.27 (0.12) | -0.32 (0.14) | -0.15 (0.08) |
| 0–0.10 vs. 0.20–0.30 | -0.42 (0.67) | -0.14 (0.07) | 0.01 (0.06) | -0.06 (0.05) |
| 0–0.15 vs. 0.20–0.30 | -0.90 (0.54) | -0.05 (0.05) | 0.05 (0.04) | 0.01 (0.03) |
| 0–0.20 vs. 0.20–0.30 | -0.74 (0.46) | -0.03 (0.04) | 0.02 (0.03) | -0.02 (0.03) |
| 0.05–0.10 vs. 0.20–0.30 | 0.55 (0.78) | -0.09 (0.07) | 0.06 (0.07) | -0.02 (0.05) |
| 0.10–0.15 vs. 0.20–0.30 | -1.16 (0.64) | 0.04 (0.06) | 0.06 (0.05) | 0.07 (0.04) |
| 0.15–0.20 vs. 0.20–0.30 | -0.45 (0.58) | -0.00 (0.05) | -0.02 (0.04) | -0.05 (0.04) |

*Notes:* Each row is one near/far ring pair: the baseline (row 1), the far ring moved out (rows 2–3), the near ring widened (rows 4–6), and an outer ring in place of the near ring (rows 7–9). Windows: homicide 10, burglary 8, narcotics 10, vandalism 6 months. $N = 242{,}815$.

## C.1 Recent versus Earlier Exposure

The timing of exposure matters most clearly for vandalism, whose discount is concentrated among incidents two to six months before sale. Estimated jointly, exposure two to six months before the sale and exposure eight to eleven months before carry different weights across offenses. The vandalism discount is concentrated in the recent block and absent from the late one, and the difference is significant ($p = 0.008$); the narcotics discount is concentrated in the late block, with the difference significant at ten percent ($p = 0.09$); for homicide and burglary the point estimates are larger in the late and early block respectively, but the two blocks are not statistically distinguishable ($p = 0.12$ and 0.20; Figure A1).

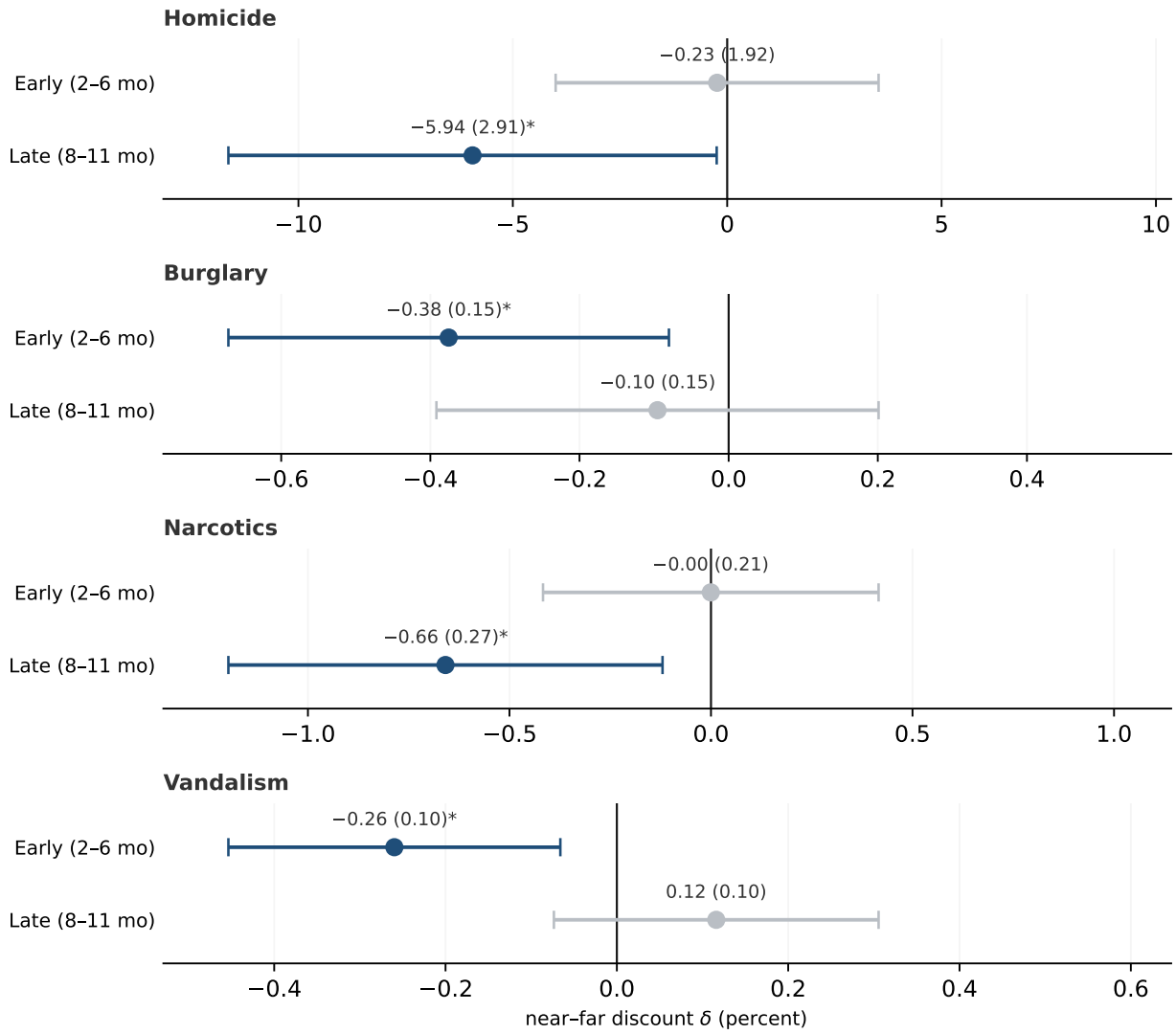



Figure A1: Early versus late exposure, estimated jointly. For each priced offense, the near–far discount $\delta$ (%) from one regression that enters an early block (incidents two to six months before the sale) and a late block (eight to eleven months before) as separate counts, each with its far-ring counterpart and separate controls for the omitted months (one, seven, twelve). Month one is omitted because it falls after the contract date for most sales: the median time from contract to close is 42 days, and for about three-quarters of sales it exceeds 30 days, so crime in the last month before the close could affect the agreed price only through renegotiation. Month seven separates the two blocks, and month twelve ends the window. A single late block of months seven to twelve, estimated alone, gives the same pattern (homicide −3.60%, s.e. 2.13; narcotics −0.47%, s.e. 0.20; burglary −0.03% and vandalism 0.06%, neither significant). The two points in each panel are the early-block and late-block estimates (significant at five percent in colour, otherwise grey), shown with 95% confidence intervals and the estimate and standard error beside each point; each panel has its own horizontal scale. The Wald test that the early and late coefficients are equal gives $p = 0.121$ (homicide), 0.200 (burglary), 0.090 (narcotics), and 0.008 (vandalism), so the early and late estimates are individually distinguishable only for vandalism. Parcel and tract×year-month fixed effects, clustered by census tract, $N = 242{,}815$.

# D Observability and Recurrence

Table A7: Where the Discount Comes From: Private versus Public Incidents

| Offense | Private (residential) | Public (street) | $p$: priv.=pub. | Windows sig. | $N$ |
|---|---|---|---|---|---|
| Homicide | −0.41 | −4.59*** | 0.137 | 1/12 | 242,815 |
| | (2.39) | (1.68) | | | |
| Burglary | −0.25* | −0.93 | 0.289 | 0/12 | 242,815 |
| | (0.14) | (0.65) | | | |
| Narcotics | −0.28 | −0.38** | 0.897 | 0/12 | 242,815 |
| | (0.65) | (0.19) | | | |
| Vandalism | −0.56*** | −0.01 | 0.018** | 7/12 | 242,815 |
| | (0.20) | (0.11) | | | |

*Notes.* Near–far close-price effect (%) of each priced offense's near-ring count, split by whether the incident was recorded at a *residential* address (private: the dwelling and its garage, porch, yard, hallways, and stairwells), in *public space* (streets, sidewalks, alleys, parking and vacant lots, parks, and vehicles and transit; Online Appendix C), or at a commercial address; residential, street, and commercial incidents are entered jointly in one regression under parcel fixed effects, and the commercial estimates are reported in Table A9. Pooled 2007–2025, parcel and tract×month fixed effects, near 0–0.05 vs. far 0.20–0.30 mi; standard errors clustered by tract in parentheses. Estimates at each offense's main window (homicide 10, burglary 8, narcotics 10, vandalism 6 months); "Windows sig." counts, of the twelve exposure windows, how many reject private=public at five percent. $N = 242{,}815$.
$^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table A8: Residential versus Street Incidents by Exposure Window, Priced Offenses

| *Panel A* | | | | | | |
|---|---|---|---|---|---|---|
| | Homicide | | | Narcotics | | |
| Window | Residential | Street | $p$ | Residential | Street | $p$ |
| 1 | 0.19 | −8.29* | 0.227 | −2.61 | −0.65 | 0.447 |
| | (5.62) | (4.29) | | (2.63) | (0.81) | |
| 2 | 3.47 | −4.75 | 0.099 | −1.10 | −0.26 | 0.647 |
| | (3.97) | (3.42) | | (1.69) | (0.50) | |
| 3 | 0.80 | −2.23 | 0.526 | −1.16 | −0.26 | 0.520 |
| | (3.36) | (3.47) | | (1.25) | (0.46) | |
| 4 | −0.58 | −2.67 | 0.624 | −0.22 | −0.30 | 0.948 |
| | (2.92) | (2.93) | | (1.09) | (0.35) | |
| 5 | −0.10 | −1.72 | 0.678 | −0.70 | −0.22 | 0.647 |
| | (2.64) | (2.65) | | (0.92) | (0.30) | |
| 6 | 0.81 | −2.33 | 0.410 | −0.34 | −0.16 | 0.843 |
| | (2.72) | (2.50) | | (0.82) | (0.25) | |
| 7 | 0.79 | −2.83 | 0.297 | −0.08 | −0.24 | 0.853 |
| | (2.50) | (2.17) | | (0.75) | (0.21) | |
| 8 | 1.93 | −2.89 | 0.136 | −0.14 | −0.47** | 0.679 |
| | (2.47) | (1.95) | | (0.71) | (0.22) | |
| 9 | 0.35 | −4.22** | 0.106 | −0.24 | −0.39** | 0.851 |
| | (2.31) | (1.73) | | (0.70) | (0.19) | |
| 10 | −0.41 | −4.59*** | 0.137 | −0.28 | −0.38** | 0.897 |
| | (2.39) | (1.68) | | (0.65) | (0.19) | |
| 11 | 0.00 | −5.04*** | 0.080 | −0.32 | −0.43** | 0.879 |
| | (2.48) | (1.64) | | (0.62) | (0.17) | |
| 12 | 1.16 | −4.53*** | 0.046 | −0.62 | −0.35** | 0.677 |
| | (2.43) | (1.65) | | (0.58) | (0.14) | |
| Sig. windows | 0/12 | 4/12 | 1/12 | 0/12 | 5/12 | 0/12 |
| *Panel B* | | | | | | |
| | Vandalism | | | Burglary | | |
| Window | Residential | Street | $p$ | Residential | Street | $p$ |
| 1 | −0.02 | 0.39 | 0.495 | −0.21 | 1.72 | 0.216 |
| | (0.53) | (0.27) | | (0.41) | (1.47) | |
| 2 | −0.20 | 0.29 | 0.270 | −0.30 | 0.36 | 0.520 |
| | (0.38) | (0.19) | | (0.27) | (1.00) | |
| 3 | −0.35 | 0.07 | 0.238 | −0.19 | −0.09 | 0.916 |
| | (0.31) | (0.15) | | (0.22) | (0.89) | |
| 4 | −0.37 | −0.01 | 0.182 | −0.22 | −0.51 | 0.712 |
| | (0.24) | (0.13) | | (0.20) | (0.78) | |
| 5 | −0.45** | −0.03 | 0.090 | −0.25 | −0.33 | 0.925 |
| | (0.21) | (0.12) | | (0.18) | (0.74) | |
| 6 | −0.56*** | −0.01 | 0.018 | −0.37** | −0.55 | 0.810 |
| | (0.20) | (0.11) | | (0.16) | (0.72) | |
| 7 | −0.62*** | 0.05 | 0.002 | −0.31** | −0.86 | 0.425 |
| | (0.18) | (0.10) | | (0.15) | (0.69) | |
| 8 | −0.61*** | 0.08 | 0.001 | −0.25* | −0.93 | 0.289 |
| | (0.18) | (0.09) | | (0.14) | (0.65) | |
| 9 | −0.53*** | 0.09 | 0.002 | −0.25* | −0.86 | 0.340 |
| | (0.17) | (0.08) | | (0.13) | (0.63) | |
| 10 | −0.51*** | 0.11 | 0.001 | −0.28** | −0.92 | 0.317 |
| | (0.16) | (0.08) | | (0.12) | (0.64) | |
| 11 | −0.45*** | 0.11 | 0.002 | −0.29** | −0.87 | 0.340 |
| | (0.16) | (0.08) | | (0.12) | (0.60) | |
| 12 | −0.47*** | 0.13* | 0.000 | −0.24** | −0.88 | 0.287 |
| | (0.15) | (0.07) | | (0.11) | (0.59) | |
| Sig. windows | 8/12 | 0/12 | 7/12 | 5/12 | 0/12 | 0/12 |

*Notes.* Window-by-window version of Table A7: near–far close-price effect (%) per incident of residential and street (public-space) incidents, estimated jointly with commercial incidents in one regression, at every exposure window 1–12 months before the sale; standard errors clustered by tract in parentheses. $p$: Wald test that the residential and street effects are equal. "Sig. windows" counts windows significant at five percent. Pooled 2007–2025, parcel and tract×month fixed effects, $N$ = 242,815. The pooled residential burglary estimate is matched by post-sale crime (Table 3 note); the frontage–interior split in Table 3 is the relevant test for burglary. Main exposure windows are 10 months for homicide and narcotics, 8 months for burglary, and 6 months for vandalism.
*$p < 0.10$, **$p < 0.05$, ***$p < 0.01$.

Table A9: Incident-Location Split, All Estimable Offenses

| Offense | Residential | Street | Commercial | $p$ res=str | $p$ str=com | $p$ all eq. | $N$ |
|---|---|---|---|---|---|---|---|
| Homicide | −0.41 | −4.59*** | −0.77 | 0.137 | 0.247 | 0.269 | 242,815 |
| | (2.39) | (1.68) | (2.76) | | | | |
| Burglary | −0.25* | −0.93 | 0.00 | 0.289 | 0.151 | 0.243 | 242,815 |
| | (0.14) | (0.65) | (0.16) | | | | |
| Narcotics | −0.28 | −0.38** | −0.10 | 0.897 | 0.766 | 0.951 | 242,815 |
| | (0.65) | (0.19) | (0.91) | | | | |
| Vandalism | −0.56*** | −0.01 | −0.23 | 0.018** | 0.338 | 0.054* | 242,815 |
| | (0.20) | (0.11) | (0.20) | | | | |
| Theft | −0.03 | 0.04*** | 0.00 | 0.326 | 0.040** | 0.105 | 242,815 |
| | (0.07) | (0.01) | (0.02) | | | | |
| Motor-vehicle theft | 0.86** | −0.04 | −0.62 | 0.042** | 0.308 | 0.069* | 242,815 |
| | (0.42) | (0.09) | (0.56) | | | | |
| Robbery | 0.47 | −0.06 | −0.06 | 0.480 | 0.994 | 0.777 | 242,815 |
| | (0.75) | (0.15) | (0.21) | | | | |
| Assault (non-dom.) | −0.28* | 0.01 | 0.03 | 0.110 | 0.849 | 0.219 | 242,815 |
| | (0.16) | (0.08) | (0.07) | | | | |
| Assault (dom.) | −0.26** | 0.04 | 0.10 | 0.296 | 0.892 | 0.453 | 242,815 |
| | (0.12) | (0.24) | (0.35) | | | | |
| Disorderly conduct | −1.58* | −0.21 | −0.85 | 0.145 | 0.625 | 0.340 | 242,815 |
| | (0.84) | (0.41) | (1.09) | | | | |
| Trespass | −0.43** | −0.24 | 0.09 | 0.478 | 0.210 | 0.071* | 242,815 |
| | (0.19) | (0.21) | (0.14) | | | | |
| Prostitution | 0.80 | 0.02 | 0.27 | 0.612 | 0.294 | 0.551 | 242,815 |
| | (1.54) | (0.16) | (0.18) | | | | |
| Fraud | −0.40*** | 0.24 | 0.02 | 0.010** | 0.324 | 0.007*** | 242,815 |
| | (0.15) | (0.24) | (0.08) | | | | |
| Sex offenses | −0.64** | −0.05 | −0.71 | 0.284 | 0.305 | 0.498 | 242,815 |
| | (0.32) | (0.47) | (0.47) | | | | |
| Forgery | −0.39 | 0.39 | 0.95*** | 0.155 | 0.289 | 0.044** | 242,815 |
| | (0.43) | (0.41) | (0.29) | | | | |
| Weapons | −1.10 | 0.19 | 1.54 | 0.311 | 0.338 | 0.376 | 242,815 |
| | (1.21) | (0.28) | (1.37) | | | | |
| Arson | −1.90 | 2.21 | 2.52 | 0.072* | 0.897 | 0.164 | 242,815 |
| | (1.96) | (1.40) | (1.95) | | | | |
| Stolen property | −0.82 | −1.58 | 1.93 | 0.858 | 0.093* | 0.231 | 242,815 |
| | (3.81) | (1.60) | (1.51) | | | | |
| Liquor | 3.24 | 0.60 | 0.31 | 0.408 | 0.812 | 0.601 | 242,815 |
| | (2.94) | (1.14) | (0.36) | | | | |
| Gambling | −5.64 | −1.69 | −1.95 | 0.706 | 0.957 | 0.931 | 242,815 |
| | (9.72) | (2.03) | (4.27) | | | | |
| Bad checks | 0.76 | 2.40 | 0.41 | 0.605 | 0.533 | 0.768 | 242,815 |
| | (0.62) | (3.14) | (0.44) | | | | |
| Family offenses | 1.00 | 0.78 | 2.36 | 0.879 | 0.484 | 0.742 | 242,815 |
| | (0.68) | (1.26) | (1.68) | | | | |

*Notes.* Three-category incident-location split (residential / street / commercial) for all 22 estimable offenses, at each priced offense's main window (homicide 10, burglary 8, narcotics 10, vandalism 6) and a 12-month window otherwise; entered jointly, pooled 2007–2025, parcel and tract×month fixed effects, standard errors clustered by tract. The last three columns are Wald $p$-values that residential equals street, that street equals commercial, and that all three are equal. This three-category detail underlies the private/public summary in Table A7. $N = 242{,}815$.
$^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table A10: Forecast of Future Near-Ring Crime by Incident Location, All Windows

| | Homicide | | Burglary | | Narcotics | | Vandalism | |
|---|---|---|---|---|---|---|---|---|
| Win. | Res | St. | Res | St. | Res | St. | Res | St. |
| 1 | 0.000 | 0.000 | 0.032 | 0.015 | 0.007 | 0.099 | 0.020 | 0.021 |
| 2 | 0.000 | 0.001 | 0.069 | 0.024 | 0.014 | 0.156 | 0.050 | 0.049 |
| 3 | 0.000 | 0.002 | 0.100 | 0.029 | 0.023 | 0.186 | 0.079 | 0.073 |
| 4 | 0.000 | 0.003 | 0.126 | 0.033 | 0.029 | 0.205 | 0.104 | 0.093 |
| 5 | 0.000 | 0.004 | 0.149 | 0.035 | 0.036 | 0.218 | 0.127 | 0.109 |
| 6 | 0.000 | 0.005 | 0.169 | 0.037 | 0.042 | 0.232 | 0.148 | 0.125 |
| 7 | 0.000 | 0.006 | 0.189 | 0.038 | 0.049 | 0.241 | 0.167 | 0.141 |
| 8 | 0.000 | 0.008 | 0.208 | 0.037 | 0.054 | 0.251 | 0.185 | 0.155 |
| 9 | 0.000 | 0.009 | 0.226 | 0.036 | 0.060 | 0.258 | 0.200 | 0.168 |
| 10 | 0.001 | 0.011 | 0.240 | 0.034 | 0.066 | 0.263 | 0.214 | 0.178 |
| 11 | 0.001 | 0.012 | 0.253 | 0.033 | 0.071 | 0.267 | 0.227 | 0.188 |
| 12 | 0.001 | 0.014 | 0.265 | 0.031 | 0.077 | 0.271 | 0.238 | 0.196 |

*Notes.* Forecast $R^2$ from regressing future near-ring crime (0–0.05 mi, after the sale) on past surrounding crime (far ring 0.20–0.30 mi, before the sale), estimated separately for residential (Res) and street (St.) incidents, no fixed effects (the informational benchmark available to a buyer at sale), all geocoded closed MLS sales before the constant-parcel and repeat-sales filters, $N = 923{,}730$, at each exposure window (Win., months). For *vandalism*, residential and street incidents forecast future near-ring crime comparably at six months, yet only residential vandalism is priced, so the difference in price effects cannot reflect a difference in objective recurrence. Homicide and narcotics forecast from the street and burglary from residential incidents, as expected from where those offenses physically occur.

Table A11: Vandalism by Target: Damage to Property versus Vehicle

| | Exposure window (months before sale) | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 |
| *Panel A. Parcel + tract×year-month fixed effects (baseline)* | | | | | | | | | | | | |
| Damage to property | 0.85 | 1.06 | 0.80 | 0.22 | 0.03 | −0.49 | −0.89** | −0.95** | −0.91** | −1.03*** | −0.92** | −0.89*** |
| | (1.12) | (0.83) | (0.63) | (0.58) | (0.51) | (0.47) | (0.45) | (0.43) | (0.41) | (0.39) | (0.36) | (0.34) |
| Damage to vehicle | 0.70 | 0.89 | 0.50 | 0.33 | 0.20 | 0.21 | 0.05 | −0.07 | −0.13 | −0.08 | −0.09 | 0.02 |
| | (0.93) | (0.67) | (0.56) | (0.53) | (0.48) | (0.45) | (0.42) | (0.40) | (0.40) | (0.38) | (0.37) | (0.34) |
| Property = vehicle ($p$) | 0.916 | 0.874 | 0.724 | 0.896 | 0.818 | 0.298 | 0.143 | 0.142 | 0.168 | 0.078 | 0.102 | 0.057 |
| *Panel B. Parcel×era + tract×year-month fixed effects* | | | | | | | | | | | | |
| Damage to property | −0.23 | 2.49* | 1.17 | 0.02 | 0.33 | 0.11 | −0.59 | −0.65 | −0.66 | −1.22* | −0.99 | −0.91 |
| | (1.97) | (1.47) | (1.17) | (1.02) | (0.92) | (0.80) | (0.73) | (0.68) | (0.64) | (0.62) | (0.60) | (0.56) |
| Damage to vehicle | 0.52 | 0.29 | 0.11 | 0.16 | 0.20 | −0.09 | −0.30 | −0.37 | −0.62 | −0.80* | −0.86* | −0.81* |
| | (1.20) | (0.80) | (0.65) | (0.67) | (0.65) | (0.60) | (0.56) | (0.52) | (0.53) | (0.48) | (0.46) | (0.42) |
| Property = vehicle ($p$) | 0.759 | 0.208 | 0.447 | 0.914 | 0.918 | 0.849 | 0.764 | 0.753 | 0.962 | 0.602 | 0.869 | 0.895 |

*Notes.* Full twelve-window version of Table 3, Panel B: vandalism frontage incidents split by target, damage to the property versus damage to a vehicle, each as its own near–far difference $\delta = (\beta_{\text{near}} - \beta_{\text{far}}) \times 100$ in one regression, pooled 2007–2025, clustered by tract; standard errors below the estimates. The "Property = vehicle" row is the Wald $p$-value for equality of the two arms. *Panel A* is the baseline design (parcel and tract×year-month fixed effects, $N = 242{,}815$). *Panel B* absorbs parcel×era instead of parcel ($N = 123{,}133$), which drops parcels that do not resell within an era.
$^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

**Forecastability and recurrence.** Crime in the far ring predicts subsequent near-ring crime for every offense at twelve months and all but one at six, although its incremental explanatory power is modest. Table A13 examines whether past crime in the far ring (0.20–0.30 miles) predicts future crime in the near ring (0–0.05 miles). In forecasting regressions without fixed effects, far-ring crime significantly predicts near-ring crime at the five-percent level for twenty-one of twenty-two offenses at six months (all except stolen property) and for all twenty-two at twelve months. Its incremental explanatory power is modest: after controlling for past near-ring crime, the far ring's partial $R^2$ is at most 0.09 for any offense at either horizon.

Table A12 reports recurrence coefficients from regressions of post-sale near-ring crime on pre-sale near-ring crime, with parcel and tract $\times$ month fixed effects. The pre- and post-sale windows have equal lengths, matching the window used to report each offense's price estimate. The coefficient $\rho$ measures how strongly past incidents predict future incidents within 0.05 miles of the same parcel, conditional on these fixed effects. It does not identify the causal effect of an incident on subsequent crime.

The forecasting and recurrence exercises measure different aspects of predictability. The first assesses the information in surrounding crime, including persistent differences across locations; the second measures recurrence after absorbing parcel and tract-by-month differences. Neither directly measures the change in buyers' expectations after learning of an incident.

Table A12: Crime Recurrence and Price Effects

| Offense | Recurrence $\rho$ (parcel + tract FE) | $\rho$, no parcel FE | $\delta$ (Table 2) [mo] |
|---|---|---|---|
| Homicide† | $-0.026^{**}$ | $0.070^{***}$ | $-3.02^{**}$ [10] |
| | (0.012) | (0.006) | (1.38) |
| Burglary† | $0.067^{***}$ | $0.362^{***}$ | $-0.24^{**}$ [8] |
| | (0.009) | (0.011) | (0.12) |
| Narcotics† | $0.397^{***}$ | $0.685^{***}$ | $-0.29^{**}$ [10] |
| | (0.038) | (0.013) | (0.14) |
| Vandalism† | $0.093^{***}$ | $0.389^{***}$ | $-0.17^{**}$ [6] |
| | (0.018) | (0.013) | (0.08) |
| Theft | 0.237 | – | 0.01 [12] |
| | (0.158) | | (0.02) |
| Motor-vehicle theft | $0.140^{***}$ | – | $-0.02$ [12] |
| | (0.031) | | (0.09) |
| Robbery | $0.038^{***}$ | – | $-0.03$ [12] |
| | (0.012) | | (0.13) |
| Assault (non-domestic) | $0.280^{***}$ | – | $-0.04$ [12] |
| | (0.029) | | (0.04) |
| Assault (domestic) | $0.118^{***}$ | – | $-0.19^{*}$ [12] |
| | (0.012) | | (0.10) |
| All crime | $0.314^{***}$ | – | – |
| | (0.113) | | |

*Notes.* Own-recurrence $\rho$ of near-ring crime: the 0–0.05 mi incident count over the window *after* the sale regressed on the count over the matching window *before*, standard errors clustered by tract in parentheses; $N = 242{,}815$. For the four priced offenses $\rho$ is measured over the same exposure window as $\delta$ (homicide and narcotics ten months, burglary eight, vandalism six; shown in the $\delta$ bracket); for the remaining offenses $\rho$ is measured over six months and $\delta$ is shown at the twelve-month window (Table A3). The first $\rho$ column uses parcel and tract × sale-month fixed effects; the second omits parcel fixed effects and is shown for the priced offenses. Parcel fixed effects bias $\rho$ toward zero in short panels, whereas the no-fixed-effect estimate includes time-invariant level differences across properties and overstates recurrence, so the two bracket the within-parcel persistence; for homicide the within-parcel estimate is slightly negative while the no-fixed-effect estimate is positive. $\rho > 0$ means the offense recurs at the same property. $\delta$ is each offense's near–far close-price discount at its main window (Table 2; window length in brackets), unpriced offenses at the twelve-month reference window. †Priced offenses.
$^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table A13: Rational Forecast: The Discount Does Not Track Forecastable Risk

| | 0–6 months | | 0–12 months | |
|---|---|---|---|---|
| Offense | Joint $F$ | Partial $R^2$ | Joint $F$ | Partial $R^2$ |
| Homicide† | $127.3^{***}$ | 0.004 | $237.0^{***}$ | 0.009 |
| Burglary† | $1580.8^{***}$ | 0.047 | $1871.1^{***}$ | 0.062 |
| Narcotics† | $411.1^{***}$ | 0.038 | $251.5^{***}$ | 0.037 |
| Vandalism† | $711.6^{***}$ | 0.045 | $627.5^{***}$ | 0.054 |
| Theft | $6.8^{***}$ | 0.035 | $9.1^{***}$ | 0.017 |
| Motor-vehicle theft | $789.0^{***}$ | 0.047 | $721.1^{***}$ | 0.052 |
| Robbery | $357.2^{***}$ | 0.045 | $365.8^{***}$ | 0.060 |
| Assault (non-dom.) | $103.6^{***}$ | 0.030 | $99.3^{***}$ | 0.025 |
| Assault (domestic) | $696.5^{***}$ | 0.062 | $497.1^{***}$ | 0.058 |
| Arson | $80.3^{***}$ | 0.001 | $204.3^{***}$ | 0.002 |
| Bad checks | $27.4^{***}$ | 0.002 | $34.0^{***}$ | 0.005 |
| Disorderly conduct | $129.5^{***}$ | 0.012 | $132.0^{***}$ | 0.020 |
| Family offenses | $109.4^{***}$ | 0.002 | $148.8^{***}$ | 0.004 |
| Forgery | $82.3^{***}$ | 0.012 | $91.1^{***}$ | 0.018 |
| Fraud | $102.1^{***}$ | 0.087 | $75.4^{***}$ | 0.075 |
| Gambling | $170.2^{***}$ | 0.013 | $261.6^{***}$ | 0.029 |
| Liquor | $30.3^{***}$ | 0.001 | $34.5^{***}$ | 0.003 |
| Prostitution | $18.0^{***}$ | 0.007 | $10.9^{***}$ | 0.007 |
| Sex offenses | $75.8^{***}$ | 0.008 | $45.2^{***}$ | 0.014 |
| Stolen property | $3.2^{*}$ | 0.000 | $5.8^{**}$ | 0.000 |
| Trespass | $22.4^{***}$ | 0.019 | $20.2^{***}$ | 0.022 |
| Weapons | $251.0^{***}$ | 0.054 | $348.9^{***}$ | 0.067 |

*Notes.* Estimated without fixed effects. Outcome: crime within 0.05 miles after the sale. Predictor: crime 0.20–0.30 miles away before the sale, controlling for the near ring's own prior crime. Windows: 0–6 and 0–12 months. $F$ tests that the far-ring term is zero; partial $R^2$ is the share of the remaining variation the far ring explains. A predictive benchmark, not a causal estimate.
$^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. † priced offense.

# E Perceived Safety

**Recorded crime and perceived safety.** Recorded crime and perceived safety do not align uniformly across offenses. I relate monthly police-sector safety scores to recorded crime during the preceding three months, using sector and calendar-month fixed effects. Figure A2 reports estimates from a regression including all twenty-two offenses jointly, scaled to a one-standard-deviation increase in each offense's log-transformed count. Homicide, burglary, and narcotics are associated with lower perceived safety, as are domestic assault and trespass, neither of which carries a consistent sale-price discount. Vandalism is positively associated with safety in the joint regression, although its association is not statistically significant when entered separately. For homicide, burglary, and narcotics, the joint estimates imply declines of approximately 0.04 points on the 0–10 safety scale per standard-deviation increase in crime. These associations describe residents' perceptions at the sector level; they do not measure buyers' awareness of individual incidents or establish why those incidents enter prices.

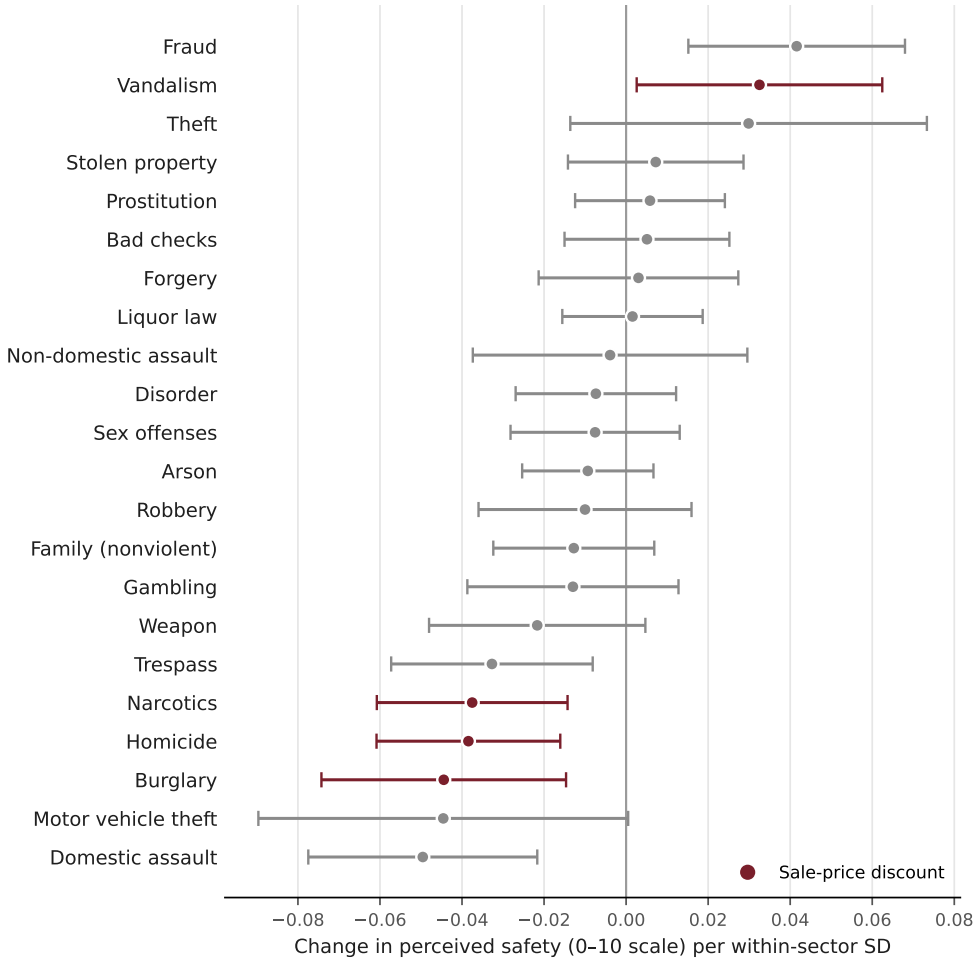



Figure A2: Recorded crime and perceived safety. The outcome is the safety score from the City of Chicago's Police Sentiment Scores (Chicago Data Portal, dataset 28me-84fj), a survey of residents' reported safety in their neighborhood, for each police sector and calendar month from November 2017 to December 2023. Scores are reported by calendar month and weighted to Census demographics; whether responses are pooled across months is not documented. All twenty-two offenses enter one regression jointly, each as $\log(1 + \text{recorded incidents})$ in the sector over the three complete months before the survey month, with sector and calendar-month fixed effects. Each coefficient is multiplied by the within-sector standard deviation of its regressor (deviations from the sector mean) and divided by ten, so it gives the change in safety on a 0–10 scale associated with a one-standard-deviation within-sector increase in that offense. Bars show 95% confidence intervals based on standard errors clustered by sector. $N = 4{,}958$ sector-months in 67 sectors. Two sectors in the survey, 2540 and 2550, correspond to no police beat and receive no recorded incidents; they are excluded. Maroon markers indicate the four offenses with sale-price discounts. Estimates are conditional associations, not causal effects.

# F Price Formation and Agent Experience

**Additional transaction-outcome results.** Asking-price responses differ across offenses and fixed-effect specifications. Burglary's asking-price estimate is −0.15 (s.e. 0.12). Disorderly conduct lowers asking prices across all listings, significantly at the five-percent level in four of twelve windows with parcel fixed effects, but in none with parcel-by-era fixed effects.

Homicide time-on-market estimates are predominantly positive under both specifications, but significant at the five-percent level only at seven to nine months with parcel-by-era fixed effects. Cumulative time-on-market estimates change sign across specifications and are never statistically significant (Table A15). Withdrawal estimates are positive in every window under both specifications, with significance only at two months (2.2 percentage points, s.e. 0.96).

For weapons violations, withdrawal estimates are significant at the five-percent level in eight windows with parcel fixed effects and three with parcel-by-era fixed effects. The post-off-market placebo is significant in at most one window (Table A16). Among listings that sell, asking-price estimates are negative in every window with parcel-by-era fixed effects and significant in three.

Table A14: Homicide and the Asking Price, Every Window

| | Sold listings | | All listings | |
|---|---|---|---|---|
| Window (months) | Parcel | Parcel×era | Parcel | Parcel×era |
| 1 | $-10.28^{**}$ | $-11.69^{*}$ | 0.02 | 0.12 |
| | (4.44) | (6.84) | (1.39) | (1.40) |
| 2 | −2.49 | −5.32 | 1.05 | 0.73 |
| | (3.55) | (5.60) | (1.12) | (1.17) |
| 3 | −0.90 | −0.42 | 0.10 | −0.33 |
| | (2.97) | (4.22) | (0.99) | (1.03) |
| 4 | −2.38 | −3.47 | 0.24 | −0.09 |
| | (2.43) | (3.29) | (0.87) | (0.92) |
| 5 | −1.70 | −4.25 | 0.00 | −0.25 |
| | (1.97) | (2.72) | (0.79) | (0.90) |
| 6 | −0.95 | −3.36 | 0.40 | 0.13 |
| | (1.88) | (2.45) | (0.74) | (0.87) |
| 7 | −0.44 | −2.38 | 0.49 | 0.41 |
| | (1.77) | (2.36) | (0.68) | (0.79) |
| 8 | −0.89 | −1.27 | 0.45 | 0.50 |
| | (1.67) | (2.10) | (0.64) | (0.75) |
| 9 | −1.04 | −1.05 | 0.16 | 0.41 |
| | (1.49) | (1.86) | (0.60) | (0.71) |
| 10 | −0.31 | −0.88 | 0.51 | 0.76 |
| | (1.38) | (1.74) | (0.58) | (0.67) |
| 11 | −0.44 | −1.63 | 0.38 | 0.51 |
| | (1.24) | (1.59) | (0.57) | (0.65) |
| 12 | −0.35 | −1.46 | 0.50 | 0.64 |
| | (1.17) | (1.53) | (0.55) | (0.62) |
| Listings | 235,030 | 119,425 | 985,586 | 820,301 |

*Notes.* Near–far estimate $\delta$ (percent) of homicide on the ln original asking price: incidents within 0–0.05 miles against 0.20–0.30 miles in the given number of months before the list date, with controls for incidents in the remaining months up to twelve. Sold listings ended in a sale; all listings add those withdrawn. Parcel: parcel and tract × list-month fixed effects; parcel×era: parcel-by-era and tract × list-month fixed effects. Standard errors clustered by census tract. Listings: estimation sample (largest across windows). Each listing counts separately, so a home relisted after a withdrawal contributes several listings; the sample is 1,140,708 listings (523,916 sold, 616,792 withdrawn). Parcel fixed effects identify from homes with more than one listing in the sample (singletons are dropped), and parcel×era from homes listed more than once within an era.

$^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table A15: Homicide and Time on Market, Every Window

| | Days on market | | Cumulative days on market | |
|---|---|---|---|---|
| Window (months) | Parcel | Parcel×era | Parcel | Parcel×era |
| 1 | 6.46 | −8.97 | 2.61 | 12.49 |
| | (11.91) | (16.44) | (13.39) | (18.83) |
| 2 | 8.97 | 5.74 | 2.54 | 19.55 |
| | (8.87) | (12.42) | (10.09) | (14.64) |
| 3 | 4.38 | 8.43 | −2.97 | 9.66 |
| | (7.79) | (10.95) | (8.63) | (12.35) |
| 4 | 3.08 | 13.63 | −3.13 | 7.21 |
| | (6.49) | (9.35) | (7.71) | (10.56) |
| 5 | −2.45 | 9.70 | −5.41 | 4.82 |
| | (6.31) | (8.82) | (7.27) | (9.51) |
| 6 | 1.99 | 13.51 | −2.56 | 5.24 |
| | (5.70) | (8.43) | (6.59) | (9.06) |
| 7 | 3.43 | 17.33** | −3.35 | 4.40 |
| | (5.52) | (8.13) | (5.98) | (8.74) |
| 8 | 2.36 | 14.54** | −2.22 | 6.68 |
| | (5.13) | (6.85) | (5.58) | (7.48) |
| 9 | 1.64 | 11.77** | −2.18 | 3.18 |
| | (4.74) | (5.95) | (5.17) | (6.63) |
| 10 | 0.35 | 9.05 | −1.86 | 1.67 |
| | (4.64) | (5.49) | (4.94) | (6.47) |
| 11 | 0.06 | 10.58* | −1.34 | 3.37 |
| | (4.44) | (5.56) | (4.87) | (6.31) |
| 12 | −0.20 | 8.43 | −1.55 | 1.38 |
| | (4.35) | (5.43) | (4.74) | (6.19) |
| Listings | 235,655 | 119,296 | 235,978 | 119,617 |

*Notes.* Near–far estimate $\delta$ (percent) of homicide on ln days on market, sold listings: incidents within 0–0.05 miles against 0.20–0.30 miles in the given number of months before the list date, with controls for incidents in the remaining months up to twelve. Days on market: the current listing (median 24 days; 16 in the parcel×era estimation sample); cumulative: summed across relistings of the same home (median 41; 27 in the parcel×era estimation sample). Parcel: parcel and tract × list-month fixed effects; parcel×era: parcel-by-era and tract × list-month fixed effects. Standard errors clustered by census tract. Listings: estimation sample (largest across windows). Each listing counts separately, so a home relisted after a withdrawal contributes several listings; the sample is 1,140,708 listings (523,916 sold, 616,792 withdrawn). Parcel fixed effects identify from homes with more than one listing in the sample (singletons are dropped), and parcel×era from homes listed more than once within an era.
$^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table A16: Weapons Violations: Every Window

| Outcome | FE | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | Placebo |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Ask (all listings) | Parcel | −0.13 | −0.38 | −0.26 | −0.25 | −0.19 | −0.13 | −0.16 | −0.14 | −0.11 | −0.11 | −0.09 | −0.14 | 1/12 |
| | | (0.41) | (0.33) | (0.28) | (0.26) | (0.24) | (0.21) | (0.20) | (0.20) | (0.19) | (0.18) | (0.17) | (0.17) | |
| | Parcel×era | 0.50 | 0.08 | 0.15 | −0.02 | 0.02 | −0.04 | −0.09 | −0.08 | −0.04 | −0.04 | −0.06 | −0.09 | 0/12 |
| | | (0.38) | (0.30) | (0.26) | (0.23) | (0.21) | (0.20) | (0.20) | (0.20) | (0.19) | (0.18) | (0.19) | (0.18) | |
| Ask (sold listings) | Parcel | −1.45 | −0.73 | −0.13 | −0.30 | −0.24 | −0.15 | −0.23 | −0.18 | 0.19 | 0.37 | 0.11 | −0.02 | 0/12 |
| | | (0.97) | (0.75) | (0.62) | (0.56) | (0.47) | (0.40) | (0.40) | (0.38) | (0.32) | (0.29) | (0.29) | (0.28) | |
| | Parcel×era | −2.88** | −1.85* | −1.29 | −1.26* | −0.97* | −1.12** | −1.22** | −1.05* | −0.72 | −0.34 | −0.50 | −0.54 | 0/12 |
| | | (1.23) | (1.05) | (0.81) | (0.67) | (0.57) | (0.57) | (0.60) | (0.58) | (0.51) | (0.46) | (0.48) | (0.45) | |
| Sale-to-list | Parcel | 0.21 | 0.11 | 0.17 | 0.11 | 0.15 | 0.13 | 0.09 | 0.08 | 0.13 | 0.06 | 0.03 | 0.06 | 0/12 |
| | | (0.34) | (0.25) | (0.20) | (0.16) | (0.14) | (0.13) | (0.12) | (0.11) | (0.10) | (0.09) | (0.08) | (0.08) | |
| | Parcel×era | 0.19 | 0.34 | 0.14 | 0.07 | 0.02 | 0.10 | −0.01 | −0.04 | 0.06 | 0.05 | 0.03 | 0.04 | 2/12 |
| | | (0.41) | (0.30) | (0.23) | (0.17) | (0.15) | (0.14) | (0.14) | (0.14) | (0.12) | (0.11) | (0.10) | (0.10) | |
| Price cut (pp) | Parcel | −2.56** | −0.99 | −0.71 | −0.55 | −0.02 | −0.09 | 0.02 | 0.15 | −0.01 | −0.03 | 0.05 | 0.05 | 0/12 |
| | | (1.25) | (0.89) | (0.73) | (0.62) | (0.61) | (0.54) | (0.48) | (0.45) | (0.43) | (0.41) | (0.41) | (0.37) | |
| | Parcel×era | −2.02 | −1.17 | −0.63 | −0.28 | 0.06 | −0.19 | 0.39 | 0.40 | 0.08 | −0.07 | 0.00 | 0.02 | 0/12 |
| | | (1.61) | (1.19) | (0.93) | (0.76) | (0.66) | (0.57) | (0.56) | (0.52) | (0.49) | (0.43) | (0.43) | (0.40) | |
| Days on market | Parcel | −5.96* | −2.61 | −2.12 | −0.62 | 0.13 | −0.78 | −0.53 | −0.52 | −1.27 | −1.10 | −0.87 | −0.79 | 0/12 |
| | | (3.21) | (2.64) | (1.94) | (1.59) | (1.28) | (1.18) | (1.08) | (1.04) | (0.95) | (0.87) | (0.79) | (0.79) | |
| | Parcel×era | −2.86 | −0.33 | 0.33 | 0.98 | 1.32 | −0.23 | 0.91 | −0.12 | −0.61 | −0.67 | −0.54 | −0.91 | 2/12 |
| | | (3.90) | (3.52) | (2.28) | (2.08) | (1.72) | (1.67) | (1.74) | (1.54) | (1.43) | (1.39) | (1.22) | (1.17) | |
| Withdrawal (pp) | Parcel | 0.94** | 0.59* | 0.66*** | 0.59*** | 0.69*** | 0.63*** | 0.50*** | 0.42** | 0.34** | 0.28* | 0.24* | 0.25* | 1/12 |
| | | (0.41) | (0.31) | (0.24) | (0.21) | (0.20) | (0.18) | (0.17) | (0.18) | (0.17) | (0.15) | (0.15) | (0.14) | |
| | Parcel×era | 0.91* | 0.56 | 0.67** | 0.44* | 0.55** | 0.46** | 0.31 | 0.27 | 0.18 | 0.11 | 0.09 | 0.13 | 0/12 |
| | | (0.47) | (0.35) | (0.27) | (0.24) | (0.22) | (0.21) | (0.20) | (0.20) | (0.19) | (0.17) | (0.17) | (0.16) | |
| Close | Parcel | −0.47 | 0.07 | 0.11 | −0.51 | −0.62 | −0.39 | −0.18 | −0.12 | −0.17 | 0.04 | 0.09 | 0.10 | 0/12 |
| | | (0.92) | (0.71) | (0.58) | (0.48) | (0.47) | (0.44) | (0.40) | (0.37) | (0.35) | (0.31) | (0.30) | (0.28) | |
| | Parcel×era | −0.99 | −0.38 | −0.51 | −0.93* | −1.11** | −0.72 | −0.59 | −0.32 | −0.32 | −0.10 | 0.02 | 0.12 | 0/12 |
| | | (1.20) | (0.84) | (0.69) | (0.55) | (0.47) | (0.45) | (0.40) | (0.35) | (0.33) | (0.30) | (0.29) | (0.27) | |

*Notes.* Near–far estimate $\delta$ for weapons violations (incidents within 0–0.05 miles against 0.20–0.30 miles) in the $w$ months before the list date ($w$ = column), with controls for incidents in the remaining months up to twelve; percent for log outcomes, percentage points for the price cut and withdrawal. Close: repeat sales, crime before the sale date. Parcel: parcel and tract × month fixed effects; parcel×era: parcel-by-era and tract × month fixed effects; standard errors clustered by tract. Placebo: windows (of twelve) significant at 5% when crime after the outcome is fixed replaces crime before it (after the list date for the ask, after the close date for sale-to-list, price cut, days on market and close, after the off-market date for withdrawal). The price cut in Table 4 is a separate specification: crime during the listing (list date to close date), sold listings, with fixed effects for the number of months listed (one to twelve), parcel and tract × list-month fixed effects: 1.35** (0.53); its placebo, crime in the same number of months after the close date with tract × close-month fixed effects, is 0.49 (0.60).
$^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table A19: Agent Local Experience, All Twenty-Two Offenses

| | | Listing agent, asking price | | | | Buyer's agent, closing price | | | |
|---|---|---|---|---|---|---|---|---|---|
| | | Local only | | + Total experience | | Local only | | + Total experience | |
| Offense | Window | Gradient | Sig. windows | Gradient | Sig. windows | Gradient | Sig. windows | Gradient | Sig. windows |
| *Priced* | | | | | | | | | |
| Homicide | 10 | −0.97*** | 7/12 | −0.79** | 1/12 | −1.61** | 6/12 | −1.99** | 5/12 |
| | | (0.34) | | (0.35) | | (0.75) | | (0.86) | |
| Burglary | 8 | 0.00 | 0/12 | 0.02 | 0/12 | 0.09 | 1/12 | 0.15 | 2/12 |
| | | (0.04) | | (0.04) | | (0.10) | | (0.12) | |
| Narcotics | 10 | −0.08*** | 7/12 | −0.02 | 0/12 | 0.08 | 1/12 | 0.10 | 1/12 |
| | | (0.03) | | (0.03) | | (0.10) | | (0.11) | |
| Vandalism | 6 | −0.03 | 3/12 | −0.04 | 3/12 | 0.14** | 2/12 | 0.15* | 1/12 |
| | | (0.03) | | (0.03) | | (0.06) | | (0.08) | |
| *Suggestive* | | | | | | | | | |
| Disorderly conduct | 12 | −0.07 | 0/12 | −0.02 | 0/12 | −0.47** | 4/12 | −0.64*** | 5/12 |
| | | (0.09) | | (0.10) | | (0.21) | | (0.20) | |
| Sex offenses | 12 | −0.08 | 0/12 | −0.04 | 0/12 | 0.03 | 0/12 | −0.01 | 0/12 |
| | | (0.06) | | (0.06) | | (0.14) | | (0.14) | |
| *Precise nulls* | | | | | | | | | |
| Theft | 12 | −0.01 | 0/12 | −0.01 | 0/12 | 0.00 | 0/12 | 0.01* | 1/12 |
| | | (0.00) | | (0.00) | | (0.01) | | (0.01) | |
| Motor vehicle theft | 12 | 0.02 | 1/12 | 0.00 | 0/12 | 0.00 | 4/12 | −0.03 | 1/12 |
| | | (0.02) | | (0.02) | | (0.06) | | (0.08) | |
| Robbery | 12 | −0.16*** | 10/12 | −0.17*** | 9/12 | 0.03 | 0/12 | 0.12 | 1/12 |
| | | (0.04) | | (0.04) | | (0.07) | | (0.09) | |
| Assault (non-domestic) | 12 | −0.07*** | 9/12 | −0.05*** | 7/12 | 0.01 | 3/12 | 0.03 | 3/12 |
| | | (0.01) | | (0.01) | | (0.03) | | (0.03) | |
| Assault (domestic) | 12 | −0.06** | 8/12 | −0.06** | 7/12 | −0.05 | 0/12 | −0.10* | 3/12 |
| | | (0.03) | | (0.03) | | (0.05) | | (0.06) | |
| Criminal trespass | 12 | 0.01 | 0/12 | 0.03 | 0/12 | 0.05 | 1/12 | 0.05 | 1/12 |
| | | (0.03) | | (0.03) | | (0.06) | | (0.07) | |
| Fraud | 12 | −0.04** | 9/12 | −0.05** | 9/12 | 0.01 | 0/12 | 0.03 | 0/12 |
| | | (0.02) | | (0.02) | | (0.04) | | (0.05) | |
| Prostitution | 12 | −0.03 | 0/12 | −0.06** | 1/12 | −0.13 | 1/12 | −0.15* | 0/12 |
| | | (0.03) | | (0.03) | | (0.08) | | (0.09) | |
| Forgery | 12 | −0.14 | 0/12 | −0.14 | 0/12 | −0.21* | 0/12 | −0.09 | 0/12 |
| | | (0.09) | | (0.09) | | (0.12) | | (0.13) | |
| Family offenses | 12 | −0.07 | 0/12 | −0.01 | 0/12 | 0.11 | 0/12 | −0.10 | 0/12 |
| | | (0.15) | | (0.15) | | (0.30) | | (0.29) | |
| Bad checks | 12 | −0.15 | 2/12 | −0.12 | 0/12 | −0.25 | 0/12 | −0.35 | 0/12 |
| | | (0.10) | | (0.10) | | (0.27) | | (0.28) | |
| Liquor law | 12 | −0.13 | 0/12 | −0.10 | 0/12 | 0.28 | 0/12 | 0.19 | 0/12 |
| | | (0.11) | | (0.13) | | (0.42) | | (0.51) | |
| *Inconclusive* | | | | | | | | | |
| Weapons | 12 | 0.11* | 3/12 | 0.10* | 0/12 | 0.11 | 0/12 | 0.07 | 0/12 |
| | | (0.06) | | (0.06) | | (0.08) | | (0.17) | |
| Arson | 12 | 0.05 | 0/12 | 0.03 | 0/12 | 1.93** | 6/12 | 2.74*** | 7/12 |
| | | (0.29) | | (0.30) | | (0.84) | | (1.01) | |
| Gambling | 12 | −1.00** | 3/12 | −0.65 | 0/12 | −0.82 | 0/12 | −0.65 | 0/12 |
| | | (0.49) | | (0.49) | | (1.80) | | (1.91) | |
| Stolen property | 12 | −0.33 | 0/12 | −0.41 | 0/12 | 0.12 | 0/12 | 0.48 | 0/12 |
| | | (0.39) | | (0.43) | | (1.16) | | (1.39) | |

*Notes.* Local-experience gradient, $(\text{near} \times E - \text{far} \times E) \times 100$, at each offense's main window (homicide 10, burglary 8, narcotics 10, vandalism 6 months; 12 months for other offenses), with standard errors clustered by tract in parentheses. $E$ is the agent's prior listings in the same tract; "+ Total experience" adds the agent's total prior listings in any tract, interacted with near and far exposure. "Sig. windows" counts the exposure windows (of twelve) in which the local-experience gradient is significant at five percent, as in Table 2; every window is reported in Tables OA11 and OA12. Design as in Table 5: parcel, tract×month, and agent fixed effects; $N = 972{,}629$ listings (listing agent) and 186,510 sales (buyer's agent). Offense groups as in Table 2. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table A17: Agent Experience Gradient on Price

| Offense | Discount (mean agent) | Experience gradient (per SD) no agent FE | agent FE | Sig. windows (no FE/FE) |
|---|---|---|---|---|
| *A. Asking price (listing agent)* | | | | |
| Narcotics | −0.01 (0.04) | $-0.15^{***}$ | $-0.09^{**}$ | 9/7 |
| | | (0.05) | (0.04) | |
| Vandalism | 0.02 (0.03) | $-0.11^{***}$ | $-0.07^{**}$ | 8/3 |
| | | (0.03) | (0.03) | |
| Homicide | 0.18 (0.59) | −0.82 | $-1.01^{**}$ | 0/7 |
| | | (0.63) | (0.49) | |
| Burglary | −0.05 (0.10) | −0.04 | 0.02 | 0/0 |
| | | (0.11) | (0.08) | |
| *B. Closing price (buyer's agent)* | | | | |
| Homicide | −0.94 (0.79) | −0.01 | $-1.14^{**}$ | 0/6 |
| | | (0.95) | (0.53) | |
| Narcotics | −0.09 (0.12) | $-0.29^{**}$ | 0.08 | 3/0 |
| | | (0.14) | (0.09) | |
| Vandalism | −0.13 (0.11) | −0.02 | 0.06 | 0/0 |
| | | (0.10) | (0.09) | |
| Burglary | −0.07 (0.08) | $0.14^{*}$ | 0.07 | 0/0 |
| | | (0.08) | (0.07) | |

*Notes.* Gradient per standard deviation of agent experience ($E = \ln(1 +$ prior same-agent same-tract listings), s.d. 1.20 for listing agents and 0.71 for buyer's agents with agent fixed effects, 1.20 and 0.67 without; each on its own estimation sample), each offense and panel at the window whose experience-gradient estimate is closest to the median of its estimates significant at five percent (of all twelve if none is): listing side homicide 6, burglary 2, narcotics 9, vandalism 12; buyer side homicide 10, burglary 9, narcotics 7, vandalism 2 months. Panel A: listing agent, asking price. Panel B: buyer's agent, closing price. Discount: near–far effect at mean experience. Gradients are estimated without and with agent fixed effects; Sig. windows counts significant windows of twelve, without/with. Sample: listings with an agent identifier.
$^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table A18: Agent Local versus Total Experience, All Twenty-Two Offenses

| Offense | Window | Local experience gradient (s.e.) | Total experience gradient (s.e.) | Sig. windows local / total |
|---|---|---|---|---|
| *Panel A. Listing agent, asking price (N = 972,629)* | | | | |
| *Priced* | | | | |
| Homicide | 10 | −0.79** (0.35) | −0.45 (0.48) | 1/12 / 3/12 |
| Burglary | 8 | 0.02 (0.04) | −0.06 (0.09) | 0/12 / 0/12 |
| Narcotics | 10 | −0.02 (0.03) | −0.21*** (0.05) | 0/12 / 12/12 |
| Vandalism | 6 | −0.04 (0.03) | 0.04 (0.04) | 3/12 / 0/12 |
| *Suggestive* | | | | |
| Disorderly conduct | 12 | −0.02 (0.10) | −0.22 (0.18) | 0/12 / 0/12 |
| Sex offenses | 12 | −0.04 (0.06) | −0.15* (0.09) | 0/12 / 0/12 |
| *Precise nulls* | | | | |
| Theft | 12 | −0.01 (0.00) | 0.00 (0.01) | 0/12 / 0/12 |
| Motor vehicle theft | 12 | 0.00 (0.02) | 0.04 (0.05) | 0/12 / 0/12 |
| Robbery | 12 | −0.17*** (0.04) | 0.00 (0.06) | 9/12 / 1/12 |
| Assault (non-domestic) | 12 | −0.05*** (0.01) | −0.04* (0.02) | 7/12 / 2/12 |
| Assault (domestic) | 12 | −0.06** (0.03) | −0.01 (0.04) | 7/12 / 0/12 |
| Criminal trespass | 12 | 0.03 (0.03) | −0.06 (0.05) | 0/12 / 0/12 |
| Fraud | 12 | −0.05** (0.02) | 0.05 (0.04) | 9/12 / 0/12 |
| Prostitution | 12 | −0.06** (0.03) | 0.30** (0.15) | 1/12 / 5/12 |
| Forgery | 12 | −0.14 (0.09) | −0.02 (0.08) | 0/12 / 1/12 |
| Family offenses | 12 | −0.01 (0.15) | −0.20 (0.31) | 0/12 / 0/12 |
| Bad checks | 12 | −0.12 (0.10) | −0.17 (0.22) | 0/12 / 0/12 |
| Liquor law | 12 | −0.10 (0.13) | −0.14 (0.22) | 0/12 / 0/12 |
| *Inconclusive* | | | | |
| Weapons | 12 | 0.10* (0.06) | 0.04 (0.11) | 0/12 / 0/12 |
| Arson | 12 | 0.03 (0.30) | 0.03 (0.41) | 0/12 / 1/12 |
| Gambling | 12 | −0.65 (0.49) | −0.11 (0.73) | 0/12 / 0/12 |
| Stolen property | 12 | −0.41 (0.43) | 0.33 (0.58) | 0/12 / 0/12 |
| *Panel B. Buyer's agent, closing price (N = 186,510)* | | | | |
| *Priced* | | | | |
| Homicide | 10 | −1.99** (0.86) | 0.41 (0.46) | 5/12 / 0/12 |
| Burglary | 8 | 0.15 (0.12) | −0.07 (0.05) | 2/12 / 1/12 |
| Narcotics | 10 | 0.10 (0.11) | −0.05 (0.07) | 1/12 / 0/12 |
| Vandalism | 6 | 0.15* (0.08) | −0.01 (0.05) | 1/12 / 0/12 |
| *Suggestive* | | | | |
| Disorderly conduct | 12 | −0.64*** (0.20) | 0.26* (0.16) | 5/12 / 0/12 |
| Sex offenses | 12 | −0.01 (0.14) | 0.05 (0.11) | 0/12 / 0/12 |
| *Precise nulls* | | | | |
| Theft | 12 | 0.01* (0.01) | −0.01*** (0.00) | 1/12 / 11/12 |
| Motor vehicle theft | 12 | −0.03 (0.08) | 0.03 (0.03) | 1/12 / 0/12 |
| Robbery | 12 | 0.12 (0.09) | −0.08 (0.05) | 1/12 / 0/12 |
| Assault (non-domestic) | 12 | 0.03 (0.03) | −0.01 (0.01) | 3/12 / 0/12 |
| Assault (domestic) | 12 | −0.10* (0.06) | 0.06* (0.04) | 3/12 / 0/12 |
| Criminal trespass | 12 | 0.05 (0.07) | 0.00 (0.05) | 1/12 / 0/12 |
| Fraud | 12 | 0.03 (0.05) | −0.04 (0.03) | 0/12 / 0/12 |
| Prostitution | 12 | −0.15* (0.09) | 0.03 (0.24) | 0/12 / 0/12 |
| Forgery | 12 | −0.09 (0.13) | −0.15 (0.11) | 0/12 / 0/12 |
| Family offenses | 12 | −0.10 (0.29) | 0.40 (0.43) | 0/12 / 0/12 |
| Bad checks | 12 | −0.35 (0.28) | 0.20 (0.18) | 0/12 / 3/12 |
| Liquor law | 12 | 0.19 (0.51) | 0.13 (0.27) | 0/12 / 1/12 |
| *Inconclusive* | | | | |
| Weapons | 12 | 0.07 (0.17) | 0.03 (0.09) | 0/12 / 0/12 |
| Arson | 12 | 2.74*** (1.01) | −0.97 (0.66) | 7/12 / 1/12 |
| Gambling | 12 | −0.65 (1.91) | −0.74 (0.82) | 0/12 / 1/12 |
| Stolen property | 12 | 0.48 (1.39) | −0.55 (0.64) | 0/12 / 0/12 |

*Notes.* Local experience $E = \ln(1 + \text{prior same-agent listings in the tract})$ and total experience $T$ (all prior listings) enter the same equation, each interacted with near and far exposure. Gradients are $(\text{near} \times E - \text{far} \times E) \times 100$ and $(\text{near} \times T - \text{far} \times T) \times 100$ per log unit of $E$ and per standard deviation of $T$, at each offense's main window (homicide 10, burglary 8, narcotics 10, vandalism 6 months; 12 months for other offenses); last column counts windows significant at five percent. Design as in Table 5. Correlation of $E$ and $T$: 0.36 (listing), 0.37 (buyer).
$^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

### F.1 Agent Experience: Further Checks

Three further checks refine these patterns. First, the asking-price response to listing-agent experience is negative for homicide, narcotics, and vandalism in nearly every window, both with parcel fixed effects and with agent fixed effects added (Table A17). Second, for narcotics it weakens once agents listing in a tract for the first time are allowed to differ, and for vandalism it comes mainly from those first listings (Table OA17).

Third, controlling for the agent's total experience (all prior listings, in any tract), the buyer's-agent homicide result remains ($-1.99$, s.e. 0.86; significant in five of twelve windows), and total experience has no effect on the buyer side. The listing-agent homicide result is significant only at the main window ($-0.79$, s.e. 0.35), and total experience lowers the homicide asking price in three of twelve windows. For narcotics, the asking price falls with the listing agent's total experience ($-0.21$, s.e. 0.05; significant in all twelve windows), and local experience has no effect once total experience is controlled for (Table A18). Table A19 reports these tests for all twenty-two offenses, and Tables OA11 and OA12 report the estimates at every window.

# G Robustness

Table A20: Priced Discounts with Prior Neighborhood and Housing-Market Controls

| | Homicide (10) | Burglary (8) | Narcotics (10) | Vandalism (6) |
|---|---|---|---|---|
| Baseline (no controls) | $-3.02^{**}$ | $-0.24^{**}$ | $-0.29^{**}$ | $-0.17^{**}$ |
| | (1.38) | (0.12) | (0.14) | (0.08) |
| + Building permits | $-3.01^{**}$ | $-0.25^{**}$ | $-0.29^{**}$ | $-0.17^{**}$ |
| | (1.38) | (0.12) | (0.14) | (0.08) |
| + Distinct locations reported vacant | $-3.01^{**}$ | $-0.24^{**}$ | $-0.30^{**}$ | $-0.17^{**}$ |
| | (1.38) | (0.12) | (0.14) | (0.08) |
| + 311 service requests (three types) | $-3.02^{**}$ | $-0.24^{**}$ | $-0.30^{**}$ | $-0.17^{**}$ |
| | (1.38) | (0.12) | (0.14) | (0.08) |
| + Active business licenses | $-3.02^{**}$ | $-0.24^{**}$ | $-0.30^{**}$ | $-0.17^{**}$ |
| | (1.38) | (0.12) | (0.14) | (0.08) |
| + Tree canopy | $-3.00^{**}$ | $-0.25^{**}$ | $-0.30^{**}$ | $-0.17^{**}$ |
| | (1.38) | (0.12) | (0.14) | (0.08) |
| All five neighborhood controls | $-2.98^{**}$ | $-0.26^{**}$ | $-0.30^{**}$ | $-0.17^{**}$ |
| | (1.38) | (0.12) | (0.14) | (0.08) |
| Five housing-market controls jointly | $-2.98^{**}$ | $-0.24^{**}$ | $-0.29^{**}$ | $-0.17^{**}$ |
| | (1.38) | (0.12) | (0.14) | (0.08) |
| All ten controls | $-2.94^{**}$ | $-0.26^{**}$ | $-0.30^{**}$ | $-0.17^{**}$ |
| | (1.37) | (0.12) | (0.14) | (0.08) |

*Notes.* Near–far estimates $\delta$ (percent per incident) at each offense's main window: homicide 10, burglary 8, narcotics 10, and vandalism 6 months. All specifications include parcel and tract×month fixed effects; standard errors clustered by census tract are in parentheses. $N = 242{,}815$ throughout. Row 1 reproduces the main-window estimates of Table A3. Subsequent rows add neighborhood controls individually and jointly, housing-market controls jointly, and all ten controls. Each control enters separately for the near (0–0.05 miles) and far (0.20–0.30 miles) rings. Permits, distinct locations reported vacant, and 311 requests are counted 361–720 days before closing; business licenses are those active at the start of that period. These measures enter as $\ln(1 + x)$; tree canopy enters as percent cover from the last annual raster completed before the crime window opens. Housing-market controls, measured over the same twelve months, comprise neighboring repeat-sale price changes, the distressed-sale share, the sale-to-list ratio, days on market, and new listings. Unavailable controls are set to zero with missing-value indicators to preserve the sample. Appendix A.2 provides construction details. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table A21: Hedonic Estimates on All Sales

| Offense | $\delta$ (%) | (s.e.) | Window |
|---|---|---|---|
| *Priced offenses* | | | |
| Homicide | $-1.86^{***}$ | (0.71) | 10 |
| Burglary | $-0.47^{***}$ | (0.13) | 8 |
| Narcotics | $-0.52^{***}$ | (0.09) | 10 |
| Vandalism | $-0.62^{***}$ | (0.11) | 6 |
| *No discount under repeat sales* | | | |
| Theft | 0.01 | (0.08) | 12 |
| Motor-vehicle theft | $-0.46^{**}$ | (0.20) | 12 |
| Robbery | $-0.72^{***}$ | (0.20) | 12 |
| Assault (non-dom.) | $-0.52^{***}$ | (0.09) | 12 |
| Assault (dom.) | $-0.70^{***}$ | (0.10) | 12 |

*Notes.* Cross-sectional hedonic near–far regression on all sold listings ($N$ = 490,563), replacing the parcel fixed effect with observed characteristics (bedrooms, bathrooms, ln living area, ln lot size, age, stories, property type, new construction, and units, with missing indicators) plus tract×month fixed effects, clustered by tract. Regressors are the 0–0.05 and 0.20–0.30 mi counts in the $w$ months before sale and the same-ring counts in months ($w$+1)–12. Each cell reports $\delta = (\beta_{\text{near}} - \beta_{\text{far}}) \times 100$ at the offense's main window (homicide 10, burglary 8, narcotics 10, vandalism 6 months; 12 for the other offenses).
$^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

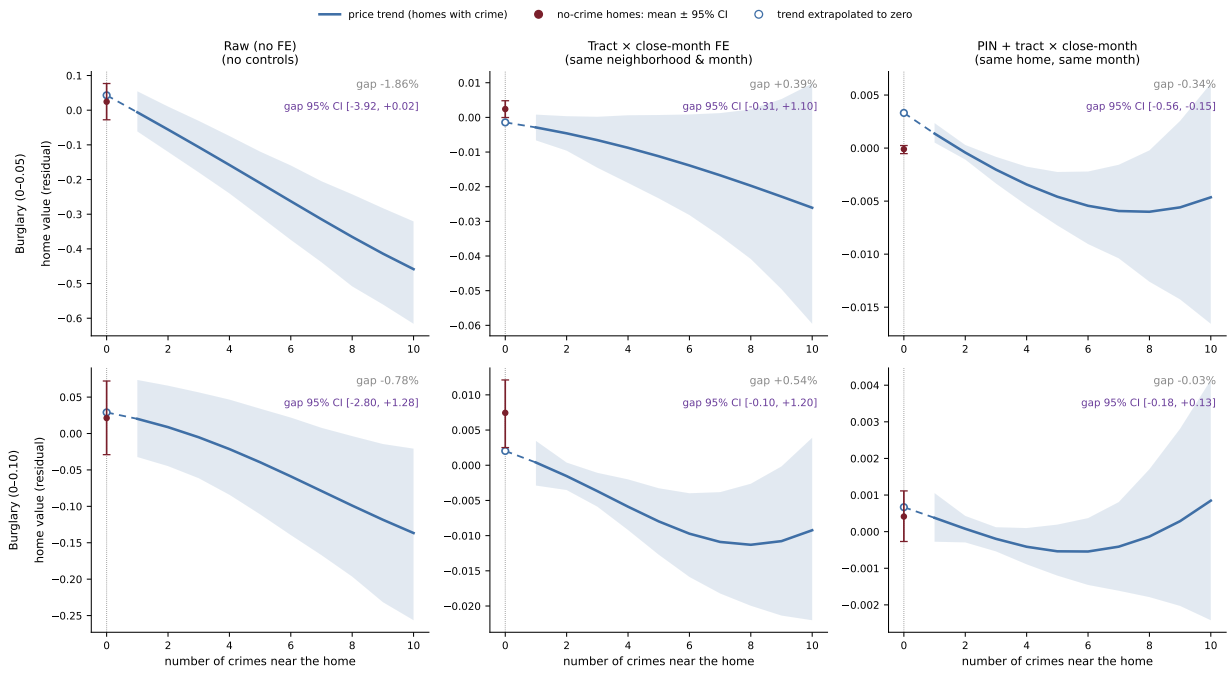



Figure A3: Burglary: Boundary Diagnostic at Two Near Rings

*Notes.* Top row: near ring 0–0.05 miles; bottom row: 0–0.10 miles. Columns add fixed effects from left to right; the right column is the baseline design (parcel and tract×close-month fixed effects). Blue: price trend for homes with nearby burglary; red: homes with none (mean and 95 percent interval); open circle: the trend extrapolated to zero. The gap shown in each panel is the mean residual price of no-crime homes minus the fitted trend extrapolated to zero; the formal test is the boundary-dummy coefficient in Table OA29 (−0.98, s.e. 0.32 at 0–0.05 miles; −0.06, s.e. 0.27 at 0–0.10 miles). Fewer than one home in ten (9 percent) has more than five incidents within 0.10 miles, so the trend beyond that point is imprecise.

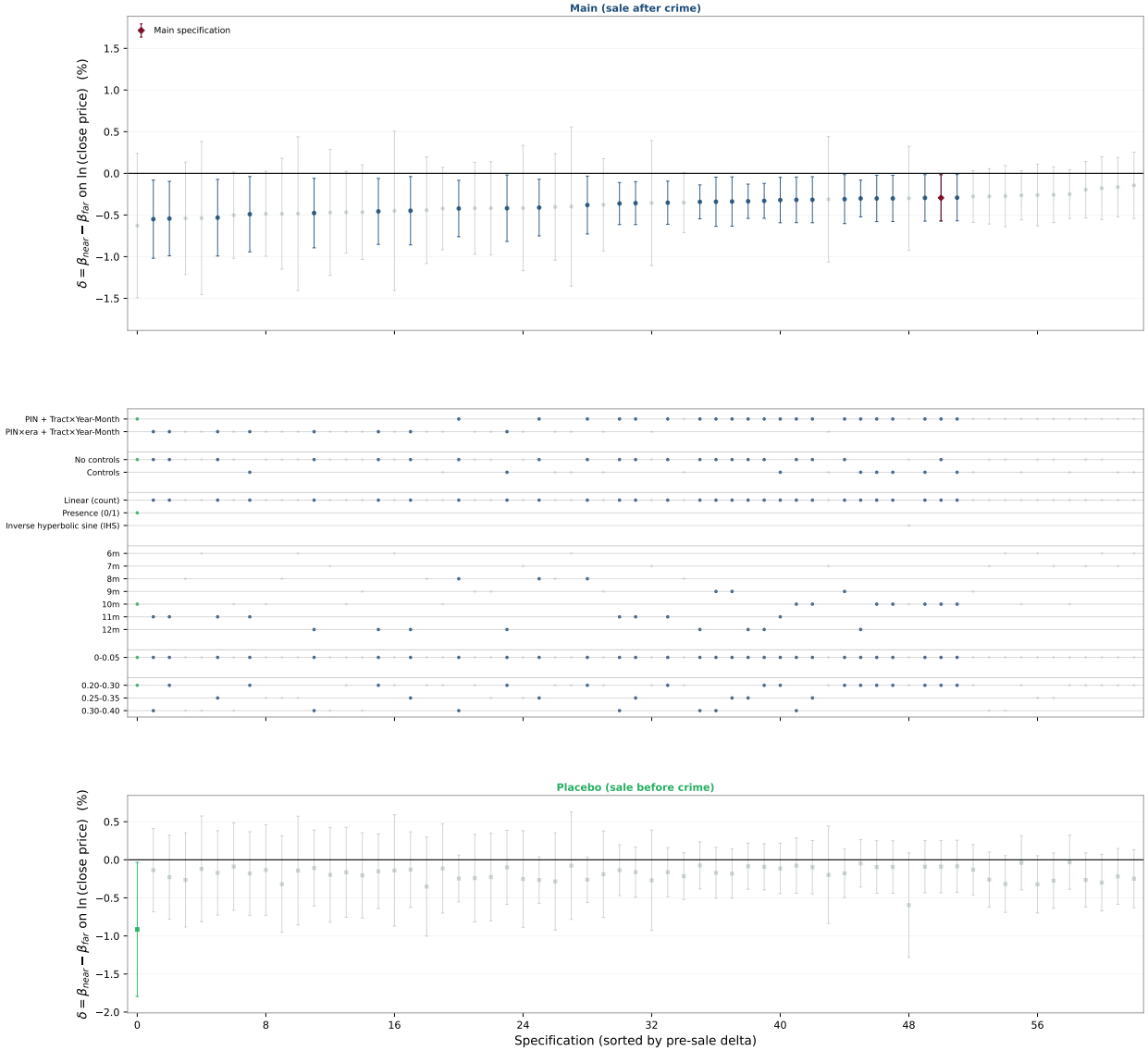



Figure A4: Narcotics: Sale-Price Discount Across Specifications

*Notes.* The 63 specifications vary the exposure window (six to twelve months), the fixed effects (parcel + tract×month, or parcel×era + tract×month), the comparison ring (0.20–0.30, 0.25–0.35, or 0.30–0.40 mile), the functional form (linear count, presence indicator, or inverse hyperbolic sine), and the predetermined controls (none, all five jointly, or one at a time); see Section 6.3. Top panel: $\delta$ (%) with 95 percent intervals, sorted from most to least negative. The baseline specification is the linear count with parcel + tract×month fixed effects, the 0.20–0.30 mile comparison ring, no controls, and a ten-month window. Middle panel: specification choices. Bottom panel: post-sale placebo. Narcotics: 29/63 significant at 5%, placebo 1/63.

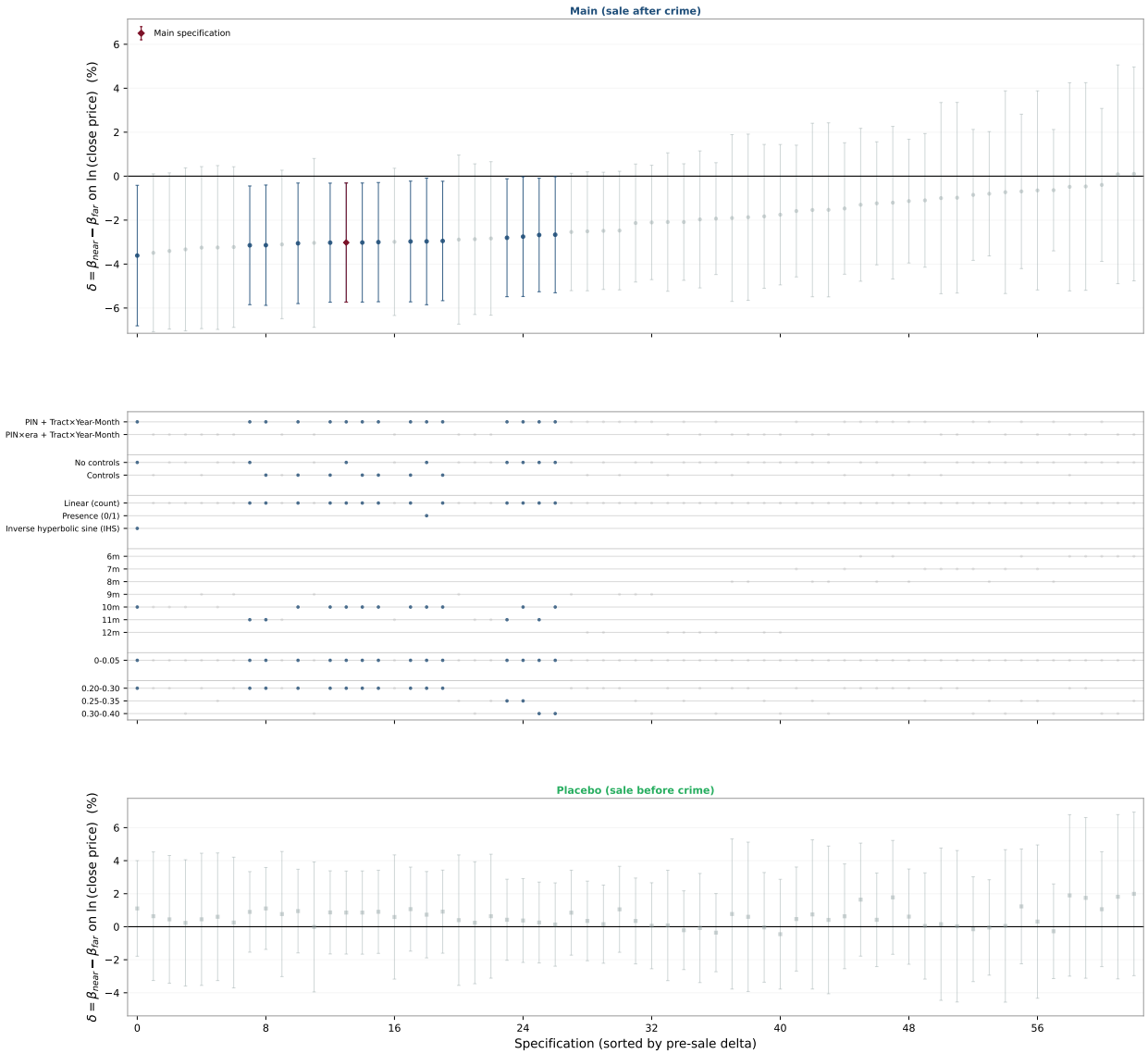



Figure A5: Homicide: Sale-Price Discount Across Specifications

*Notes.* The 63 specifications vary the exposure window (six to twelve months), the fixed effects (parcel + tract×month, or parcel×era + tract×month), the comparison ring (0.20–0.30, 0.25–0.35, or 0.30–0.40 mile), the functional form (linear count, presence indicator, or inverse hyperbolic sine), and the predetermined controls (none, all five jointly, or one at a time); see Section 6.3. Top panel: $\delta$ (%) with 95 percent intervals, sorted from most to least negative. The baseline specification is the linear count with parcel + tract×month fixed effects, the 0.20–0.30 mile comparison ring, no controls, and a ten-month window. Middle panel: specification choices. Bottom panel: post-sale placebo. Homicide: 15/63 significant at 5%, placebo 0/63.

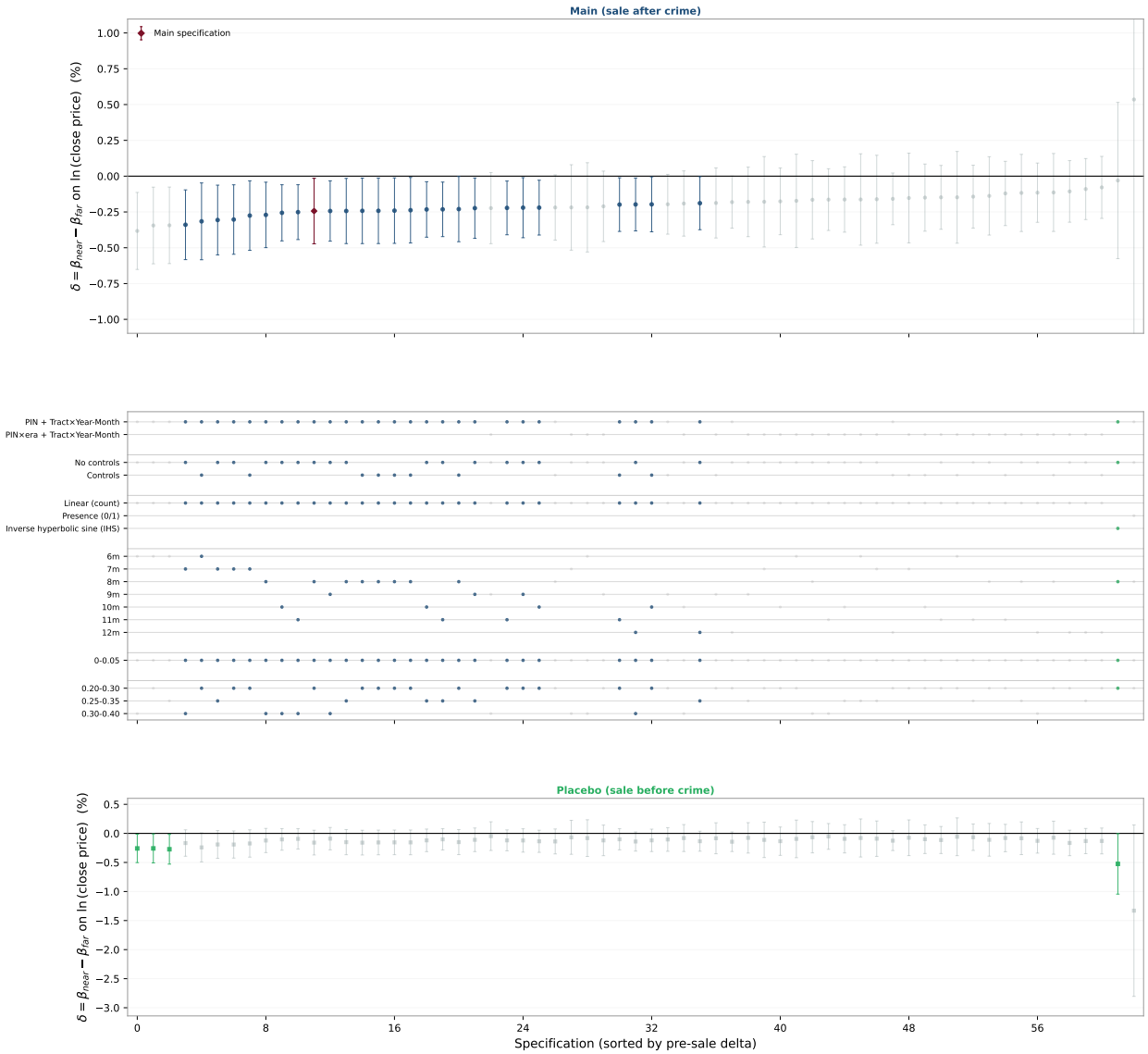



Figure A6: Burglary: Sale-Price Discount Across Specifications

*Notes.* The 63 specifications vary the exposure window (six to twelve months), the fixed effects (parcel + tract×month, or parcel×era + tract×month), the comparison ring (0.20–0.30, 0.25–0.35, or 0.30–0.40 mile), the functional form (linear count, presence indicator, or inverse hyperbolic sine), and the predetermined controls (none, all five jointly, or one at a time); see Section 6.3. Top panel: $\delta$ (%) with 95 percent intervals, sorted from most to least negative. The baseline specification is the linear count with parcel + tract×month fixed effects, the 0.20–0.30 mile comparison ring, no controls, and an eight-month window. Middle panel: specification choices. Bottom panel: post-sale placebo. Burglary: 29/63 significant at 5%, placebo 4/63.

# H Rental Prices

I apply the same near–far design to lease listings, with the log asking rent as the outcome (Table A22). Three offenses lower the asking rent, but the estimates are sensitive to the fixed effects: burglary under parcel fixed effects (6 of 12 windows at five percent), sex offenses consistently under parcel-by-era fixed effects (7 of 12 windows), but rarely under parcel fixed effects (1 of 12), and liquor violations only under parcel-by-era fixed effects (4 of 12); all three survive the post-date placebo (0 of 12 windows). Narcotics leaves rents unchanged under parcel fixed effects, where its post-date placebo is clean, though the placebo fails under parcel-by-era fixed effects; homicide is estimated too imprecisely in the lease sample to sign, in line with Klimova and Lee (2014), who find no effect of nearby murders on Sydney rents. Narcotics instead appears in the leasing frictions of Panel B: it raises the probability a listing is a relisting (the unit had an earlier spell on the market) in 11 of 12 windows under each specification, and the result holds across all twenty-two robustness specifications (Figure OF2), with a null future-crime placebo. The full offense battery is in Table A23.

Table A22: The Rental Market

| | PIN | | | PIN×era | | |
|---|---|---|---|---|---|---|
| Offense | Min (%) | Max (%) | Sig. (plac.) | Min (%) | Max (%) | Sig. (plac.) |
| *A. Asking rent (percent)* | | | | | | |
| *Priced (asking rent lower; identifying FE in note)* | | | | | | |
| Burglary | −0.34*** (0.12) | −0.07 (0.06) | 6/12 (0) | −0.30** (0.12) | −0.04 (0.07) | 1/12 (0) |
| Sex offenses | −0.29* (0.15) | −0.21* (0.13) | 1/12 (0) | −0.32 (0.27) | −0.16 (0.14) | 7/12 (0) |
| Liquor law | −0.23 (0.26) | −0.01 (0.13) | 0/12 (0) | −0.30*** (0.07) | 0.09 (0.49) | 4/12 (0) |
| *Precise nulls (rent effect near zero, tightly estimated)* | | | | | | |
| Narcotics† | 0.01 (0.06) | 0.10 (0.08) | 0/12 (0) | 0.02 (0.17) | 0.14 (0.09) | 0/12 (4) |
| Vandalism | −0.06 (0.06) | 0.05 (0.07) | 0/12 (0) | −0.03 (0.04) | 0.09 (0.07) | 0/12 (1) |
| Prostitution | −0.16 (0.10) | 0.00 (0.17) | 0/12 (1) | −0.33*** (0.12) | −0.03 (0.14) | 1/12 (0) |
| *Imprecise / underpowered* | | | | | | |
| Homicide | −0.55 (0.61) | 1.03 (1.20) | 0/12 (0) | −0.59 (0.59) | 1.16 (1.22) | 0/12 (0) |
| *B. Leasing frictions* | | | | | | |
| Narcotics, relisting (pp) | 0.18** (0.08) | 0.97*** (0.34) | 11/12 (0) | 0.27*** (0.08) | 1.28*** (0.40) | 11/12 (0) |
| Homicide, time on market (%) | 1.01 (4.65) | 12.19 (11.00) | 0/12 (0) | 1.54 (3.98) | 19.63* (11.70) | 4/12 (0) |
| Burglary, withdrawn (pp)† | −1.06** (0.46) | −0.02 (0.16) | 3/12 (0) | −1.12** (0.53) | −0.08 (0.15) | 5/12 (2) |

*Notes.* Near–far estimates on lease listings, 2007–2025, tract × listing-month fixed effects, clustered by tract (118,849 listings with parcel fixed effects; 98,813 with parcel × era). For each specification, Min and Max are the most negative and most positive estimates across the twelve windows, followed by the number significant at 5%; in parentheses, windows with a significant post-listing placebo; †, placebo fails in two or more. Panel A: log asking rent. Panel B: relisting (the unit had an earlier spell on the market) and withdrawal (percentage points), time on market (percent). $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Listing agents' local experience is associated with shorter marketing times near narcotics incidents. Table OA19 shows how the listing agent's experience in the tract moderates the effect of nearby crime on rental listing outcomes. Near narcotics incidents, more experienced agents lease units faster, with days on market about 0.6 percent shorter per standard deviation of experience, and set slightly lower asking rents; the average rental discount stays small. The tenant's agent's experience is unrelated to asking rents or days on market in any window.

**Specification curves: rents** The narcotics association with relisting persists across all twenty-two rental specifications. The rental specification curves cover twenty-two specifications: eleven exposure windows (two to twelve months) under each of two fixed-effect designs (parcel, and parcel by era). The narcotics estimate on the probability a lease listing is a relisting (the unit had an earlier spell on the market) is positive and significant in all twenty-two, with no significant placebo (Figure OF2). The sex-offense estimate on the asking rent is significant in eight of the same twenty-two specifications, with no significant placebo (Figure OF3).

Table A23: The Rental Market: Full Offense Battery

| | PIN | | | PIN×era | | |
|---|---|---|---|---|---|---|
| Offense | Min (%) | Max (%) | Sig. (plac.) | Min (%) | Max (%) | Sig. (plac.) |
| *A. Asking rent (percent)* | | | | | | |
| *Priced (asking rent lower; identifying FE in note)* | | | | | | |
| Burglary | −0.34*** (0.12) | −0.07 (0.06) | 6/12 (0) | −0.30** (0.12) | −0.04 (0.07) | 1/12 (0) |
| Sex offenses | −0.29* (0.15) | −0.21* (0.13) | 1/12 (0) | −0.32 (0.27) | −0.16 (0.14) | 7/12 (0) |
| Liquor law | −0.23 (0.26) | −0.01 (0.13) | 0/12 (0) | −0.30*** (0.07) | 0.09 (0.49) | 4/12 (0) |
| *No consistent rent effect (near zero; prostitution's parcel×era minimum is significant at one window)* | | | | | | |
| Narcotics† | 0.01 (0.06) | 0.10 (0.08) | 0/12 (0) | 0.02 (0.17) | 0.14 (0.09) | 0/12 (4) |
| Vandalism | −0.06 (0.06) | 0.05 (0.07) | 0/12 (0) | −0.03 (0.04) | 0.09 (0.07) | 0/12 (1) |
| Larceny/theft† | −0.01** (0.00) | 0.00 (0.01) | 6/12 (2) | −0.02 (0.01) | 0.00 (0.01) | 0/12 (0) |
| Assault (non-dom.) | −0.06 (0.04) | −0.03 (0.02) | 2/12 (0) | −0.05 (0.04) | −0.01 (0.02) | 0/12 (0) |
| Motor-vehicle theft | −0.12 (0.08) | −0.01 (0.04) | 0/12 (0) | −0.13 (0.09) | −0.01 (0.03) | 0/12 (0) |
| Robbery | 0.01 (0.04) | 0.07 (0.06) | 0/12 (0) | 0.04 (0.05) | 0.12 (0.11) | 0/12 (0) |
| Assault (dom.)† | −0.03 (0.05) | 0.02 (0.09) | 0/12 (5) | −0.06 (0.11) | 0.03 (0.08) | 0/12 (0) |
| Fraud | −0.01 (0.04) | 0.08 (0.06) | 0/12 (0) | 0.03 (0.04) | 0.07*** (0.03) | 0/12 (1) |
| Trespass | −0.17* (0.10) | 0.01 (0.09) | 0/12 (0) | −0.13 (0.10) | 0.05 (0.10) | 0/12 (0) |
| Forgery | −0.08 (0.15) | 0.12 (0.10) | 0/12 (0) | 0.06 (0.16) | 0.22 (0.16) | 0/12 (0) |
| Prostitution | −0.16 (0.10) | 0.00 (0.17) | 0/12 (1) | −0.33*** (0.12) | −0.03 (0.14) | 1/12 (0) |
| *Imprecise / underpowered (too sparse near leases to sign)* | | | | | | |
| Homicide | −0.55 (0.61) | 1.03 (1.20) | 0/12 (0) | −0.59 (0.59) | 1.16 (1.22) | 0/12 (0) |
| Disorderly conduct | −0.40 (0.30) | 0.15 (0.25) | 0/12 (0) | −0.25 (0.16) | 0.40 (0.29) | 0/12 (0) |
| Weapons | 0.02 (0.26) | 0.17 (0.22) | 0/12 (0) | −0.08 (0.23) | 0.22 (0.22) | 0/12 (0) |
| Arson | −0.21 (0.47) | 0.20 (0.47) | 0/12 (0) | −0.27 (0.43) | 0.54 (0.85) | 0/12 (0) |
| Bad checks† | −0.03 (0.18) | 0.82** (0.37) | 0/12 (3) | −0.16 (0.17) | 0.67 (0.43) | 0/12 (0) |
| Family (non-violent) | 0.07 (0.22) | 0.36* (0.21) | 0/12 (0) | 0.11 (0.38) | 0.68 (0.50) | 0/12 (0) |
| Kidnapping† | −0.70* (0.36) | −0.30 (0.27) | 0/12 (3) | −0.62 (0.72) | 0.05 (0.29) | 0/12 (0) |
| Embezzlement | −0.39 (0.34) | 0.96 (0.94) | 0/12 (0) | −0.06 (0.48) | 0.94 (0.83) | 0/12 (0) |
| Child pornography/obscenity | −0.02 (0.56) | 0.40 (0.74) | 0/12 (0) | −0.34 (1.06) | 0.34 (0.51) | 0/12 (0) |
| Extortion† | −1.30* (0.70) | −0.28 (0.33) | 2/12 (3) | −1.13** (0.51) | −0.19 (0.26) | 2/12 (5) |
| Gambling† | −1.50 (1.44) | 3.54 (2.25) | 0/12 (4) | −1.25 (1.51) | 2.88 (4.05) | 0/12 (12) |

*Notes.* Asking-rent near–far estimates on lease listings for the full offense battery, the near ring against the 0.20–0.30 mile far ring, pooled 2007–2025 with tract×listing-month fixed effects, clustered by tract (118,849 listings under parcel fixed effects; 98,813 under parcel×era). Under each fixed-effect specification, parcel (PIN) and parcel×era, Min and Max are the most negative and most positive estimate across the twelve exposure windows (standard error in parentheses), and Sig. counts windows significant at five percent; the parenthetical is the number of windows in which the post-date placebo (crime after the listing) is significant. A dagger (†) marks an offense whose placebo fails in two or more windows, shown only for completeness. Priced: asking rent lower, two or more significant windows under an identifying fixed-effect specification with a clean placebo. The compact version is Table A22. "Child pornography/obscenity" is predominantly child pornography (IUCR 1582). Non-domestic assault meets the window rule (2/12, clean placebo), but its estimates are small (about −0.06%) and it is not counted as priced.

$^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

# I Heterogeneity across Eras

The point estimates differ across macroeconomic eras, but the differences are not statistically significant (Table A24). Burglary and vandalism discounts are larger in the 2009–2013 foreclosure

crisis, homicide's in the 2014–2025 recovery, and narcotics is negative in each era (crisis $-0.33$, s.e. 0.23; recovery $-0.52$, s.e. 0.31). For no priced offense does a Wald test reject equal crisis and recovery discounts ($p = 0.41$ for homicide, 0.26 for burglary, 0.63 for narcotics, and 0.10 for vandalism). The crisis estimates for rare offenses rest on few within-crisis repeat sales and are imprecise, as their standard errors show.

Table A24: Price Effects of Crime by Era

| Offense | Identifying parcels | Pooled 2007–25 | Crisis 2009–13 | Recovery 2014–25 |
|---|---|---|---|---|
| *Priced* | | | | |
| Homicide | 2,630 | $-3.02^{**}$ | $-0.07$ | $-4.12^{***}$ |
| | | (1.38) | (4.73) | (1.47) |
| Narcotics | 32,563 | $-0.29^{**}$ | $-0.33$ | $-0.52^{*}$ |
| | | (0.14) | (0.23) | (0.31) |
| Burglary | 55,782 | $-0.24^{**}$ | $-0.49^{*}$ | $-0.18$ |
| | | (0.12) | (0.25) | (0.14) |
| Vandalism | 64,964 | $-0.17^{**}$ | $-0.54^{**}$ | $-0.11$ |
| | | (0.08) | (0.25) | (0.10) |
| *Suggestive* | | | | |
| Disorderly conduct | 4,436 | $-0.65$ | $-1.77$ | 0.50 |
| | | (0.63) | (1.21) | (0.74) |
| Sex offenses | 22,139 | $-0.26$ | 0.84 | $-0.27$ |
| | | (0.25) | (0.83) | (0.24) |
| *Precise nulls* | | | | |
| Larceny/theft | 82,677 | 0.02 | 0.00 | $0.03^{***}$ |
| | | (0.01) | (0.05) | (0.01) |
| Motor vehicle theft | 61,445 | 0.07 | 0.00 | 0.09 |
| | | (0.09) | (0.27) | (0.10) |
| Assault (non-dom.) | 46,861 | $-0.03$ | $-0.21$ | 0.02 |
| | | (0.09) | (0.26) | (0.09) |
| Family (non-violent) | 7,862 | $1.49^{**}$ | 0.99 | $2.05^{**}$ |
| | | (0.60) | (1.08) | (0.86) |
| Assault (domestic) | 54,331 | $-0.03$ | 0.18 | $-0.02$ |
| | | (0.12) | (0.28) | (0.14) |
| Trespass | 7,409 | $-0.09$ | $-0.49$ | 0.05 |
| | | (0.38) | (0.93) | (0.39) |
| Robbery | 46,644 | 0.01 | 0.01 | $-0.04$ |
| | | (0.13) | (0.30) | (0.14) |
| Liquor law | 4,308 | 0.51 | 0.19 | 0.40 |
| | | (0.36) | (0.80) | (0.31) |
| Parcel FE | | Yes | Yes | Yes |
| Tract × year-month FE | | Yes | Yes | Yes |
| Crime × era interaction | | No | Yes | Yes |

*Notes.* $\delta$ as in Table 2. Offense groups as in Table 2: *priced*, a significant discount at two or more windows; *suggestive*, a significant discount at only one window; *precise nulls*, no significant discount and intervals that rule out a one-percent discount in at least nine of twelve windows; *inconclusive*, no significant discount but intervals too wide to rule out a one-percent discount. Crisis (2009–2013) and recovery (2014–2025) effects come from an era-interacted model with 2007–2008 as the base; the pooled column is the full-sample effect and need not lie between them. The priced offenses are shown at their main windows (homicide 10, burglary 8, narcotics 10, vandalism 6 months) and the other offenses at the window of their median estimate in Table 2 (the sixth-lowest of the twelve pooled estimates), held fixed across the three columns; identifying parcels (exposure varies within parcel) are counted at that window, whereas Table 2 counts them at twelve months. The crisis and recovery columns report per-era near–far levels, not deviations from the pooled column.
$^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

## Supplementary Material

The *Online Appendix* contains additional tables, figures, and data details.